Finite Prior-Particle Likelihood Objectives Can Reward Risk-Increasing Dependence

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Abstract

Latent-variable models are often fitted by estimating the marginal likelihood with an average of K conditional likelihoods at independent prior draws and minimizing the expected negative logarithm of that estimate. For fixed K this is a criterion of its own, not a noisier copy of the log likelihood: estimating it more precisely does not turn it into one. We ask which dependence structures it favours, and whether they agree with the KL risk of the integrated model. In a Gaussian–logistic model we append n observed fair coins. After integration they leave every candidate's risk unchanged; inside the particle average, their per-particle likelihoods change which particles dominate. As n grows, with the active Gaussian marginals fixed and $K \geq 2$ at most polynomial in n, we identify the first term that distinguishes correlations, uniformly and for the actual non-Gaussian weights. It is a strictly concave potential that rewards larger variance of the active log-likelihood across prior draws. Population near-minimizers that resolve this shrinking term approach the spectral boundary of the correlation class and retain positive risk. With no appended block and all parameters free, a computer-assisted proof gives an eight-particle population local minimum whose single-variable marginals are all correct but whose joint KL is about 0.0104; from that point, every sufficiently small nonzero step along a marginal-preserving path lowers the risk and raises the objective. Fitting the finite objective accurately, or fitting the marginals correctly, therefore does not settle which dependence is learned.

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1. Introduction

Many latent-variable models have an intractable marginal likelihood $p_\theta(y) = E_Z\, p_\theta(y \mid Z)$. A simple replacement draws K independent prior particles $Z_1, \ldots, Z_K$ and minimizes the expected negative logarithm of their average likelihood. It appears in simulated maximum likelihood (Lee, 1995), in prior-sampled learning of binary latent models (Tipping, 1998), and as the importance-weighted bound with its proposal fixed to the prior (Burda et al., 2016). Because the logarithm is concave, the objective is biased, and the bias depends on the parameters (Maddison et al., 2017; Sung & Geyer, 2007). At fixed K it is therefore a different criterion from the likelihood. More outer repetitions estimate the same finite-K objective more precisely; they do not turn it into integrated log likelihood.

This paper asks a question that the general bias statement leaves open: *which dependence structure does a finite prior-particle objective prefer, and when does that preference disagree with the KL risk $D(P\|Q_\theta)$ of the integrated model Q_θ?*

A controlled comparison. Take three binary observations whose latent Gaussian logits are independent, and use $K = 100$ particles. Now append n further observed binary variables that are fair coins independent of everything else. Each coin has its own latent logit $G_j \sim N(0,1)$ and emission probability $\sigma(u_j G_j)$ for outcome $u_j \in \{-1, 1\}$. Averaged over the prior, every coin has probability $1/2$ under every candidate. Each candidate's integrated model is therefore its old model times 2^{-n}, and every risk is unchanged. Inside the particle average the coins act differently. Particle k carries the factor $\prod_j \sigma(u_j G_{kj})$, which varies from particle to particle even though its prior mean is 2^{-n}. These factors decide which particles dominate the average, and hence how the active likelihoods compete. Integrating the coins out *before* averaging would instead contribute the common constant 2^{-n} and change nothing.

Figure 1 compares $n = 0$ with $n = 256$ on the same active observations, active particles and candidates. Without coins, the finite objective prefers the true independent model to an equicorrelated candidate C_b with all correlations .9, and the truth is a strict local minimum. With 256

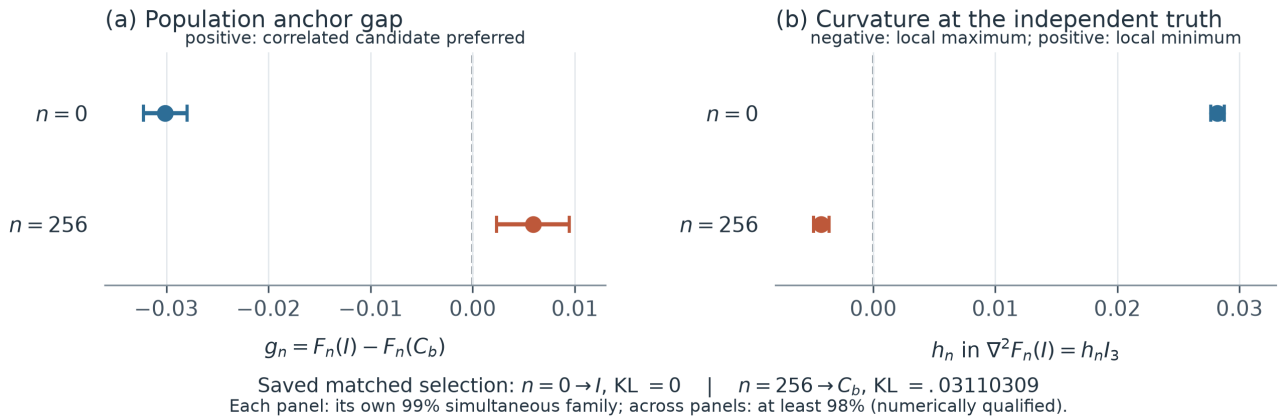


Figure 1. The fair-coin control. Three binary variables with independent latent logits ($C_* = I$), $K = 100$ particles per logarithm. (a) Population gap $g_n = F_n(I) - F_n(C_b)$ between the truth and the equicorrelated candidate C_b; positive values mean the finite objective prefers C_b. (b) Curvature h_n of the objective at the truth, where $\nabla^2 F_n(I) = h_n I_3$ in the three free correlations; negative values make the truth a local maximum. At $n = 256$, the appended fair coins' per-particle likelihoods stay inside the particle average; active observations, active particles and candidates are shared with $n = 0$. Bars are 99% simultaneous intervals for each panel under ideal iid sampling and stated numerical allowances (Appendix C.2, C.4); positivity of h_0 is also proved analytically (C.3). The footer reports one matched 143-point grid selection (C.1).

coins the preference reverses: C_b scores better, the truth becomes a local maximum, and a matched grid search returns C_b, whose risk is .031 nats.

We regard this construction as a control, not as a recommended practice. The appended block is known to be independent, and its likelihood factorizes exactly. Its value is that it changes the estimator while preserving the active likelihood ratios and every candidate's KL risk. Whatever changes in Figure 1 is therefore the criterion's own preference.

Why a fine scale is needed. As n grows, nuisance log-weights spread on a scale $\sqrt{n}$, and the average is dominated by the particle with the largest nuisance weight. With fixed active marginals, the expected loss of one prior particle is the same for every candidate correlation matrix, so the leading term of the objective does not see dependence at all. Heuristically, the leading selection term in this regime comes from events where the two leading particles are nearly tied, and such events have frequency of order $w_{n,K} \asymp \sqrt{\log K/n}$. Selection therefore happens at a scale that shrinks with n, and this has two consequences. An approximation that gets the objective right to $o(1)$ can be silent about which correlations are chosen; it must be accurate relative to $w_{n,K}$. And identifying the preference requires controlling near-ties for the actual, non-Gaussian log-likelihood weights, uniformly over candidates. A Gaussian model of the log-weights formally predicts the large-K coefficient but does not supply this control (Section 6).

Theorem 1 supplies it. The resulting potential is strictly concave and has a direct reading: it rewards correlation matrices under which the active log-likelihood of an observation varies more across prior draws. For two variables with true correlation zero, the exact likelihood pulls a fitted correlation back toward zero, while the limiting particle objective pushes it away. Theorem 2 combines concavity with the classical geometry of correlation matrices. Near-minimizers that resolve the scale $w_{n,K}$ are driven to the spectral boundary of the class and keep their risk bounded away from zero.

Without an appended block. The coins make the mechanism visible, but they leave a natural question. Suppose there is no designated independent block, every bias and loading is learnable, and a local minimum of the finite objective already reproduces every observation marginal. Does the objective then at least not oppose dependence changes that would lower the risk? Theorem 3 answers no for a specific eight-item, two-factor model with $K = 8$. A computer-assisted proof certifies a rank-one strict local minimum of the exact population objective in all 24 parameters, modulo common rotations. All marginals are correct and the joint KL is about .0104. Along a path that rotates loading rows without changing their lengths, every sufficiently small nonzero step lowers the risk and raises the objective. The two parts thus answer successive questions about one criterion: an asymptotic statement about *selection* in a controlled family, then a finite, *local* statement about geometry without that control. Neither is derived from the other.

Contributions.

- A uniform selection-scale theorem for the actual non-Gaussian weights, covering every fixed $K \geq 2$

and K growing polynomially in n, together with a conditional-variance reading of the limiting potential (Theorem 1).
- Its geometric consequence: a spectral-boundary preference and positive limiting risk for population near-minimizers with regret $o(w_{n,K})$ (Theorem 2).
- A finite-scale analysis: a signed fixed-K correction isolates the same-particle coupling that scalar overlap misses, and explicit bounds certify the leading sign of one coordinate at $n = 4096$ (Section 4).
- A computer-assisted certificate of a marginal-correct, risk-positive local minimum with a same-root, marginal-preserving conflict, persisting over a small explicit teacher family (Theorem 3 and its corollary).

The main theory concerns population objectives with particles drawn from the prior; Section 4.3 adds guarantees for one specified finite program. None describes where a stochastic optimizer converges.

2. Prior-Particle Objectives and a Controlled Augmentation

For latent $Z \sim p(z)$ and conditional likelihood $p_\theta(y \mid z)$, the K-particle objective is $E_{Y\sim P} E_{Z_{1:K}}[-\log K^{-1}\sum_k p_\theta(Y \mid Z_k)]$, with iid prior particles independent of Y. At $K = \infty$ it becomes the integrated cross entropy. Our first model makes the active correlations the only unknowns. Fix $m \geq 2$, a finite mean μ and $0 < \delta < 1$:

$$\begin{aligned} \mathcal{E}_\delta^{(m)} &= \{C = C^\top : \operatorname{diag} C = \mathbf{1},\ C \succeq \delta I\}, \\ X_A &= \mu + L_C Z, \qquad L_C L_C^\top = C, \\ X_{m+j} &= \tau_j G_j, \\ P(Y_i = y_i \mid X_i) &= \sigma(y_i X_i), \qquad y_i \in \{-1, 1\}. \end{aligned} \tag{1}$$

All Gaussian inputs are independent standard normals before applying L_C, and observations are conditionally independent given their logits. Active variances and the mean are known, and active and nuisance logits are uncorrelated. We write U for the nuisance observations and assume the teacher C_* lies in the relative interior, $C_* \succ \delta I$. Because each nuisance coin is fair after integration,

$$Q_{C,\tau}^{(n)}(y, u) = 2^{-n} Q_C^A(y), \qquad \mathcal{R}(C) = D(Q_{C_*}^A \| Q_C^A). \tag{2}$$

This is also the KL risk of the complete observation. Changing nuisance scales at fixed n leaves the whole observation law unchanged; changing n leaves active likelihood ratios and the KL unchanged. With K iid prior particles (Z_k, G_k) independent of the observation,

$$\begin{aligned} \widehat{Q}_{C,\tau,K}^{(n)}(y, u) &= K^{-1} \sum_{k=1}^{K} p_{C,\tau}(y, u \mid Z_k, G_k), \\ F_{n,K}(C; \tau) &= E_{Y,U,Z,G}[-\log \widehat{Q}_{C,\tau,K}^{(n)}(Y, U)]. \end{aligned} \tag{3}$$

Each realized mixture $\widehat{Q}$ is a normalized distribution, so $F - H(P) = E_{Z,G} D(P \| \widehat{Q})$. The finite objective scores a random mixture, while the risk scores its average. Suppressing fixed arguments, the gap between the two is the classical Jensen term:

$$\begin{aligned} F_P(C) &= H(P) + \mathcal{R}(C) + \mathcal{J}_P(C), \\ \mathcal{J}_P(C) &= E_P[\log Q_C \\ &\qquad - E_{Z,G} \log \widehat{Q}_C] \geq 0. \end{aligned} \tag{4}$$

Here $P = Q_{C_*}^{(n)}$. Nonnegativity of $\mathcal{J}_P$ is standard; what matters for learning is how $\mathcal{J}_P$ varies with C, which nonnegativity alone does not determine. This is consistent with properness of the logarithmic score, because F is the expected score of a random predictive rule rather than of Q_C. Two further facts fix what the augmentation does and does not do. Transporting the prior, emissions and particles by an invertible latent map leaves every realization of $\widehat{Q}$ unchanged (Appendix A.1), whereas appending observed coordinates changes the estimator. And at $K = 1$ the objective depends only on the fixed marginals, so it is constant in C: any dependence selection requires $K \geq 2$.

3. Which Correlations the Finite Objective Prefers

3.1. The Selection Scale

Write $b_i = EY_i$ and $\kappa_{ij}(c)$ for the covariance of Y_i, Y_j when their latent correlation is c. Fixed marginals make b_i independent of C, and Price's identity (Price, 1958) gives

$$\begin{aligned} \kappa_{ij}(0) &= 0, \\ \kappa_{ij}'(c) &= \tfrac{1}{4} E_C[\operatorname{sech}^2(X_i/2) \\ &\qquad \times \operatorname{sech}^2(X_j/2)] > 0. \end{aligned} \tag{5}$$

So observed covariances identify latent correlations. Differentiating the objective weights a per-particle *dependence mark* by how much that particle shares the average. With π_k the normalized **complete** particle likelihoods, which include the nuisance factors,

$$\begin{aligned} \partial_{c_{ij}} F_{n,K} &= -E \sum_k \pi_k(1-\pi_k) H_{k,ij}, \\ H_{k,ij} &= Y_i Y_j \sigma(-Y_i X_{ki}) \sigma(-Y_j X_{kj}), \\ E H_{k,ij} &= \{\kappa_{ij}(C_*) + \kappa_{ij}(C)\}/4. \end{aligned} \tag{6}$$

The mean of the mark follows because labels and candidate particles are independent. At finite n, however, a particle's weight and its mark are coupled, which will matter in Section 4.

Set nuisance scales to one, write $F^*_{n,K}(C) = F_{n,K}(C; \mathbf{1})$, and define

$$\begin{aligned} v = \operatorname{Var} \log \sigma(G), \qquad J_K = E \max_{k \le K} G_k, \\ w_{n,K} = J_K / \sqrt{nv}. \end{aligned} \tag{7}$$

Here J_K is the expected maximum of K standard Gaussians, with $J_K \sim \sqrt{2 \log K}$ and $J_K^2 \asymp 1 + \log K$ for integers $K \ge 2$ (Appendix A.2). Hence $w_{n,K} \asymp \sqrt{\log K / n}$ in the polynomial range below, and $w_{n,K} \asymp n^{-1/2}$ at fixed K. A heuristic explains the scale. If the K nuisance log-weights were Gaussian with standard deviation $\sqrt{nv}$, the density at zero of the gap between the largest two would be $K(K-1) \int \phi^2 \Phi^{K-2} / \sqrt{nv} = w_{n,K}$. An active log-likelihood difference of order one changes the dominant particle only on such near-ties.

3.2. The Limiting Potential

Theorem 1 (uniform direction and potential). Fix m, μ, δ, $C_* \in \operatorname{relint} \mathcal{E}^{(m)}_\delta$ and $C_0, p > 0$. As $n \to \infty$, uniformly over $C \in \mathcal{E}^{(m)}_\delta$ and integers $2 \le K \le C_0 n^p$,

$$\begin{aligned} \frac{\partial_{c_{ij}} F^*_{n,K}(C)}{w_{n,K}} &\longrightarrow -\frac{\kappa_{ij}(C_*) + \kappa_{ij}(C)}{4}, \\ \frac{F^*_{n,K}(C) - F^*_{n,K}(I)}{w_{n,K}} &\longrightarrow V(C), \\ V(C) &= -\frac{1}{4} \sum_{i<j} \Big[\kappa_{ij}(C_*) c_{ij} \\ &\qquad + \int_0^{c_{ij}} \kappa_{ij}(u)\, du \Big]. \end{aligned} \tag{8}$$

The proof (Appendix A.1–A.2) works with the actual log-sigmoid nuisance weights. It controls nearly tied pairs of nuisance log-weights through uniform moderate-tail and density estimates valid for polynomial K, together with a common envelope for active scores over the fixed-dimensional correlation class. Compactness alone does not give control at a shrinking scale. Integrating the gradient along the feasible segment from I to C gives V.

Three readings make (8) concrete.

The Jensen gap absorbs the risk. Differences of $F^*_{n,K}$ vanish at rate $w_{n,K}$, while risk differences $\mathcal{R}(C) - \mathcal{R}(I)$ do not depend on n. By (4), the C-dependence of the Jensen term therefore cancels the risk differences to leading order. What survives at scale $w_{n,K}$ is V, which can rank correlations differently from observation risk.

What V rewards. Let $A_C(y, Z) = \sum_i \log \sigma(y_i X_i)$ be the active log-likelihood of label y at one prior draw, and $T(C) = E_Y \operatorname{Var}_Z A_C(Y, Z)$ its variance across prior draws, averaged over true labels. Price's identity gives $\partial_{c_{ij}} T = \{\kappa_{ij}(C_*) + \kappa_{ij}(C)\}/2$, so

$$V(C) = -\tfrac{1}{2}\{T(C) - T(I)\}.$$

The limiting objective prefers correlation matrices under which an observation's active log-likelihood is more variable across prior draws. This is the conditional variance of the active log-likelihood, not the variance of the likelihood itself, of the nuisance weights, or of a stochastic gradient (Appendix B.1).

Two variables. With centred variables, true correlation θ and spin moment $r(\rho)$, the exact cross entropy has derivative $r'(\rho)\{r(\rho) - r(\theta)\}/\{1 - r(\rho)^2\}$. Its descent moves ρ toward θ. At $\theta = 0$, the limiting particle gradient $-\kappa(\rho)/4$ has the opposite sign, so descent moves a nonzero correlation away from zero. More generally, by (5) the Hessian of V in correlation coordinates is diagonal with strictly negative entries: V is strictly concave.

3.3. Spectral Boundary and Risk

A strictly concave function on a compact convex set attains its minimum only at extreme points. For correlation matrices, those points are constrained by a classical rank bound (Li & Tam, 1994).

Theorem 2 (spectral preference and risk). Under Theorem 1, let $\mathcal{M} = \arg\min_{\mathcal{E}^{(m)}_\delta} V$ and $r_{\max} = \lfloor (\sqrt{8m+1} - 1)/2 \rfloor$. Every $C \in \mathcal{M}$ satisfies

$$\operatorname{rank}\left(\frac{C - \delta I}{1 - \delta} \right) \le r_{\max}. \tag{9}$$

At least $m - r_{\max}$ eigenvalues therefore equal δ, while C remains positive definite. For integer sequences $2 \le K_n \le C_0 n^p$ eventually, if $\widehat{C}_n \in \mathcal{E}^{(m)}_\delta$ has population regret $F^*_{n,K_n}(\widehat{C}_n) - \inf_C F^*_{n,K_n}(C) = o(w_{n,K_n})$, then

$$\begin{aligned} &\operatorname{dist}(\widehat{C}_n, \mathcal{M}) \to 0, \\ &\liminf_n \mathcal{R}(\widehat{C}_n) \geq \varepsilon_*, \\ &\varepsilon_* := \min_{C \in \mathcal{M}} \mathcal{R}(C) > 0. \end{aligned} \tag{10}$$

The constant depends on (m, μ, δ, C_*). The affine map in (9) sends $\mathcal{E}_\delta^{(m)}$ to a set of correlation matrices, where an extreme point of rank r satisfies $r(r+1)/2 \leq m$. The monotonicity in (5) identifies latent correlations from observed moments, and compactness separates $\mathcal{M}$ from the interior teacher in observation KL (Appendix A.3). For example, in the four-variable fixture (A.21), with $\mu = (-.6, -.2, .2, .6)$, $\delta = .1$ and teacher eigenvalues $(.5, 1, 1.2, 1.3)$, at least two limiting eigenvalues equal .1 and $\liminf_n \mathcal{R}(\widehat{C}_n) \geq 9/12800$.

"Spectral boundary" here means eigenvalues at the floor δ. The selected matrix is positive definite, and individual correlations need not all grow. For random outputs with $o_p(w_{n,K_n})$ regret, the distance to $\mathcal{M}$ tends to zero in probability and $P\{\mathcal{R}(\widehat{C}_n) < \varepsilon_* - \eta\} \to 0$ for every $\eta > 0$. The theorem concerns population regret, not every finite dataset: an empirical label distribution changes the linear term of V, and a one-label dataset can share its exact and particle minimizers (Appendix A.6).

Other parameter regimes. Two variants show that the regime matters. If active means and scales are also free, with means in a compact box containing the logits of the true marginal probabilities, scales in $[0, S]^m$ and the coins still at unit scale, then $o(1)$ population regret already drives the active scales to zero. The observation law converges to the product of the true marginals, and the risk tends to the teacher's total correlation (Appendix A.8). If instead the common nuisance scale is learnable and may reach zero, joint minimization switches the coins off: the extra nuisance penalty disappears, leaving the active finite-K objective with its own bias (Appendix A.9). Theorems 1–2 hold nuisance scales at one, which is exactly what makes the coins a control.

4. How the Asymptotic Picture Meets Finite Scales

Theorem 1 gives no remainder bound at any finite n. We therefore examine one instance closely: $m = 3$, $\mu = 0$, $C_* = I$, $\delta = .1$ and $K = 100$. Let $C(\rho)$ have all off-diagonals ρ and $C_b = C(.9)$. At $n = 64, 256, 1024, 4096$ we compare the gap $g_n = F_n(I) - F_n(C_b)$ and the free c_{12} derivative D_n at $C(.45)$ with their leading predictions, using $F_n = F^*_{n,100}$ and $w_n = w_{n,100}$. The ratios $A_n = g_n/\{w_n[V(I) - V(C_b)]\}$ and $B_n = -D_n/(m_H^{\text{ref}} w_n)$ equal one when the leading term is exact; m_H^{ref} is a fixed numerical reference for $\kappa(.45)/4$.

4.1. Weight Overlap Is Not the Learning Signal

A natural shortcut is to track scalar weight overlap $\omega_n = E\sum_k \pi_k(1-\pi_k)$, which measures how much of the average is shared between particles. If marks were independent of weights, (6) would give $D_n = -m_H \omega_n$. At $n = 64$ overlap matches its leading scale closely, $\omega_n/w_n \simeq .992$. Yet $A_n \simeq -.143$ and $B_n \simeq -.149$: both the gap and the derivative have the *opposite* sign to their leading predictions, and the conditional intervals exclude zero. The identity $D_n = -m_H^{\text{ref}} \omega_n - \mathcal{C}_n$, with $\mathcal{C}_n = E\sum_k \pi_k(1-\pi_k)(H_{k,12} - m_H^{\text{ref}})$, separates the overlap error O_n from weight–mark coupling J_n (Figure 2b). Coupling dominates in magnitude at 64, 256 and 1024. By 4096 the point estimates of A_n and B_n are about .83, but the intervals do not certify 10% relative accuracy. Complete weights also contain active scores, so the split is an identity, not a causal decomposition (Appendix C.6).

4.2. A Signed Correction and a Certified Direction

What scalar overlap misses can be identified at fixed K. Write $L = \log\sigma(G)$, $\kappa_3 = E(L - EL)^3$, $s_n = \sqrt{nv}$ and $B_K = EM_K^2 - 1$, where $M_K = \max_{k \leq K} G_k$. For one candidate particle, let $A_Y = \sum_i \log\sigma(Y_i X_i)$, $H_Y = H_{1,12}$ from (6), $m_H = EH_Y$ and $\bar{c} = E_Y \operatorname{Cov}_X(H_Y, A_Y)$.

Proposition (fixed-K correction). In the centred model (1) with unit nuisance scales, fix m, a positive definite candidate C, a teacher label law and an integer $K \geq 2$. Then, as $n \to \infty$,

$$\begin{aligned} D_n := \partial_{c_{12}} F^*_{n,K}(C) = &-\frac{m_H J_K}{s_n} \\ &+ \frac{B_K}{s_n^2}\left(\frac{m_H \kappa_3}{3v} - \bar{c}\right) \\ &+ O_{K,m,C}(n^{-3/2}). \end{aligned} \tag{11}$$

The new ingredient is $\bar{c}$: a covariance *within one particle*, between its dependence mark and its active log-likelihood, computed given the label and then averaged over labels. It is not a covariance between particles or between training gradients. Particles with high active likelihood receive more weight; when their marks differ systematically, the weighted mark is shifted. For the instance above at $C = C(.45)$, the n^{-1} coefficient is $b_2 \simeq 1.026$. It splits into a nuisance-skewness part $\simeq -.077$ and a mark–score part $\simeq 1.103$ (Appendix C.13). The correction thus opposes the leading term and comes almost entirely from same-particle coupling. These coefficients are model integrals rather than fits to the gradient panel, although the panel was available when the correction was derived. The proposition holds at

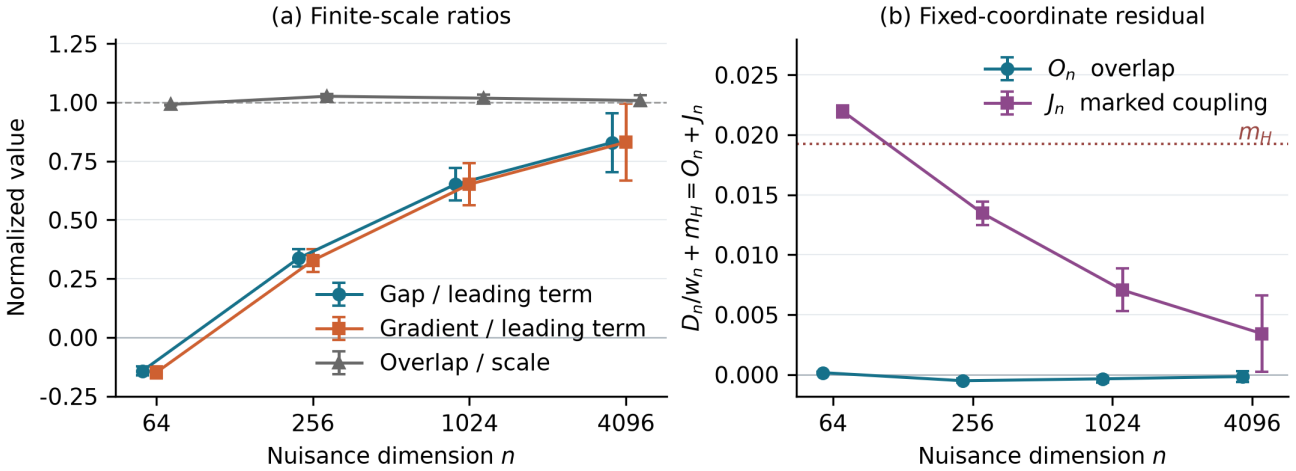


Figure 2. Finite scales with 100 particles; n counts appended fair coins. (a) Gap, free-coordinate gradient and scalar overlap relative to their leading predictions; negative ratios reverse the predicted sign. (b) Gradient residuals at $C(.45)$, separated into overlap error O_n and weight–mark coupling J_n; the label m_H denotes the fixed numerical reference m_H^{ref}. Bars are conditional 99% simultaneous intervals for 26 targets under ideal iid sampling and stated numerical allowances; the 131,072 banks are shared across n, and the last paired increase in B_n is unresolved (Appendix C.6).

fixed K and does not replace the uniform statement of Theorem 1.

Finite-scale corollary. At $m = 3$, $\mu = 0$, $C_* = I$, $K = 100$, $C = C(.45)$ and the single free c_{12} coordinate, explicit error bounds give

$$\begin{aligned} -.002045393 < D_{4096} < -.0003519373, \\ |D_{4096} - D^{(0)}_{4096}| < .001097138798 \\ < |D^{(0)}_{4096}|, \qquad D^{(0)}_n = -m_H J_{100}/s_n. \end{aligned} \tag{12}$$

At $n = 4096$ the leading term has the correct sign for this coordinate. The error bound is about 75.7% of $|D^{(0)}_{4096}|$, so the certificate is about sign, not 10% accuracy. At 64, 256 and 1024 the same recipe leaves the sign unresolved. The enclosures are deterministic and use the actual Gumbel-smoothed nuisance noise and certified model moments (Appendix C.7 onward).

The certified sign has a risk consequence. By (5) and the three-variable law, $\partial_{c_{12}}\mathcal{R} = \kappa'(.45)\kappa/\{(1-\kappa)(1+3\kappa)\} > 0$ with $\kappa = \kappa(.45) > 0$. Every sufficiently small feasible increase of c_{12} therefore lowers $F^*_{4096,100}$ while raising the risk. This is a finite, coordinate-level objective–risk conflict, established by its own explicit remainder bounds rather than by Theorem 2.

4.3. Finite-Program Consequences at $n = 256$

The instance of Figure 1 also yields finite-program statements at $n = 256$, $K = 100$ (Appendix C.15, D, E). On a 99% audit event under ideal iid sampling and declared numerical conditions, every correlation matrix in the class with population regret below .00134, on or off the grid, has risk above 10^{-6}; this excludes a continuous low-risk region under a near-optimality condition that no algorithm is shown to attain. On the same event, a fresh run of the fixed 143-point grid program selects risk at least $0.09/(32e)$ with probability at least .7035, a bound on a probability for that program rather than an observed frequency.

5. Without an Appended Block: A Certified Local Conflict

5.1. A Two-Factor Model with All Parameters Free

We now remove the control. There is no designated independent block; all biases and loadings are learned:

$$\begin{aligned} Z \sim N(0, I_2), \qquad \theta = (b, W), \\ Y_j \mid Z \sim \mathrm{Bernoulli}\{\sigma(b_j + w_j^\top Z)\}, \\ F^q_K(\theta) = E_{Y\sim q, Z_{1:K}}\Big[-\log\Big\{K^{-1}\sum_{k=1}^{K} p_\theta(Y \mid Z_k)\Big\}\Big]. \end{aligned} \tag{13}$$

Here b collects logit biases, $Y_j \in \{0, 1\}$, and the prior particles $Z_{1:K}$ are iid and independent of Y. Items are conditionally independent given Z. The target q is empirical or the teacher law, and F^q_∞ is the integrated cross entropy. The teacher has eight items, zero biases and rows of length .8 at angles $(-.7, -.5, -.3, -.1, .1, .3, .5, .7)$, so all pairwise loading inner products are positive. This class differs from Section 3: the logit covariance $WW^\top$ has rank at most two and no spectral floor, and loading rank is not the rank

of $\mathrm{Cov}(Y)$.

5.2. Correct Marginals, Positive Risk, and an Opposed Improvement

Theorem 3 (computer-assisted local dependence conflict). For the exact teacher in (13), let q_* be its observation law and $\mathcal{R}(\theta) = D(q_*\|Q_\theta)$. There is a rank-one interior parameter $\theta_* = (0, a_*, 0)$, with $a_{*,i} > 0$ and $\nabla F_8^{q_*}(\theta_*) = 0$, that is a strict local minimum of $F_8^{q_*}$ in the full 24-dimensional parameter space modulo common latent rotations. Every observation marginal is correct, $Q_{\theta_*}(Y_j = 1) = q_*(Y_j = 1) = 1/2$, yet $\mathcal{R}(\theta_*) \simeq .01043 > 0$.

There is a fixed $d = d_{\mathrm{ang}}^{(8)} \in \mathbf{1}^\perp \subset \mathbb{R}^8$, $\|d\|_2 = 1$, such that the path $\theta_{\mathrm{ang}}(t) = (0, W(t))$ satisfies

$$\begin{aligned} W_i(t) &= a_{*,i}\big(\cos(td_i), \sin(td_i)\big), \\ r(t) &= \mathcal{R}(\theta_{\mathrm{ang}}(t)), \\ r'(0) &= 0, \qquad r''(0) < -.00450377. \end{aligned} \tag{14}$$

There exists $\varepsilon > 0$ such that, for every $0 < |t| < \varepsilon$,

$$\begin{aligned} F_8^{q_*}(\theta_{\mathrm{ang}}(t)) &> F_8^{q_*}(\theta_*), \\ \mathcal{R}(\theta_{\mathrm{ang}}(t)) &< \mathcal{R}(\theta_*), \\ \operatorname{rank} W(t) &= 2. \end{aligned} \tag{15}$$

The path changes only dependence. Row lengths are fixed, so each candidate logit keeps its Gaussian marginal $N(0, a_{*,i}^2)$ and each observation stays a fair coin, while $\mathrm{Cov}(X_i(t), X_j(t)) = a_{*,i}a_{*,j}\cos\{t(d_i - d_j)\}$ changes. These preserved logit variances are the candidate's own, not the teacher's $.8^2$. Since the root is a strict local minimum and a nonconstant angle path leaves the rotation orbit, the objective rises along it. The risk falls because its curvature along the path is negative. The conflict is second order and local: every sufficiently small nonzero step along this path lowers risk and raises the objective (Appendix I.10). Fixed row lengths define this comparison path; they are not a constraint on the learning problem. The full statement in Appendix I.8 also certifies a direction d_{risk} with a negative first-order risk derivative; a certified step of size 10^{-7} along it raises F_8, lowers the risk and keeps the observation marginals fair, while changing row lengths.

For scale, in four training streams of 4096 teacher samples, independent-Bernoulli fits have KL about .107 and Adam training with $K = 8$ ended near .011 (Appendix I.4); a refitted stationary point of the integrated objective within the collinear submodel has KL about .0070 (I.6). These are numerical comparisons, not certified optima.

Certificate outline. Symmetry reduces root finding to four coordinates, and interval root inclusion proves an exact stationary point near a dyadic centre. Full-space stability needs every direction. All 16 bias/radial and seven non-rotation transverse directions have certified positive curvature; an angular check alone would not suffice (Appendix I.2, I.8). The risk curvature along (14) includes the path acceleration. It is enclosed over all 256 labels by Gaussian integration by parts, analytic quadrature error bounds (Trefethen & Weideman, 2014) and midpoint-radius arithmetic (Johansson, 2017), then transferred from the centre to the root by a Lipschitz bound in the row lengths (I.10).

Corollary (quantitative local persistence). Keep all eight teacher biases zero and row lengths $4/5$, changing only its first angle to $-.7 + \lambda$, $|\lambda| \le .001$. With observation law q_λ, there is a smooth branch $\theta_\lambda = (0, a(\lambda), 0)$ connected to Theorem 3's root, each a full-space strict local minimum of its own $F_8^{q_\lambda}$ modulo rotation. Using the same fixed d and replacing a_* by $a(\lambda)$ in (14) gives, for each teacher and every sufficiently small nonzero $|t| < \varepsilon_\lambda$, a rank-two path that raises its finite objective and lowers its own risk $D(q_\lambda\|Q_\theta)$, preserving candidate Gaussian logit marginals and correct fair observation marginals.

This teacher family breaks the symmetry of the original teacher, but it is small: endpoint observation laws differ by about 1.6×10^{-4} in L^1 (Appendix I.11). For broader, unquantified perturbations of the teacher, Appendix I.9 shows that a nearby rank-one strict local minimum survives with a first-order risk-decreasing direction.

5.3. Weak-Loading Intuition for Alignment

A weak-loading expansion at fixed $K > 2$ gives intuition, though not a quantitative account (Appendix I.1). With row lengths fixed and loadings of order ε, the finite objective is, up to angle-independent terms, $\varepsilon^4 a_K\|c - \{K/(K-2)\}c_*\|^2 + O(\varepsilon^6)$, where $a_K > 0$ is given in (I.2) and c collects pairwise row inner products after removing the common scale ε. It rewards inner products *amplified* relative to the teacher's. With fixed lengths, rows can amplify their inner products only by aligning. In the independent-teacher case of Section 3, the limiting potential likewise favours dependence that the teacher lacks. The two expansions are derived separately, and neither explains the other. The weak-loading stability boundary of the collinear configuration, $K \simeq 4.33$, does not predict behaviour at loading .8.

Numerical learning is consistent with this picture without establishing it (Appendix I.4–I.7). In four streams, $K = 8$ training produced nearly collinear loadings, whereas $K = 64$ and the integrated reference did not. Refitted at each K, the collinear point numerically changes local type between $K = 11$ and 12; angle probes there change the risk by only about .02%, an effect size that belongs to those probes, not to the certified $K = 8$ conflict.

6. Related Work

Monte Carlo objectives and simulated likelihood. The logarithm of an unbiased likelihood estimator is biased, and the bias depends on the parameters. This is established for simulated maximum likelihood in discrete-choice models (Lee, 1995), for Monte Carlo likelihood inference, where increasing simulation accuracy inside the logarithm and averaging a fixed-K log objective are distinct limits (Sung & Geyer, 2007), and for Monte Carlo variational objectives (Maddison et al., 2017). For a nonnegative unbiased normalizing-constant estimator, AESMC's Claim 1 writes the expected-log gap as a KL divergence on an extended sampling space (Le et al., 2018); that divergence is not the observation risk of the integrated model studied here. For dependence specifically, the abstract of Geweke et al. (1997) reports downward bias of serial-correlation estimates in GHK simulated-likelihood experiments; we treat this as an empirical precedent, not as something our mechanism explains.

Weight degeneracy and gradient asymptotics. High-dimensional importance sampling (Agapiou et al., 2017; Chatterjee & Diaconis, 2018), particle-filter collapse (Bengtsson et al., 2008; Bickel et al., 2008; Snyder et al., 2008) and multi-sample gradient estimators (Rainforth et al., 2018; Tucker et al., 2019) are well studied. The large-K scale $w_{n,K}$ matches the expected maximum-weight excess in Bickel et al. (2008); dependence selection requires control at that scale. The nearest result to Theorem 1 is Daudel & Roueff (2026) (Section D.4.1, equation (208)). Under a Gaussian random-field condition on the log-weights, and with derivative–expectation interchange, the subleading mean gradient follows the derivative of the log-weight standard deviation. Through the conditional-variance identity of Section 3, that formula formally predicts our large-K coefficient. Its proof uses independence between the standardized log-weight field and its parameter derivative, which approximate Gaussianity does not preserve (their Section 3.4). Daudel et al. (2023) (Theorem 6) treats non-Gaussian iid log-weight sums and gives scalar objective asymptotics; that scalar remainder cannot simply be differentiated into a correlation field that is uniform at the smaller selection scale. What Theorem 1 adds is this control for the actual non-Gaussian weights, uniform over the correlation class and over $2 \le K \le C_0 n^p$, including every fixed $K \ge 2$. Theorem 2 then combines it with the classical extreme-point geometry of correlation matrices (Li & Tam, 1994) to reach an observation-risk conclusion. Appendix B gives a theorem-level comparison.

Latent-variable learning and local geometry. Learning binary latent models by prior-sampled likelihood predates this work (Tipping, 1998). Importance-weighted autoencoders change their learned representations when the objective changes (Burda et al., 2016, Section 5.2 and Table 2); our criterion is the negative of their bound with the proposal fixed to the prior. A learned-proposal version is used for item factor analysis (Urban & Bauer, 2021). Collapse and local stability have been analysed in linear latent-variable models (Lucas et al., 2019; Wang & Ziyin, 2022); for linear VAEs, Lucas et al. (2019) (Theorem 1) rule out additional ELBO local maxima relative to the marginal likelihood, and fixing the observation variance affects pPCA stability. Theorem 3 instead certifies, in a nonlinear model, a full-space local minimum of a finite-particle objective with a marginal-preserving direction along which it opposes a risk improvement; it is not a claim about low rank in general.

7. Discussion

What the results change. At fixed K the prior-particle objective has its own dependence preferences, and estimating it more accurately does not move them toward the integrated likelihood. In the controlled family these preferences are of order $w_{n,K}$, yet they decide which correlations are selected and yield positive limiting risk (Theorems 1–2). In the finite-scale example, scalar overlap can track its leading prediction while the correlation gradient has the opposite sign to its own (Section 4). Correct marginals at a full-space local minimum do not mean that the objective welcomes risk-reducing dependence changes (Theorem 3).

Scope. The results carry different kinds of guarantee: Theorems 1–2 are uniform asymptotic statements; the corollary in (12) is a deterministic enclosure at one n; Section 4.3 rests on a 99% audit event with declared numerical conditions; Theorem 3 is a computer-assisted local proof. All use prior particles, and apart from the grid-program bound of Section 4.3 they concern population objectives. The first part requires a known independent appended block with unit scales, fixed active dimension and marginals, and regret small relative to $w_{n,K}$. The second part is one model with a small explicit teacher family. A natural next question is whether learned, posterior-adapted proposals (Burda et al., 2016) remove, shrink or reverse this preference.

Reproducibility. The accompanying anonymous supplement provides saved interval inputs, arithmetic checks for the certificates and finite-program bounds, and data and scripts for Figures 1–2. Its README gives the commands and verification scope. These checks reconstruct saved evidence; they do not rerun production quadrature or training.

Impact Statement

This work aims to improve understanding of simulation-based probabilistic learning. Its potential benefit is more reliable separation of a random training criterion from the

risk of the integrated model. The studies use controlled synthetic observations, without establishing deployment benefits. Applying the results requires retaining their assumptions and finite-budget limitations, including the reported recovery boundaries and negative controls.

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Appendix reading guide

The retained appendices supply the proofs and evidence for the main-text results. The table maps each main-text item to its support, with continuous appendix lettering.

Main text	Where it is established	Type of support
Figure 1, fair-coin control (Section 1)	C.1 matched selection; C.2 gap audit; C.3 analytic curvature at $n = 0$; C.4 curvature audit at $n = 256$	Analytic bound at $n = 0$; 99% audit families with stated numerical allowances; one saved matched run
Model, decomposition and representation facts (Section 2)	A.1	Analytic
Selection scale, Theorem 1 (Section 3.1–3.2)	A.2; scale heuristic and its relation to particle-filter collapse in B.2	Uniform asymptotic proof
Conditional-variance reading of V	B.1	Exact identity
Theorem 2 and the four-variable example	A.3 (A.21–A.22)	Analytic
Other parameter regimes	A.8 (free active scales), A.9 (free nuisance scale); further boundaries in A.4–A.7	Analytic
Figure 2, overlap versus coupling (Section 4.1)	C.5–C.6	Conditional 99% simultaneous intervals, 26 targets
Fixed-K correction and finite-scale corollary (Section 4.2)	C.7 onward: proof C.8, budgets C.9–C.12, constants and synthesis C.13, risk derivative C.14	Asymptotic proof; deterministic enclosure at $n = 4096$
Finite-program consequences (Section 4.3)	C.15; D (fresh-run bound); E (saved-input arithmetic)	99% audit event, ideal iid sampling, declared numerical conditions
Model (13) and weak-loading heuristic (Sections 5.1, 5.3)	I.1	Expansion with qualitative remainder
Theorem 3 (Section 5.2)	I.2 criterion (full derivatives); I.3 representation; I.8 root and full-space statement; I.10 same-root angular path	Computer-assisted proof
Quantitative persistence corollary (Section 5.2)	I.11 and full family statement	Computer-assisted proof over $\lvert\lambda\rvert \le .001$
Qualitative persistence (Section 5.2)	I.9	Implicit-function argument; neighbourhood not quantified
Numerical learning and K11/K12 probes (Section 5.3)	I.4–I.7	Numerical, with stated quadrature qualifications

Material reported only in the appendices. Known-mean robustness and observation-equivalent scales (F); a two-particle counterexample showing that matched means need not preserve finite-particle stationarity (G); and the earlier finite selection and its negative control (Appendix H).

A.2 retains the full uniform proof. Saved-endpoint and certificate-composition checks do not recompute probability, gradient or Hessian grids.

Appendix A. Proofs and representation boundaries

Theorem 1 is proved in A.1–A.2 and Theorem 2 in A.3. Here the star on F denotes the fixed teacher: the appendix notation F with star and nuisance scales is the same population score as in the main text. The constant ε_* in Theorem 2 is the minimum risk on the potential minimizing set; R_* in (A.19) is an explicit lower bound for it. A.4–A.9 give the additional representation and scale boundaries.

All logarithms are natural. Risk is the KL divergence for a complete observation, without division by dimension. Throughout A.1–A.6, the active dimension $m \geq 2$, mean vector $\mu \in \mathbb{R}^m$, and spectral floor $0 < \delta < 1$ are fixed. Spin labels are $Y_i \in \{-1, 1\}$; native API bits are $Y_i^{(01)} = (Y_i + 1)/2$. Particle samples are independent of the observation being scored. The population statements below do not assume that a finite optimization run attains their required regret.

A.1. Model, exact task and the finite-particle derivative

Learn the upper-triangular entries c_{ij}, $i < j$, of

$$\begin{aligned} \mathcal{E}_\delta^{(m)} &= \{C = C^\top : \operatorname{diag} C = \mathbf{1},\ C \succeq \delta I\}, \\ |c_{ij}| \leq 1 - \delta, \qquad & \delta \leq \lambda_{\min}(C) \leq \lambda_{\max}(C) \leq m - (m-1)\delta. \end{aligned} \tag{A.1}$$

This is a compact convex set with nonempty relative interior in the affine space of unit-diagonal symmetric matrices. Let $X \sim N(\mu, C)$, with conditionally independent emissions $P(Y_i = y_i \mid X_i) = \sigma(y_i X_i)$. Independent nuisance logits are $\tau_j G_j$, $G_j \sim N(0,1)$, with the same logistic emission. Symmetry gives

$$\begin{aligned} Q_C^A(y) &= E_C \prod_{i=1}^m \sigma(y_i X_i), & Q_{C,\tau}^{(n)}(y,u) &= 2^{-n} Q_C^A(y), \\ \widehat{Q}_{C,\tau,K}^{(n)}(y,u) &= \frac{1}{K} \sum_{k=1}^K \prod_{i=1}^m \sigma(y_i X_{ki}) \prod_{j=1}^n \sigma(u_j \tau_j G_{kj}), \\ F_{n,K}^{*,\tau}(C) &= E_{(Y,U)\sim Q_{C_*}^{(n)}, Z}[-\log \widehat{Q}_{C,\tau,K}^{(n)}(Y,U)], & \mathcal{R}_*(C) &= D(Q_{C_*}^A \| Q_C^A). \end{aligned} \tag{A.2}$$

The true and candidate means and marginal variances agree. At a fixed n, changing nuisance scales preserves the whole observation law. Across different n, sample spaces differ; active likelihood ratios and complete joint KL remain unchanged. Each realized particle predictor is normalized, and its expectation is Q_C; its expected log score nevertheless differs from the log score of the integrated model.

This is distinct from a pure latent-coordinate change. For an invertible measurable T_C, push the prior forward, transport the emission by $p'_C(y \mid z') = p_C(y \mid T_C^{-1} z')$, and pair the particles. For every realization and C,

$$\frac{1}{K} \sum_k p'_C(y \mid T_C(Z_k)) = \frac{1}{K} \sum_k p_C(y \mid Z_k). \tag{A.3}$$

Differentiable versions preserve path derivatives with respect to the same physical C when all transport derivatives are included. Changing optimization coordinates still requires the chain rule and need not preserve Adam updates. Without paired samples, the estimator laws agree rather than their realizations. Changing Gaussian nuisance scales while retaining the uncompensated logistic emission is an observation-equivalent representation change, not (A.3).

Write $b_i = EY_i = E\tanh((\mu_i + G)/2)$ and $\kappa_{ij}(c) = E_C Y_i Y_j - b_i b_j$. Fixed marginal means and variances make this a function only of the pair correlation c. Gaussian integration by parts gives

$$\kappa_{ij}(0) = 0, \qquad \kappa'_{ij}(c) = \frac{1}{4} E_c[\operatorname{sech}^2(X_i/2) \operatorname{sech}^2(X_j/2)] > 0. \tag{A.4}$$

For one complete bank, let π_k be normalized complete particle likelihoods and $\gamma_k = \pi_k(1-\pi_k)$. Its mixed logit Hessian and the off-diagonal Gaussian Price identity imply

$$\begin{aligned} \partial_{c_{ij}} F^{*,\tau}_{n,K}(C) &= -E\sum_k \gamma_k H_{k,ij}, \\ H_{k,ij} &= Y_iY_j\sigma(-Y_iX_{ki})\sigma(-Y_jX_{kj}) \\ &= (Y_i^{(01)} - \sigma(X_{ki}))(Y_j^{(01)} - \sigma(X_{kj})), \\ EH_{k,ij} &= \frac{\kappa_{*,ij} + \kappa_{ij}(c_{ij})}{4}, \qquad \kappa_{*,ij} = \kappa_{ij}(c_{*,ij}). \end{aligned} \tag{A.5}$$

Indeed the mixed derivative for particle k is $-\gamma_k H_{k,ij}$; varying both symmetric covariance entries cancels the factor $1/2$ in Price's formula. Gaussian linear envelopes make the loss integrable and justify differentiation; the mixed derivatives are bounded. Expanding H as $[Y_iY_j - Y_iT_j - Y_jT_i + T_iT_j]/4$, $T_i = \tanh(X_{ki}/2)$, and using observation–particle independence proves its mean. A Price sample is not generally equal to a realized Cholesky path derivative.

A.2. Uniform near-tie analysis and the matrix potential

We prove the direction result slightly more generally for prescribed nuisance arrays $a_{nj} \in [a_-, a_+]$, where $0 < a_- \le a_+ < \infty$ are fixed. The theorem's unit scales are included. Define

$$\begin{gathered} L_a = \log\sigma(aG), \quad v(a) = \operatorname{Var} L_a, \quad s_n^2 = \sum_{j=1}^n v(a_{nj}), \\ I_K = \int_{\mathbb{R}} \varphi(z)^2\Phi(z)^{K-2}\,dz, \qquad J_K = K(K-1)I_K = E\max_{k\le K} G_k, \qquad w_{n,K} = J_K/s_n. \end{gathered} \tag{A.6}$$

Since $L_a = aG/2 - \log 2 - \log\cosh(aG/2)$, odd/even orthogonality gives $v(a) \ge a^2/4$, while a Gaussian envelope gives $s_n^2 \asymp n$ uniformly. Integration by parts proves the identity for J_K. The real-exponent argument below supplies $J_K \sim \sqrt{2\log K}$ and $J_K^2 \asymp 1 + \log K$ for integer $K \ge 2$.

The following steps keep the constants uniform. In the density estimates they may depend on fixed a_-, a_+ and the displayed truncation levels, but not on n, the nuisance array or its ordering. In the pair estimate they may additionally depend on fixed C_0, p, C_1. Only the active-score envelope will introduce dependence on fixed m, μ, δ. Each $o(1)$ has this uniform meaning. No explicit convergence modulus for (A.13) is asserted.

STEP 1. UNIFORM TILTED DENSITIES: LOW AND HIGH FREQUENCIES

The density of L_a, extended by zero for $x \ge 0$, is

$$f_a(x) = \frac{\varphi(z_a(x))}{a(1-e^x)}, \qquad z_a(x) = \frac{x - \log(1-e^x)}{a}, \qquad x < 0. \tag{A.7}$$

These densities and each fixed-order derivative have uniform rapidly decreasing envelopes for $a \in [a_-, a_+]$. The left tail is Gaussian up to polynomial factors. Near zero set $t = -\log(-x)$; each fixed number of derivatives contributes at most e^{ct} times a polynomial, dominated by $\exp[-t^2/(2a_+^2) + O(t)]$. The zero extension is therefore smooth. The same bounds hold after an exponential tilt in a sufficiently small fixed interval.

Let $k_a(\lambda) = \log Ee^{\lambda(L_a - EL_a)}$. Its fixed-order derivatives are uniformly bounded for $|\lambda| \le \lambda_0 < 1/4$; the tilted variances are bounded away from zero by continuity and nondegeneracy on this compact interval. For the centered tilted characteristic functions $\psi_{a,\lambda}$, a nonzero frequency has modulus strictly below one because the tilted law has a density. Continuity makes this strict bound uniform on compact frequency annuli. Uniform integration by parts gives decay at large frequencies, joining the two bounds and yielding

$$0 < c \le k_a''(\lambda) \le C, \qquad \sup_{a,\lambda,|u|\ge\varepsilon} |\psi_{a,\lambda}(u)| \le q_\varepsilon < 1, \qquad \sup_{a,\lambda}\int |\psi_{a,\lambda}(u)|^2\,du < \infty. \tag{A.8}$$

Let $S_n = \sum_j (L_{a_{nj},j} - EL_{a_{nj}})$ and add an independent centered standard Gumbel E. Its log MGF g_0 is analytic for $\lambda < 1$. Put $\mathcal{K}_n = \sum_j k_{a_{nj}} + g_0$, and let $f_{n,\lambda}$ denote the density of the exponentially tilted $S_n + E$. Write $B_{n,\lambda}^2 = \mathcal{K}_n''(\lambda) \asymp n$, and let $\Psi_{n,\lambda}$ be the characteristic function after centering at $\mathcal{K}_n'(\lambda)$ and dividing by $B_{n,\lambda}$. Uniform third cumulant bounds give

$$\log \Psi_{n,\lambda}(u) = -u^2/2 + R_{n,\lambda}(u),$$
$$|R_{n,\lambda}(u)| \le C|u|^3/\sqrt{n},$$
$$\int_{|u|\le \varepsilon\sqrt{n}} |\Psi_{n,\lambda}(u) - e^{-u^2/2}|\, du \le Cn^{-1/2}.$$

For sufficiently small fixed ε, the real part of the exponent, and of its interpolation with $-u^2/2$, is at most $-cu^2$. Thus the integrand difference is bounded by $C|u|^3 e^{-cu^2}/\sqrt{n}$, proving the integral estimate even though its range grows. In the complementary range, $|u|/B_{n,\lambda}$ is bounded away from zero. Retain two distinct nuisance factors, use (A.8) on the other $n-2$, and apply Cauchy–Schwarz:

$$\int_{|u|>\varepsilon\sqrt{n}} |\Psi_{n,\lambda}(u)|\, du \le B_{n,\lambda} q^{n-2} \int_{\mathbb{R}} |\psi_{a_{n1},\lambda}(t)\psi_{a_{n2},\lambda}(t)|\, dt \le C\sqrt{n}\, q^{n-2}.$$

The tilted Gumbel factor has modulus at most one, so it does not affect this upper bound. The normal integral outside the low-frequency range is exponentially small. Fourier inversion now gives

$$f_{n,\lambda}(\mathcal{K}_n'(\lambda)) = \frac{1 + O(n^{-1/2})}{\sqrt{2\pi \mathcal{K}_n''(\lambda)}}, \qquad \sup_x s_n f_{n,\lambda}(x) \le C. \tag{A.9}$$

Indeed the standardized density differs from φ by $O(n^{-1/2})$ uniformly in its argument; at the centered argument this is a relative error because $\varphi(0) > 0$. The same Fourier integral bounds the density everywhere. This establishes both parts of (A.9) uniformly over the prescribed arrays and small tilts.

STEP 2. RELATIVE DENSITY AND UPPER-TAIL ESTIMATES

For $x = s_n z$, $|z| \le M\sqrt{\log n}$, strict convexity gives a unique small solution of $\mathcal{K}_n'(\lambda) = x$. Taylor expansion at zero, using $\mathcal{K}_n'(0) = 0$, $\mathcal{K}_n''(0) = s_n^2 + \operatorname{Var} E$ and uniformly $O(n)$ third derivatives, gives $\lambda = z/s_n + O_M((1+z^2)/n)$, $\lambda x - \mathcal{K}_n(\lambda) = z^2/2 + O_M((1+|z|^3)/\sqrt{n})$, and $s_n/\sqrt{\mathcal{K}_n''(\lambda)} = 1 + O_M((1+|z|)/\sqrt{n})$. Let q_n, H_n be the density and CDF of $(S_n + E)/s_n$. Exact untilting, namely $q_n(z) = s_n e^{\mathcal{K}_n(\lambda) - \lambda s_n z} f_{n,\lambda}(s_n z)$, and (A.9) prove the first line of

$$\begin{aligned} \sup_{|z|\le M\sqrt{\log n}} \left| \frac{q_n(z)}{\varphi(z)} - 1 \right| &= O_M\left(\frac{1 + (\log n)^{3/2}}{\sqrt{n}} \right), \\ \sup_{-1\le z\le M\sqrt{\log n}} \left| \frac{1 - H_n(z)}{1 - \Phi(z)} - 1 \right| &= o(1). \end{aligned} \tag{A.10}$$

To prove the second line, an absolute CLT is insufficient. Tilt the global density bound in (A.9) at $h = M\sqrt{\log n}/s_n$. The same cumulant expansion gives, for $z \ge M\sqrt{\log n}$,

$$q_n(z) \le C_M n^{-M^2/2} \exp\{-M\sqrt{\log n}(z - M\sqrt{\log n}) + o(1)\}. \tag{A.11}$$

Fix $M' > M$, put $T = M\sqrt{\log n}$, $T' = M'\sqrt{\log n}$, and let $\rho_n(M')$ denote the relative density error on $[-T', T']$. Integrating that estimate up to T', then using (A.11) with M' beyond T', gives

$$\left| \frac{1 - H_n(z)}{1 - \Phi(z)} - 1 \right| \le \rho_n(M') + \frac{\int_{T'}^{\infty} [q_n(t) + \varphi(t)]\, dt}{1 - \Phi(T)} = o(1), \qquad -1 \le z \le T.$$

Here Mills' lower bound gives $1-\Phi(T) \geq c_M n^{-M^2/2}/\sqrt{\log n}$, whereas the numerator from the omitted tails is at most $C_{M'} n^{-M'^2/2}/\sqrt{\log n}$. The strict inequality $M' > M$ makes their ratio vanish. This proves the relative upper-tail statement, uniformly down to $z = -1$. The zero-tilt part of (A.9) also gives $\sup_z q_n(z) \leq C$. Moreover the standardized variable is centered and has second moment $1 + \operatorname{Var} E/s_n^2$, uniformly bounded. Integrating the relative density estimate on fixed compact intervals and using this moment bound gives compact-uniform CDF convergence to Φ.

STEP 3. CONDITIONAL PAIR IDENTITY AND SMALL-SHIFT CDF POWERS

For fixed labels absorb nuisance signs into the Gaussian expectation. Conditional on active log scores $A_1, \ldots, A_K$, the weights are the softmax of $S_{nk} + A_k$. The Gumbel maximum representation, differentiated in a second score, gives

$$E_S[\pi_k \pi_\ell \mid A] = \frac{1}{s_n} \int q_n(z-d_k) q_n(z-d_\ell) \prod_{j \neq k,\ell} H_n(z-d_j)\, dz, \qquad d_j = A_j/s_n. \tag{A.12}$$

The derivative of a softmax choice probability is $-\pi_k \pi_\ell$; differentiating the corresponding maximum integral differentiates one CDF. Bounded softmax derivatives and an integrable density product justify both operations. We will prove, for every fixed $C_1, C_0, p > 0$,

$$\sup_{\substack{2 \leq K \leq C_0 n^p \\ \max_j |A_j| \leq C_1 \sqrt{\log n} \\ k \neq \ell}} \left| \frac{s_n E_S[\pi_k \pi_\ell \mid A]}{I_K} - 1 \right| \longrightarrow 0. \tag{A.13}$$

Set $D_n = C_1 \sqrt{\log n}/s_n$, so $|d_j| \leq D_n$, and choose fixed M with $M^2 > 4p + 4$. On $0 \leq z \leq T = M\sqrt{\log n}$, the identity $\log[\varphi(z-d)/\varphi(z)] = zd - d^2/2$ is uniformly $O(\log n/\sqrt{n})$. Mills bounds give $\varphi(t)/(1-\Phi(t)) \leq C(1+t_+)$ for $t \geq -1$, so integrating this log-tail derivative controls the normal upper-tail shift by the same order. Applying (A.10) on a slightly larger fixed window yields

$$\sup_{\substack{0 \leq z \leq T \\ |d| \leq D_n}} \left| \frac{q_n(z-d)}{\varphi(z)} - 1 \right| = o(1), \qquad \sup_{\substack{0 \leq z \leq T \\ |d| \leq D_n}} \left| \frac{1-H_n(z-d)}{1-\Phi(z)} - 1 \right| = o(1).$$

The passage from upper tails to powers is explicit. Put $u = 1-\Phi(z)$ and $u' = 1-H_n(z-d)$. Here $0 < u \leq 1/2$ and $u'/u = 1 + o(1)$ uniformly. On, for example, $0 < u \leq 1/2$ and $|u'/u - 1| \leq 1/4$, the derivative of $\log(1-u)$, together with $u \leq -\log(1-u)$, bounds $|\log(1-u')/\log(1-u) - 1|$ by a constant times $|u'/u - 1|$. Consequently there is a deterministic $\varepsilon_n \to 0$, uniform over all arrays and shifts, such that

$$\Phi(z)^{1+\varepsilon_n} \leq H_n(z-d) \leq \Phi(z)^{1-\varepsilon_n}, \qquad \Phi(z)^{(K-2)(1+\varepsilon_n)} \leq \prod_{j \neq k,\ell} H_n(z-d_j) \leq \Phi(z)^{(K-2)(1-\varepsilon_n)}.$$

Both inequalities hold for all $0 \leq z \leq T$ and $|d| \leq D_n$. The reversal on multiplying by the negative $\log \Phi(z)$ explains the order of the powers. Multiplication changes the exponent; it does not require a pointwise error small compared with $1/K$.

STEP 4. REAL-EXPONENT NORMAL PAIR INTEGRALS AND UNIFORM RATIOS

For real $x > 1$, extend (A.6) by

$$I_x = \int_{\mathbb{R}} \varphi(z)^2 \Phi(z)^{x-2}\, dz, \qquad J_x = x \int_{\mathbb{R}} z\varphi(z)\Phi(z)^{x-1}\, dz = x(x-1)I_x.$$

The CDF Φ^x defines a variable M_x whose mean is J_x. Integration by parts has zero boundary terms since $\varphi(z)\Phi(z)^{x-1} \to 0$ at both ends. On every compact subinterval of $x > 1$, Mills bounds supply an integrable envelope for $\varphi^2 \Phi^{x-2}$; hence I_x is continuous and strictly positive there.

For completeness, put $b_x = \sqrt{2\log x}$, take fixed $0 < \eta < 1$, and write $M_x^+ = \max(M_x, 0)$, $M_x^- = \max(-M_x, 0)$. Mills bounds and $1 - \Phi(t)^x \le x[1 - \Phi(t)]$ give

$$\begin{aligned} EM_x^- &\le 2^{-(x-1)} \int_0^\infty \Phi(-t)\, dt, \\ EM_x^+ &\le b_x + x \int_{b_x}^\infty [1 - \Phi(t)]\, dt = b_x + O(b_x^{-2}), \\ P(M_x \le (1-\eta)b_x) &\le \exp\{-x[1 - \Phi((1-\eta)b_x)]\} \longrightarrow 0. \end{aligned}$$

The second line uses $x\varphi(b_x) = (2\pi)^{-1/2}$. The last line follows because the Mills lower bound makes $x[1-\Phi((1-\eta)b_x)] \to \infty$. Thus $J_x \le b_x + O(b_x^{-2})$ and $J_x \ge (1-\eta)b_x(1 - o(1)) - o(1)$, proving $J_x \sim \sqrt{2\log x}$ as real $x \to \infty$.

These facts give the ratio needed in Step 3 uniformly in the allowed integer K. If uniformity failed, choose a violating sequence $n \to \infty$, K_n and θ_n, with an error bounded below by a fixed positive number. A bounded subsequence of K_n has a further subsequence equal to one integer K; then $x_n \to K$, and continuity gives $I_{x_n}/I_K \to 1$. Otherwise take a subsequence with $K_n \to \infty$. Since $x_n/K_n \to 1$, the real-exponent asymptotic gives

$$\frac{I_{x_n}}{I_{K_n}} = \frac{J_{x_n}}{J_{K_n}} \frac{K_n(K_n-1)}{x_n(x_n-1)} \longrightarrow 1, \qquad x_n = 2 + (K_n - 2)(1 + \theta_n), \quad |\theta_n| \le \varepsilon_n.$$

Both alternatives contradict the fixed positive error. In particular the conclusion holds for $\theta_n = \pm\varepsilon_n$. No rate in the growth of K_n is required for this argument.

STEP 5. RELATIVE PAIR TAILS AND THE BOUNDED-PARTICLE CASE

The remote positive tail is uniformly negligible relative to I_K. Exact untilting at $M\sqrt{\log n}/s_n$, as in (A.11), also bounds the shifted density for $z \ge T$ with the factor e^{TD_n}. Integrating the product of the two density bounds gives

$$\int_T^\infty q_n(z - d_k) q_n(z - d_\ell) \prod_{j \ne k,\ell} H_n(z - d_j)\, dz \le \frac{C n^{-M^2} e^{2TD_n + o(1)}}{T} = o(I_K), \qquad I_K \ge c\frac{\sqrt{1 + \log K}}{K^2}.$$

The lower bound follows from $J_K^2 \asymp 1 + \log K$ and (A.6), with the finitely many small integers absorbed into the constant. Because $K \le C_0 n^p$ and $TD_n = o(1)$, the displayed positive-tail ratio is at most $C n^{2p - M^2}/\sqrt{\log n} \to 0$. The normal pair integrals with exponents $(K-2)(1 \pm \varepsilon_n)$ have the same upper-tail bound, since their CDF factors are at most one.

For the negative half-line, compact CDF convergence and $D_n \to 0$ imply that $H_n(z - d_j) \le q_* < 1$ for every $z \le 0$, every allowed shift and all sufficiently large n. Using $\sup q_n \le C$ and integrating the other density gives

$$\begin{aligned} \int_{-\infty}^0 q_n(z - d_k) q_n(z - d_\ell) \prod_{j \ne k,\ell} H_n(z - d_j)\, dz &\le C q_*^{K-2}, \\ \frac{C q_*^{K-2}}{I_K} &\le C \frac{K^2 q_*^{K-2}}{\sqrt{1 + \log K}} \longrightarrow 0 \quad (K \longrightarrow \infty). \end{aligned}$$

This limit depends only on K, so it also covers arbitrarily slow diverging sequences K_n. The negative normal pair integral is similarly bounded by $C 2^{-(x-2)}$; it is $o(I_K)$ when $x = 2 + (K-2)(1 \pm \varepsilon_n)$ and $K \to \infty$. For such sequences, the positive middle interval can therefore be replaced by the whole normal pair integral. The density estimate in Step 3 and the CDF-power bounds sandwich it between $(1 + o(1))I_{2+(K-2)(1+\varepsilon_n)}$ and $(1 + o(1))I_{2+(K-2)(1-\varepsilon_n)}$. Step 4 and the two tails prove (A.13) on every diverging sequence.

For bounded K, the negative half-line cannot be discarded: it contributes to I_K, especially at $K = 2$. Pass to a fixed-integer subsequence. On each $[-B, B]$, densities and CDFs with shifts converge uniformly to their normal counterparts. For $B > D_n$, the remaining tails obey

$$\int_{|z|>B} q_n(z-d_k)q_n(z-d_\ell)\prod_{j\neq k,\ell} H_n(z-d_j)\,dz \le CP(|W_n+d_k|>B) \le \frac{C}{(B-D_n)^2},$$
$$W_n = (S_n+E)/s_n.$$

The second inequality is Chebyshev's inequality and the uniform second-moment bound from Step 2. The corresponding normal tails also vanish as $B\to\infty$. First let $n\to\infty$ for fixed B, then $B\to\infty$, to obtain convergence of the complete integral to the fixed positive I_K. Finally, a violating sequence for (A.13) has either a fixed-integer subsequence or a diverging one; both have just been excluded. All estimates are uniform in the arrays and the score bound, establishing the stated supremum rather than only a pointwise limit.

STEP 6. ACTIVE-SCORE TRUNCATION, REMOVAL AND MATRIX UNIFORMITY

For $X_k=\mu+C^{1/2}Z_k$, put $\Lambda=m-(m-1)\delta$. Uniformly in C, labels and particles,

$$|A_k| \le m\log 2+\sqrt{m}\,\|\mu\|+\sqrt{m\Lambda}\,\|Z_k\|, \qquad Ee^{\|Z_k\|^2/4}=2^{m/2}. \tag{A.14}$$

Indeed $-\log\sigma(t)\le\log 2+|t|$, the Euclidean-to-coordinate bound contributes $\sqrt{m}$, and $\|C^{1/2}\|_{\mathrm{op}}\le\sqrt{\Lambda}$. Markov's inequality and a union bound show that the event $\mathcal{B}_n=\{\max_k\|Z_k\|\le 2\sqrt{(p+3)\log n}\}$ has complement probability at most $C_0 2^{m/2}n^{-3}$. On this event (A.14) implies $\max_k|A_k|\le C_1\sqrt{\log n}$ for one fixed C_1 depending only on m,μ,δ,p, for all sufficiently large n. This is the source of uniformity in C; compactness of the matrix class alone would not upgrade a pointwise pair estimate.

Condition on all active particles and labels. They determine both the scores and the marks, whereas the nuisance sums remain independent. Rewrite (A.5) as $-E\sum_{k<\ell}\pi_k\pi_\ell(H_{k,ij}+H_{\ell,ij})$. On $\mathcal{B}_n$, (A.13) says that each conditional pair mass is $(I_K/s_n)(1+r_{k\ell}(A))$. Since $|H_{k,ij}|\le 1$, the total error is bounded by

$$\left|\frac{I_K}{s_n}E\left[\mathbf{1}_{\mathcal{B}_n}\sum_{k<\ell} r_{k\ell}(A)(H_{k,ij}+H_{\ell,ij})\right]\right| \le \eta_n\frac{K(K-1)I_K}{s_n}=\eta_n w_{n,K}, \qquad \sup_{A,k\neq\ell}|r_{k\ell}(A)|\le\eta_n\longrightarrow 0.$$

Without the event, the main term is exactly $-K(K-1)I_K EH_{k,ij}/s_n=-w_{n,K}EH_{k,ij}$, by particle exchangeability. Removing the indicator from this main term costs at most $w_{n,K}P(\mathcal{B}_n^c)$. For the actual derivative off the event, $\sum_k\pi_k(1-\pi_k)\le 1$ gives an absolute error at most $P(\mathcal{B}_n^c)=O(n^{-3})$. Uniformly in $K\ge 2$, $J_K\ge J_2>0$ and $s_n\le C\sqrt{n}$, so $w_{n,K}\ge c/\sqrt{n}$; hence this last error is $o(w_{n,K})$. Substitution of the mark mean in (A.5) proves

$$\sup_{\substack{\mathbf{a}_n\in[a_-,a_+]^n,\ 2\le K\le C_0n^p\\ C\in\mathcal{E}_\delta^{(m)}}}\left|\frac{\partial_{c_{ij}}F_{n,K}^{*,\mathbf{a}_n}(C)}{w_{n,K}}+\frac{\kappa_{*,ij}+\kappa_{ij}(c_{ij})}{4}\right|\to 0. \tag{A.15}$$

The K supremum is over integers. This is an absolute error after scaling, including at gradient zeros. Specialize to unit nuisance scales, so $s_n=\sqrt{nv(1)}$. Integrating along $I+t(C-I)$, which stays in the convex feasible set, proves uniformly in C and allowed K

$$\frac{F_{n,K}^*(C)-F_{n,K}^*(I)}{w_{n,K}}\longrightarrow V(C),$$
$$V(C)=-\frac{1}{4}\sum_{i<j}\left[\kappa_{*,ij}c_{ij}+\int_0^{c_{ij}}\kappa_{ij}(u)\,du\right]. \tag{A.16}$$

The finite number of coordinates and their uniform bounds make the integrated uniform error vanish. This proves the gradient and value conclusions only: it does not differentiate the remainder to infer Hessian convergence. No assertion here allows m or μ to grow with n.

A.3. Extreme-point spectrum and observation risk

The Hessian of V in the free coordinates is diagonal with entries $-\kappa'_{ij}(c_{ij})/4 < 0$. Thus V is strictly concave. A minimizer of a strictly concave function on a convex set must be an extreme point: a nontrivial convex combination would have value strictly above the average of its endpoint values. The affine bijection $R = (C - \delta I)/(1 - \delta)$ maps the feasible set to the real correlation matrices.

For completeness, the required classical rank bound has a short proof. If $R = UU^\top$ has rank r, take $U \in \mathbb{R}^{m\times r}$ of full column rank, with unit-length rows u_i. If $r(r+1)/2 > m$, the linear equations $u_i^\top S u_i = 0$ admit a nonzero symmetric S. For sufficiently small $\varepsilon > 0$, the matrices $U(I \pm \varepsilon S)U^\top$ are distinct correlation matrices averaging to R; distinctness follows from injectivity of $S \mapsto USU^\top$. Hence an extreme R must satisfy

$$\begin{aligned} \mathcal{M} = \operatorname*{argmin}_{\mathcal{E}_\delta^{(m)}} V, \qquad r_{\max} = \left\lfloor \frac{\sqrt{8m+1}-1}{2} \right\rfloor, \\ C \in \mathcal{M} \quad \Longrightarrow \quad \operatorname{rank} \frac{C - \delta I}{1-\delta} \le r_{\max}, \qquad \lambda_1(C) = \cdots = \lambda_{m-r_{\max}}(C) = \delta. \end{aligned} \tag{A.17}$$

Eigenvalues are ordered increasingly. The rank restriction is the known extreme-correlation geometry of (Li & Tam, 1994); its application here is to the particle-induced potential. C remains positive definite and full rank.

All active pattern probabilities are positive and continuous in C, hence uniformly bounded away from zero on the compact class. Equation (A.4) means that the pair observation moments identify every c_{ij}; therefore $Q_C^A = Q_{C_*}^A$ only at $C = C_*$. Every relative-interior C_* is excluded from $\mathcal{M}$, including $C_* = I$, for which the limiting gradient is zero but strict concavity still excludes a minimum.

An explicit general lower bound is available. Using $\log\cosh x \le x^2/2$ and Jensen,

$$\kappa'_{ij}(c) \ge \frac{1}{4} E e^{-(X_i^2+X_j^2)/4} \ge \frac{1}{4} e^{-(1+\|\mu\|_\infty^2)/2} =: \beta_\mu > 0. \tag{A.18}$$

Put $q = m - r_{\max} \ge 1$ and $\Delta_* = \lambda_q(C_*) - \delta > 0$. For $C \in \mathcal{M}$, Weyl's inequality and the zero-diagonal row-sum bound give $\Delta_* \le \|C - C_*\|_{\mathrm{op}} \le (m-1)\max_{i<j} |c_{ij} - c_{*,ij}|$. One spin-product mean thus differs by at least $\beta_\mu \Delta_*/(m-1)$. Map the full observation to that $\{-1,1\}$-valued statistic and apply Bernoulli Pinsker:

$$\inf_{C\in\mathcal{M}} \mathcal{R}_*(C) \ge R_* := \frac{1}{2}\left[\frac{\beta_\mu \Delta_*}{m-1}\right]^2 > 0. \tag{A.19}$$

No overlapping pair risks are added, and no latent-Gaussian KL lower bound is transferred to observations.

For any prescribed allowed K_n, compactness and the uniform limit (A.16) give

$$F^*_{n,K_n}(\widehat{C}_n) - \inf_C F^*_{n,K_n}(C) = o(w_n) \quad \Longrightarrow \quad \operatorname{dist}(\widehat{C}_n, \mathcal{M}) \to 0, \quad \liminf_n \mathcal{R}_*(\widehat{C}_n) \ge R_*. \tag{A.20}$$

Indeed the potential regret is bounded by the scaled objective regret plus twice the uniform approximation error. Outside every neighborhood of $\mathcal{M}$, the continuous potential has a positive gap. For random $o_p(w_n)$ regret, distance converges in probability; for each $\varepsilon > 0$, $P(\mathcal{R}_*(\widehat{C}_n) < R_* - \varepsilon) \to 0$, and $\liminf E\mathcal{R}_*(\widehat{C}_n) \ge R_*$. Multiple minimizers need not share one KL value. Ordinary $o(1)$ objective regret is insufficient because $w_n \to 0$.

FOUR-ACTIVE NUMERICAL FIXTURE

For the controlled experiments,

$$\mu = (-.6, -.2, .2, .6), \qquad C_* = \begin{pmatrix} 1 & .25 & -.15 & .10 \\ .25 & 1 & .10 & -.15 \\ -.15 & .10 & 1 & .25 \\ .10 & -.15 & .25 & 1 \end{pmatrix}, \tag{A.21}$$
$$\delta = .1, \qquad \operatorname{spec}(C_*) = (.5, 1, 1.2, 1.3).$$

Here $r_{\max} = 2$, so every $C \in \mathcal{M}$ has two eigenvalues equal to .1. Its second eigenvalue differs from truth by .9, forcing some correlation error to be at least .3. Equation (A.18) gives $\beta_\mu = \frac{1}{4}e^{-.68} > 1/8$; the final inequality is analytic, since $\log 2 > 2(1/3 + 1/81) = 56/81 > .68$. Therefore

$$\mathcal{R}_*(C) \geq \frac{1}{2}(.3/8)^2 = \frac{9}{12800} \quad (C \in \mathcal{M}), \qquad \liminf_n \mathcal{R}_*(\widehat{C}_n) \geq \frac{9}{12800} \tag{A.22}$$

under (A.20), with the corresponding random-regret statement. This does not assert two exact floor eigenvalues for finite-n population minima, or a spectral/risk guarantee for the reported 512-step Adam programs.

Initialization also matters to interpretation. At truth the scaled particle gradient tends to $-\kappa_{*,ij}/2$, whereas exact likelihood has zero gradient. At independence, with $h_i = E\sigma'(\mu_i + G) > 0$,

$$\partial_{c_{ij}} E_*[-\log Q_C^A(Y)]\big|_{C=I} = -\frac{4h_i h_j \kappa_{*,ij}}{(1-b_i^2)(1-b_j^2)}. \tag{A.23}$$

Factorization at I gives the corresponding score $4h_ih_j(Y_i - b_i)(Y_j - b_j)/[(1-b_i^2)(1-b_j^2)]$, proving (A.23). Its sign agrees with the leading raw direction at I. Initial movement toward the true signs is not itself a wrong direction; the potential's global preference is constrained by matrix feasibility, not by independent coordinatewise endpoints.

A.4. Two-active warm-up and matched-variance support

For $m = 2, \mu = 0, \rho \in [-R, R]$, $0 < R < 1$, let $\theta \in [-R, R]$ be the fixed true correlation and define $r(\rho) = E_\rho[\tanh(X_1/2)\tanh(X_2/2)]$. Then

$$Q_\rho^A(y) = \frac{1 + y_1 y_2 r(\rho)}{4}, \qquad \mathcal{R}_\theta(\rho) = \operatorname{kl}\left(\frac{1+r(\theta)}{2} \middle\| \frac{1+r(\rho)}{2}\right),$$
$$\ell_s'(\rho) = \frac{r'(\rho)[r(\rho) - s]}{1 - r(\rho)^2}, \qquad \frac{(F_{n,K}^\theta)'(\rho)}{w_{n,K}} \to -\frac{r(\theta) + r(\rho)}{4}. \tag{A.24}$$

Here s is an empirical agreement moment, not the radial parameter of the older four-active experiments. The function r is odd and strictly increasing by (A.4). Consequently the limiting raw potential is $-[r(\theta)\rho + \int_0^\rho r(v)\,dv]/4$, with minimum $\operatorname{sign}(\theta)R$ for $\theta \neq 0$, or the endpoint set for $\theta = 0$. Interior true parameters have positive near-optimal risk; true boundary points are not asserted to fail.

Equal total log-weight variance does not force the same limit. Let $v(a) = \operatorname{Var} \log \sigma(aG)$. The identity $v(a) = a^2/4 + \operatorname{Var} \log\cosh(aG/2)$ and Gaussian Poincaré give $a^2/4 \leq v(a) \leq a^2/2$. Continuity and $v'(a) = a/2 + 2\operatorname{Cov}(\log\cosh(a|G|/2), |G|\tanh(a|G|/2)/2) \geq a/2$ give a unique positive A_n satisfying

$$v(A_n) + (n-1)v(1/n) = nv(1), \qquad A_n = \Theta(\sqrt{n}),$$
$$W_n = 2\sigma(A_nG_n)\prod_{j<n} 2\sigma(G_j/n), \qquad EW_n = 1, \quad EW_n^2 \leq 2e^{1/(4n)}, \quad \|\log W_n\|_2 \leq A_n + 1. \tag{A.25}$$

The second moment follows from $E[(2\sigma(aG))^2] = 1 + E\tanh^2(aG/2) \leq 1 + \min(1, a^2/4)$. The log bound follows from $|\log(2\sigma(x))| \leq |x|$ and the L^2 triangle inequality. The observation law and total log variance match the diffuse unit-scale representation exactly.

The negative log tail needs explicit control. If iid $T_k > 0$ satisfy $ET = 1$, $ET^2 \le M$, $\|\log T\|_2 \le L$, then

$$\begin{aligned} P(\bar{T}_K < 1/2) &\le e^{-K/(8M)}, \\ 0 \le E[-\log \bar{T}_K] &\le \frac{2(M-1)}{K} + (\log K + L)e^{-K/(16M)} =: \Delta_K, \\ E|\log \bar{T}_K| &\le 2\sqrt{\frac{M-1}{K}} + (\log K + L)e^{-K/(16M)} =: \varepsilon_K. \end{aligned} \tag{A.26}$$

Use $e^{-x} \le 1 - x + x^2/2$ and a negative Laplace bound at $1/(2M)$ for the first line. The log is integrable because $\bar{T}_K \ge T_1/K$ and $(\log \bar{T}_K)^+ \le \bar{T}_K$. On $\bar{T}_K \ge 1/2$, use $-\log x + x - 1 \le 2(x-1)^2$ and $|\log x| \le 2|x-1|$; on its complement use $-\log \bar{T}_K \le \log K - \log T_1$ and Cauchy–Schwarz. Finally $E[-\log \bar{T}_K] = E[-\log \bar{T}_K + \bar{T}_K - 1]$. Positivity and a second moment alone would not control this log tail.

For any fixed m, μ, δ, write $f_C(y \mid Z) = \prod_i \sigma(y_i[\mu + C^{1/2}Z]_i)$ and apply (A.26) to $T = W_n f_C(y \mid Z)/Q_C^A(y)$. Compactness gives $\min_{C,y} Q_C^A(y) > 0$, $f^2 \le f$, and the Gaussian log envelope gives uniform constants $M = O(1)$, $L = O(\sqrt{n})$. Thus

$$\begin{aligned} 0 \le F_{n,K}^{*,\mathrm{conc}}(C) - n\log 2 - E_*[-\log Q_C^A(Y)] \le \Delta_{n,K}, \\ \Delta_{n,n+2} = O(n^{-1}) + O((\sqrt{n} + \log n)e^{-cn}). \end{aligned} \tag{A.27}$$

For a population η-near-minimizer, comparison with C_* gives $\mathcal{R}_*(\widehat{C}) \le \Delta_{n,K} + \eta$. This proves concentrated population recovery, including on the full matrix class, while (A.20) proves diffuse population failure. The concentrated scales leave every fixed positive compact window.

For clarity, the previously established actual shared-bank separation remains one-dimensional. Here $\widehat{F}$ averages the loss over N complete rows and B complete prior banks shared across all rows and candidates. For any such dataset and grid $\mathcal{G}_n$ of size M_n, write $\ell_{\mathcal{D}}^A = N^{-1}\sum_j [-\log Q_\rho^A(Y_j)]$. Equation (A.26), triangle inequality and maximum bounded by sum give $E[\max_{\rho\in\mathcal{G}_n} |\widehat{F}^{\mathrm{conc}}(\rho) - n\log 2 - \ell_{\mathcal{D}}^A(\rho)| \mid \mathcal{D}] \le M_n \varepsilon_{n,K}$, for any prescribed $B \ge 1$; no row or candidate independence is needed. At $K = n+2$, $M_n = O(n^{1/4})$, mesh tending to zero, iid data $N_n \to \infty$ independent of the banks, and grid regret $o_p(1)$, concentrated learning is consistent. Indeed the four active pattern probabilities are uniformly positive, so empirical frequencies give a uniform exact-likelihood law of large numbers; grid approximation and the displayed maximum-error bound complete the argument. Risk is bounded, so convergence also holds in expectation.

The diffuse side requires finer accuracy. Conditional only on nuisance weights a_k, the two-active path gradient h obeys

$$h^2 \le (\sum a_k f_k (X'_{k2})^2)(\sum a_k / f_k).$$

Splitting equal and unequal particle indices, using $f \le 1$, $E(X'_2)^2 = (1-\rho^2)^{-1}$, and $E(1/f) \le C_R = 1 + 2e^{1/2} + e^{1+R}$, gives $Eh^2 \le C_R/(1-R^2)$. The bound $|h| \le \max_k \|(Z_{k1}, Z_{k2})\|/\sqrt{1-R^2}$ justifies differentiation. Jensen over arbitrary data rows and integral Cauchy–Schwarz therefore give

$$E[(\mathrm{osc}_{[-R,R]}\, e_B)^2 \mid \mathcal{D}] \le \frac{4R^2 C_R}{B(1-R^2)}, \qquad e_B = \widehat{F} - E(\widehat{F} \mid \mathcal{D}). \tag{A.28}$$

Sign symmetry makes the conditional mean affine in the empirical agreement s; apply (A.15) at two distinct fixed generating correlations to identify its two coefficients uniformly for $s \in [-1, 1]$. With $B = n, K = n+2$, (A.28) gives $\mathrm{osc}\, e_B = o_p(w_n)$ because $nw_n^2 \asymp \log n$. On a prescribed grid containing the endpoints, diffuse grid regret $o_p(w_n)$ consequently gives the endpoint limit in (A.24), while concentrated grid regret $o_p(1)$ gives zero risk under the same data and bank counts. This is the limited shared-bank learning comparison; (A.28) is not a multidimensional supremum bound or an Adam theorem.

A.5. Observation-preserving scale paths and mechanism identities

Condition on active particles and an observation; let $\alpha_k > 0$ be normalized active likelihoods. Independent fair nuisance coordinates have mean-one weights $W_k(\tau) = \prod_j 2\sigma(u_j\tau_j G_{kj})$. Two applications of Jensen give

$$\begin{gathered} 0 \le E[-\log \sum_k \alpha_k W_k(\tau)] \le \sum_j E \log\cosh(\tau_j G/2) \le \frac{1}{8}\|\tau\|_2^2, \\ \widehat{F}^{\mathrm{fact}}_{n,K} = n\log 2 + \widehat{F}^{\mathrm{raw}}_{0,K}. \end{gathered} \tag{A.29}$$

The final identity is per realization with paired active particles; factorization retains active finite-K bias. To prove strict increase in one positive scale, condition on all other inputs and magnitudes $r_k = |G_k|$. Its sum has the form $\sum_k c_k(1 + \epsilon_k u_k)$, $u_k = \tanh(\tau r_k/2)$, with independent fair signs. Holding other signs fixed, $A = c_k + \sum_{\ell\neq k} c_\ell(1+\epsilon_\ell u_\ell) > c_k u_k$, and

$$E_{\epsilon_k}[-\log(A + \epsilon_k c_k u_k)] = -\tfrac{1}{2}\log(A^2 - c_k^2 u_k^2), \qquad \partial_{u_k} = \frac{c_k^2 u_k}{A^2 - c_k^2 u_k^2} > 0.$$

For $\tau > 0$, $du_k/d\tau > 0$ almost surely. Sum partial derivatives and average; a Gaussian maximum bounds the path derivative and justifies exchange with expectation. Each positive nuisance scale is therefore a population descent opportunity toward zero without changing the observation model.

This boundary is not confined to fair marginals. For any fixed $p \in (0,1)$, strict monotonicity in μ gives a unique smooth even $\mu_p(\tau)$ with $E\sigma(\mu_p(\tau) + \tau G) = p$. Put $X_\tau = \mu_p(\tau) + \tau G$. For $\tau \ge 0$ and differentiable convex ψ satisfying the integrability needed for differentiation, the implicit function theorem and Gaussian integration by parts yield

$$\begin{gathered} \mu_p(0) = \log\frac{p}{1-p}, \qquad \mu_p'(\tau) = -\tau\frac{E\sigma''(X_\tau)}{E\sigma'(X_\tau)}, \qquad |\mu_p'(\tau)| \le \tau, \\ \frac{d}{d\tau}E\psi(\sigma(X_\tau)) = a_\tau \operatorname{Cov}_{\nu_\tau}(\psi'(\sigma(X_\tau)), G) \ge 0, \\ d\nu_\tau = \frac{\sigma'(X_\tau)}{a_\tau}\, d\Phi, \quad a_\tau = E\sigma'(X_\tau). \end{gathered} \tag{A.30}$$

The mean constraint centers G under the tilted law. For convex ψ, the covariance is nonnegative because both factors are increasing functions of G, and is positive for strictly convex ψ when $\tau > 0$. Apply this separately to each particle in $-\log\sum_k \alpha_k W_{y,k}$, where $W_1 = \sigma(X)/p$ and $W_0 = (1-\sigma(X))/(1-p)$, then sum partial derivatives along the common-scale path. The conditional functions are strictly convex; their differentiated integrands are bounded by $\tau + \max_k |G_k|$. Thus the expected score increases strictly with positive scale while preserving Bernoulli p. This is a compensated total derivative, not a fixed-mean partial derivative or an optimizer guarantee. Nonfair nuisance labels do not acquire fair-label canonicalization.

Locally, fix p, K and all inputs other than the changing nuisance coordinate; let ω_k be their normalized likelihood weights, $q = \sum_k \omega_k^2$, $c_1 = 1-p, c_0 = -p$, and $h_y = -\log\sum_k \omega_k W_{y,k}$. Mean preservation and $W_y'(0) = c_y G$ give $Eh_y(\tau) = qc_y^2\tau^2/2 + O(\tau^4)$, $Eh_y'(\tau) = qc_y^2\tau + O(\tau^3)$, and $\operatorname{Var} h_y'(0) = qc_y^2$. Evenness and Gaussian moment envelopes justify the remainders. Under $\tau = \mathrm{softplus}(u) + \varepsilon$, both mean and SD acquire the exact factor $1 - e^{-(\tau-\varepsilon)}$. With zero floor or $\varepsilon \ll \tau \ll 1$, the raw-gradient mean is of order τ^2 and SD of order τ; a small raw gradient does not establish recovery. Fixed positive floor and growing nuisance dimension require separate control.

Finally, for any fixed reference center h_{ij}^{ref}, the direction experiment uses the exact identity

$$\begin{gathered} G_{ij} = -h_{ij}^{\mathrm{ref}} A - \sum_k \gamma_k (H_{k,ij} - h_{ij}^{\mathrm{ref}}), \\ A = \sum_k \gamma_k = 1 - \frac{1}{\mathrm{ESS}}, \qquad EA = 1 - E(1/\mathrm{ESS}). \end{gathered} \tag{A.31}$$

These are full-likelihood weights, not noise-only weights; $EA \neq 1 - 1/E(\mathrm{ESS})$ in general. A changed reference center shifts the two terms oppositely and leaves their sum unchanged, so their proportions are not causal contributions. At fixed

n and truth, $K \to \infty$ makes the expected score gradient tend to zero while $EA \to 1$: the centered coupling cancels the mean-mark term. This follows from the bounded-likelihood law of large numbers and $\sum_y \partial_{c_{ij}} Q_C(y) = 0$. It explains why overlap alone cannot determine a gradient's direction, without turning the decomposition into a new sign criterion.

A separate two-active control remains available for finite $n \geq 0$ and $K \geq 2$: writing the conditional bank mean as $F_s = U + sV$, the Price marks have the fixed agreement sign and $\pi_k(1-\pi_k) > 0$, giving $F'_+ < 0 < F'_-$ and $b = -V' > 0$. The deterministic additive correction $g' = -U' - rV'$ yields $(F_s + g)' = b(r - s)$, hence the exact clipped-likelihood minimizer. This is optimizer agreement, not an unbiased log score or a normalized density. Replacing the whole criterion by an integral of any positive weight times $r - s$ has the same minimizer; neither construction supplies a universal simulation budget or a matrix-learning algorithm.

A.6. Conditional label weights and a finite-data boundary

Keep fixed m, μ, δ and unit independent fair nuisance scales. Let q be any probability vector on the 2^m active spin patterns, including zero components. It remains fixed when comparing candidate matrices and is independent of the new scoring particles. For fixed active pattern s, let $\phi_{s,n,K}(C)$ be the prior expectation of the raw score with any fixed nuisance label pattern, and put $F_{q,n,K} = \sum_s q_s \phi_{s,n,K}$. Gaussian reflection removes nuisance label signs in this expectation; it does not replace the inputs of an actual training run. With $b_{q,i} = E_q S_i$, the correct centered moment is

$$\begin{aligned} A_{q,ij} &= E_q[(S_i - b_i)(S_j - b_j)] \\ &= \mathrm{Cov}_q(S_i, S_j) + (b_{q,i} - b_i)(b_{q,j} - b_j), \\ E_{q,C} H_{k,ij} &= \frac{A_{q,ij} + \kappa_{ij}(c_{ij})}{4}. \end{aligned} \tag{A.32}$$

For empirical frequencies, Cov_q has denominator N, and empirical marginal offsets cannot in general be omitted. Expansion of the mark in (A.5) and label–particle independence prove (A.32).

The near-tie argument is uniform over labels: nuisance sums have the same law after reflection, the envelope (A.14) is independent of the label, and $|H_{k,ij}| \leq 1$. The same truncation has complement probability $O(n^{-3})$ for every label. On the good event the summed relative pair error is $o(w_{n,K})$; off it, $\sum_k \gamma_k \leq 1$ bounds the derivative error. Prove the bound for each of the finitely many labels and then average with q; no division by a label probability is needed. Hence, uniformly over the probability simplex, $C \in \mathcal{E}_\delta^{(m)}$ and integers $2 \leq K \leq C_0 n^p$,

$$\begin{aligned} &\frac{\partial_{c_{ij}} F_{q,n,K}(C)}{w_{n,K}} + \frac{A_{q,ij} + \kappa_{ij}(c_{ij})}{4} \longrightarrow 0, \\ &\frac{F_{q,n,K}(C) - F_{q,n,K}(I)}{w_{n,K}} - V_q(C) \longrightarrow 0, \\ &V_q(C) = -\frac{1}{4} \sum_{i<j} \left[A_{q,ij} c_{ij} + \int_0^{c_{ij}} \kappa_{ij}(u)\, du \right]. \end{aligned} \tag{A.33}$$

The value limit follows by line integration as in (A.16). Only the linear term depends on q. For one common q, convex weights α_ℓ, a candidate panel C_ℓ and its physical mean $\bar{C}$, write ρ_ℓ for the vector of free correlations and $H_{ij}(c) = \int_0^c \kappa_{ij}(u)\, du$. Then

$$\begin{aligned} \nabla^2 V_q &= -\tfrac{1}{4} \operatorname{diag}(\kappa'_{ij}) \preceq -\tfrac{\beta_\mu}{4} I, \\ \mathcal{J}_{V_q} &:= \sum_\ell \alpha_\ell V_q(C_\ell) - V_q(\bar{C}) \\ &= -\tfrac{1}{4} \sum_{i<j} \left[\sum_\ell \alpha_\ell H_{ij}(c_{ij}^{(\ell)}) - H_{ij}(\bar{c}_{ij}) \right] \\ &\leq -\tfrac{\beta_\mu}{8} \sum_\ell \alpha_\ell \|\rho_\ell - \bar{\rho}\|^2. \end{aligned} \tag{A.34}$$

The gap for this fixed panel is independent of q, but minimizers and panels produced by training can depend on q. Different label weights for different candidates do not cancel the linear terms. This is a statement about the limiting potential, not global concavity or Hessian convergence of the finite-particle objective.

For each fixed q, strict concavity and (A.17) place the minimizers of V_q at extreme points with the same floor-eigenvalue bound. An $o(w)$-near-minimizer of the conditional prior expectation approaches that fixed minimizer set. Uniform value convergence does not imply uniform distance to a moving minimizer set for arbitrary q_n: nearly tied minima can switch. If $q_n \to q$, the compactness argument gives distance to the limiting set for q, not necessarily to each moving set.

Extreme-point geometry alone also does not establish an additional simulation error relative to exact empirical likelihood. To see this, concentrate q on one spin pattern s. For fixed positive normalized weights a_k, independent of C and active particles, define $F_s^a(C) = E[-\log \sum_k a_k \prod_i \sigma(s_i X_{ki})]$, with iid $X_k \sim N(\mu, C)$. When $K \geq 2$, the Price derivative of F_s^a has strictly the sign $-s_i s_j$, since $\gamma_k > 0$. The exact pattern probability satisfies

$$\begin{gathered}\partial_{c_{ij}} Q_C^A(s) = s_i s_j E_C \left[\sigma'(s_i X_i)\sigma'(s_j X_j) \prod_{\ell \neq i,j} \sigma(s_\ell X_\ell)\right], \\ C_s = \delta I + (1-\delta) s s^\top, \\ \operatorname*{argmin}_{C \in \mathcal{E}_\delta^{(m)}} [-\log Q_C^A(s)] = \operatorname*{argmin}_{C \in \mathcal{E}_\delta^{(m)}} F_s^a(C) = \{C_s\}.\end{gathered} \tag{A.35}$$

Indeed C_s simultaneously maximizes every signed correlation $s_i s_j c_{ij}$. Along the feasible segment from any other C to C_s, all coordinate contributions decrease weakly and at least one decreases strictly. Conditioning on positive nuisance weights and integrating preserves this conclusion for the present finite-n logistic model. Arbitrary random-weight extensions additionally require the compared objectives or their differences to be defined. At $K = 1$ the expected raw score depends only on fixed marginals and is constant in C, so raw uniqueness fails. None of these signs is a claim about each realized path gradient.

Finally, nonnormalized numerical weights require a mass factor. If $\widetilde{q} \geq 0$ has total mass $a > 0$ and $\widetilde{A}_{ij} = \sum_s \widetilde{q}_s (s_i - b_i)(s_j - b_j)$, linearity gives

$$\begin{gathered}\frac{\partial_{c_{ij}} F_{\widetilde{q},n,K}(C)}{w_{n,K}} \longrightarrow -\frac{\widetilde{A}_{ij} + a\kappa_{ij}(c_{ij})}{4}, \\ \nabla^2 V_{\widetilde{q}} = -\frac{a}{4} \operatorname{diag}(\kappa'_{ij}).\end{gathered} \tag{A.36}$$

The fixed-panel gap also scales by a. These identities do not authorize changing frozen numerical weights, asserting exact machine-level equality across differently normalized panels, or reinterpreting finite-data boundary preferences as universal excess risk over empirical maximum likelihood.

A.7. Free active means and scales: a finite curvature threshold

We now release the active means and marginal scales. In A.7–A.9, $Y \in \{0,1\}^m$ denotes **bits**, and the teacher P may be any law with marginals $p_i \in (0,1)$. Write

$$\begin{gathered}X_i = \mu_i + s_i Z_i, \quad Z \sim N(0, C), \quad s_i \geq 0, \quad C \in \mathcal{E}_\delta^{(m)}, \\ \mu^0 = \operatorname{logit} p, \quad D = \operatorname{diag}\{p_i(1-p_i)\}, \quad \Gamma = \operatorname{Cov}_P(Y), \quad R_Y = D^{-1/2} \Gamma D^{-1/2}, \\ f_\theta(y, Z) = \prod_i \sigma\{(2y_i - 1)(\mu_i + s_i Z_i)\}, \qquad \overline{F}_K(\theta) = E[-\log \sum_k \pi_k f_\theta(Y, Z_k)].\end{gathered} \tag{A.37}$$

Here $\theta = (\mu, s, C)$, the active particles are iid, and the entire positive nuisance-weight vector $\pi_k = W_k / \sum_\ell W_\ell$ is independent of $(Y, Z_1, \ldots, Z_K)$. The weights need not be independent across particles for the next identity. Set $\chi = E \sum_k \pi_k^2$, $\omega = 1 - \chi$. These are nuisance-only weights, not the full weights in (A.31). The centered objective avoids requiring an arbitrary nuisance normalizing constant to be integrable.

At $(\mu, s) = (\mu^0, 0)$, the model is the product of the true marginals, denoted Q_0, and both the centered score and exact risk have zero mean/scale gradients. For fixed C,

$$\begin{aligned} \nabla^2_{\mu\mu}\overline{F}_K = D, \quad \nabla^2_{\mu s}\overline{F}_K = 0, \quad \nabla^2_{ss}\overline{F}_K = D - \omega(\Gamma \circ C), \\ \nabla^2_{\mu\mu}\mathcal{R} = D, \quad \nabla^2_{\mu s}\mathcal{R} = 0, \quad \nabla^2_{ss}\mathcal{R} = D - \Gamma \circ C. \end{aligned} \tag{A.38}$$

To verify the scale block, put $r_i = Y_i - p_i$. The log-kernel derivatives are $r_i Z_{ki}$ and $-D_{ii}Z^2_{ki}\mathbf{1}_{i=j}$. The Hessian of negative log-sum-exp subtracts their weighted covariance. Conditional on Y, π, independence of active particles gives $E_Z[\sum_k \pi_k Z_{ki}Z_{kj} - (\sum_k \pi_k Z_{ki})(\sum_k \pi_k Z_{kj})] = (1 - \sum_k \pi_k^2)c_{ij}$. Averaging proves the first line. Mean scores are identical across particles, and the mixed block vanishes by Gaussian centering. Exact integration gives the second line with coefficient one. Gaussian polynomial envelopes justify differentiation. One may differentiate in signed scales and then restrict to the nonnegative cone. This generalizes the uniform-weight local formula, in which $\chi = 1/K$; the uniform-weight specialization is not claimed as a separate novelty.

There is a sharp feasible threshold over the whole correlation class. Let

$$\lambda = \lambda_{\max}(R_Y), \qquad B = \delta + (1-\delta)\lambda, \qquad \sup_{C \in \mathcal{E}^{(m)}_\delta} \lambda_{\max}(R_Y \circ C) = B. \tag{A.39}$$

Indeed write $C = \delta I + (1-\delta)T$, with T a correlation matrix. The Schur product theorem applied to $(\lambda I - R_Y) \circ T$ bounds $R_Y \circ T \preceq \lambda I$. Equality is attained at $T = qq^\top$, since this conjugates R_Y by $\operatorname{diag} q$. If v is a unit top eigenvector, choose

$$q_i = \operatorname{sign}(v_i), \quad C^\dagger = \delta I + (1-\delta)qq^\top, \quad N = \|D^{-1/2}|v|\|_2, \quad a = D^{-1/2}|v|/N \geq 0. \tag{A.40}$$

Either sign may be used at zero coordinates. This choice makes the negative-curvature candidate a physically feasible nonnegative scale activation. Along $\mu = \mu^0, s = ta, C = C^\dagger$,

$$\overline{F}''_K(0) = \frac{1 - (1-\chi)B}{N^2}, \qquad \mathcal{R}''(0) = \frac{1-B}{N^2}, \qquad \chi > 1 - 1/B \iff (1-\chi)B < 1. \tag{A.41}$$

If $\lambda > 1$ and the strict inequality holds, the whitened score Hessian is uniformly positive over all C, with gap at least $1 - (1-\chi)B$. Compactness and Gaussian derivative envelopes extend this to a neighborhood: the product-marginal manifold is a local minimum transverse to C, while sufficiently small positive activation (A.40) lowers observation risk and raises the score. The mean block remains positive, so releasing means does not remove the conclusion. A smooth marginal-preserving compensation $\mu(t) = \mu^0 + O(t^2)$ has the same scale curvature because the mean gradient is zero; its existence follows from the implicit function theorem.

If $(1-\chi)B > 1$, the constructed activation lowers both quantities, without proving global recovery. Equality is inconclusive. $\lambda = 1$ means pairwise uncorrelated observations, not necessarily joint independence; pure higher-order dependence is invisible to this second-order test. At zero scales C is unidentifiable, so the minimum is a manifold, not a unique full parameter. A negative unconstrained eigenvector at another fixed C need not lie in the scale cone. Nor does nuisance collision determine general gradients away from this base point.

A.8. A leading-order collapse theorem for free active scales

Fix m, δ, a compact mean box containing μ^0, and $s \in [0, S]^m$, with $|\mu_i| \leq U$. The key bound is elementary. For positive iid active kernels independent of π, weighted Jensen and $\log x \leq x - 1$ give

$$\begin{aligned} 0 \leq L(\mu, s) - \overline{F}_K(\theta) \leq \omega M(\theta), \\ L(\mu, s) = \sum_i \{E\,\mathrm{softplus}(\mu_i + s_i G) - p_i\mu_i\}, \\ M(\theta) = E_Y[(E_Z f_\theta(Y,Z))(E_Z f_\theta(Y,Z)^{-1}) - 1]. \end{aligned} \tag{A.42}$$

For clarity, the inverse moment is $E(1/f)$, not $1/Ef$. With $S_f = \sum_k \pi_k f_k$, the pathwise Jensen gap is at most $S_f \sum_k \pi_k/f_k - 1$. In its expectation, equal-index terms give $\sum_k \pi_k^2$; unequal-index terms give $(1 - \sum_k \pi_k^2) Ef\, E(1/f)$. This proves (A.42).

For unit-scale independent Gaussian-logistic nuisance weights, (A.12)–(A.13) with every active log score zero give the overlap asymptotic. Expanding f^{-1} over coordinate subsets and using the Gaussian MGF supplies the required uniform moment bound:

$$\begin{aligned} \sup_{y,\theta} E_Z f_\theta(y, Z)^{-1} &\le 2^m e^{mU + m^2 S^2/2} < \infty, \\ \omega_{n,K} = (1 + o(1)) w_{n,K}, \qquad \sup_{2 \le K \le C_0 n^b,\, \theta} &|\overline{F}_{n,K}(\theta) - L(\mu, s)| = O\Big(\sqrt{\log n / n}\Big) \to 0. \end{aligned} \tag{A.43}$$

The constants $C_0, b > 0$ are fixed and K is integer. The overlap error is uniform in this range. This argument uses the actual non-Gaussian nuisance weights through the already proved near-tie estimate; it does not substitute Gaussian log weights. The inverse-moment bound is an asymptotic qualification, not a useful numerical error certificate for every finite activation.

Strict Jensen and Bernoulli cross-entropy minimization show $L \ge \sum_i [\mathrm{softplus}(\mu_i) - p_i \mu_i] \ge \sum_i h(p_i)$, with equality exactly at $\mu = \mu^0, s = 0$. Thus, for any allowed sequence K_n,

$$\begin{aligned} & F_{n,K_n}(\theta_n) - \inf_\theta F_{n,K_n}(\theta) = o(1) \\ \Longrightarrow \quad & \mu_n \to \mu^0,\ s_n \to 0,\ Q_{\theta_n} \to Q_0, \qquad \mathcal{R}(\theta_n) \to D(P \| Q_0) = \sum_i h(p_i) - H(P). \end{aligned} \tag{A.44}$$

The nuisance-only constant cancels from regret. Compactness, uniform convergence, and the unique mean/scale minimum imply convergence to the product manifold. The finite pattern probabilities are positive and continuous with a uniform lower bound, so KL is continuous. For random $o_p(1)$ regret the conclusion holds in probability and, since risk is bounded, also in expectation. If zero scales are only in the closure, the same assertion uses the closure infimum and permits arbitrarily small positive scales.

The limiting risk is total correlation and is positive for every dependent P, including purely higher-order dependence. To call this a failure to attain otherwise realizable zero risk, one must additionally assume $P = Q_{\theta_*}$ for a feasible θ_*. This leading-order theorem requires $o(1)$ regret, unlike the $o(w)$ resolution of the fixed-scale spectral preference. It neither identifies a finite-n global minimizer nor proves optimizer convergence. In particular, the nuisance scales are fixed at one throughout this theorem.

For active scales $s \in [\varepsilon, S]^m$ with $0 < \varepsilon \le S$, assume the mean box contains the unique roots $E\sigma(\mu_{p_i}(\varepsilon) + \varepsilon G) = p_i$. Envelope differentiation gives $d \min_\mu L_i / ds = sE\sigma'(\mu_{p_i}(s) + sG) > 0$. Hence near-optimal outputs approach $s_i = \varepsilon, \mu_i = \mu_{p_i}(\varepsilon)$, while C remains free. On that limiting manifold, Price's formula and $\sigma' \le 1/4$ imply

$$\begin{aligned} & |\operatorname{Cov}_Q(Y_i, Y_j)| \le \varepsilon^2 (1 - \delta)/16, \\ \liminf_n \mathcal{R}(\theta_n) \ge \tfrac{1}{2} & \left(\max\{|\kappa_{*,ij}| - \varepsilon^2(1-\delta)/4, 0\}\right)^2, \quad \kappa_{*,ij} = \operatorname{Cov}_P(2Y_i - 1, 2Y_j - 1). \end{aligned} \tag{A.45}$$

The risk bound applies Pinsker to a single spin-product moment after the marginals converge. One may maximize over pairs, but not add overlapping pair bounds. It is an asymptotic floor-class statement, not a bound for arbitrary finite outputs with unmatched marginals.

A.9. When nuisance scales are also free

Fix finite $K \ge 2$ in this subsection; the $K = 1$ observation below is a separately noted extension.

Let all $n \ge 1$ independent fair nuisance coordinates share $\tau \ge 0$. Write $F_{A,K}(\theta)$ for the active finite-particle score. The conditional Jensen bound and strict scale monotonicity in A.5 hold for every active θ, so

$$\begin{aligned} B_{n,K}(\theta,\tau) &:= F_{n,K}(\theta,\tau) - n\log 2 - F_{A,K}(\theta), \quad 0 \le B_{n,K} \le n\tau^2/8, \quad \partial_\tau F_{n,K} > 0 \ (\tau > 0), \\ \inf_{\theta,\tau\ge 0} F_{n,K} &= n\log 2 + \inf_{\theta} F_{A,K}, \qquad r(\theta,\tau) = r_A(\theta) + B_{n,K}(\theta,\tau). \end{aligned} \tag{A.46}$$

Here r, r_A are the respective population regrets, both nonnegative. Thus a joint global minimizer has $\tau = 0$ and an active finite-K minimizer, when the minimum exists. For fixed n, K, compact active parameters and bounded τ, vanishing joint regret forces $\tau \to 0$: continuity makes B uniformly positive on each compact subset $\tau \ge a > 0$. There is no rate uniform in growing n, K. If $\tau \ge \varepsilon > 0$, the boundary becomes ε; a softplus parameter with a positive floor may attain only an infimum at finite raw coordinates.

Under correct specification $P = Q_{\theta_*}$ for a feasible θ_*, Jensen gives a useful but standard risk comparison:

$$e_K(\theta_*) = F_{A,K}(\theta_*) - H(P) \ge 0, \qquad \mathcal{R}(\theta) \le r + e_K(\theta_*) \quad \text{or} \quad r + e_K(\theta_*) + n\varepsilon^2/8 \tag{A.47}$$

for zero or positive nuisance floor, respectively; each r is defined in its own feasible class. Compare the joint optimum with $(\theta_*, 0)$, or (θ_*, ε), and use $F \ge H(P) + \mathcal{R} + n\log 2$. Positive kernels, finite active alphabet and Gaussian log moments allow (A.26) at fixed θ_*, giving $e_K \to 0$. Hence $K \to \infty, r \to 0$ suffices for ideal population risk recovery at zero nuisance floor; adding $n\varepsilon^2 \to 0$ suffices for the positive-floor bound. Fixed K leaves active simulation bias, and at $K = 1$ its optimum is the product model itself. These are not finite-training guarantees.

At $(\mu, s, \tau) = (\mu^0, 0, 0)$, the complete Hessian in these coordinates is

$$\nabla^2_{(\mu,s,\tau)} F_{n,K} = \operatorname{blockdiag}\{D,\ D - (1 - 1/K)(\Gamma \circ C),\ n/(4K)\}. \tag{A.48}$$

The C directions and their mixed blocks are zero. For each separate nuisance coordinate, its centered log-kernel derivative is $G/2$, yielding curvature $1/(4K)$; independence kills cross terms and a common scale sums the n entries. Exact observation risk has the scale block $D - \Gamma \circ C$ and zero nuisance block.

The finite-n endpoints of nuisance collision also matter. For iid Gaussian G_{kj},

$$\begin{aligned} \chi_{n,K}(\tau) &= E\sum_k \left[\frac{\prod_j \sigma(\tau G_{kj})}{\sum_\ell \prod_j \sigma(\tau G_{\ell j})}\right]^2, \quad \chi_{n,K}(0) = 1/K, \\ \lim_{\tau\to\infty} \chi_{n,K}(\tau) &= \chi_\infty(n,K) = P(M=0) + \sum_{r=1}^{K} \frac{P(M=r)}{r}, \quad M \sim \text{Binomial}(K, 2^{-n}). \end{aligned} \tag{A.49}$$

Continuity follows by dominated convergence, and finite positive τ gives $1/K < \chi < 1$. If $M \ge 1$, exactly the all-positive rows retain limiting weight, shared equally. If $M = 0$, the sums $\sum_j \min(G_{kj}, 0) < 0$ have an almost surely unique maximum, so one row dominates. The collision is bounded, proving (A.49). In particular,

$$\chi_\infty(n,K) \ge 1 - \binom{K}{2} 4^{-n}. \tag{A.50}$$

This uses $1 - \chi_\infty \le P(M \ge 2) \le E\binom{M}{2}$. The endpoint is generally not one: fixed n, infinite scale differs from increasing n at fixed positive scale.

At the product base point and arbitrary finite τ, (A.38) uses $\chi_{n,K}(\tau)$. Therefore continuity alone yields

$$\begin{aligned} (1 - 1/K)B > 1 &\implies \exists \eta > 0 : \overline{F}''_K(0;\tau) < 0 \quad (0 \le \tau < \eta), \\ \chi_\infty(n,K) > 1 - 1/B &\implies \exists T < \infty : H^F_s(C;\tau) \succ 0 \quad (C \in \mathcal{E}^{(m)}_\delta,\ \tau > T). \end{aligned} \tag{A.51}$$

The first curvature is along (A.40); exact-risk curvature remains negative. If both conditions hold, the threshold is crossed at least once. Neither monotonicity of χ, a unique crossing, nor a finite-amplitude guarantee follows. At $\tau > 0$ the joint objective still has a nuisance descent direction, so positive active curvature is not a joint stationary point.

Finally, local topology must be specified. At $C = I, \tau = 0$, the active score Hessian is $D/K \succ 0$, even when (A.51)'s first condition holds elsewhere in the class. A small Euclidean neighborhood in (μ, s, C, τ) can therefore have a local minimum manifold. All C represent the same product model at $s = 0$; a flat path can move to $C^\dagger$, followed by descent. In covariance-factor coordinates $\Sigma = AA^\top$, the path $A = t\,\mathrm{diag}(a)L_{C^\dagger}$ already lies in every small neighborhood of $A = 0$, excluding that same local-minimum claim when its curvature is negative. The observation distributions also approach Q_0 along descent. This is a coordinate-dependent boundary, not a universal optimizer trap. The experiments use positive physical scales and a specified repair rule; they do not inherit a convergence or trapping theorem from these local identities.

Appendix B. Positioning and the conditional-variance interpretation

B.1. An exact interpretation of the limiting potential

Keep the fixed Gaussian marginals and teacher of Section 2. Write

$$A_C(y,Z)=\sum_{i=1}^{m}\log\sigma(y_iX_i),\qquad T(C)=E_{Y\sim Q^A_{C_*}}\left[\mathrm{Var}_Z\,A_C(Y,Z)\right]. \tag{B.1}$$

For each fixed y, the mean $E_Z A_C(y,Z)$ depends only on the fixed one-dimensional marginals. Thus its derivative with respect to a free off-diagonal covariance vanishes. The mixed derivative of A_C^2 is $2(\partial_{x_i}A_C)(\partial_{x_j}A_C)$, since A_C is a coordinatewise sum. Gaussian integration by parts gives Price's identity (Price, 1958). The square and derivative terms are dominated by Gaussian-integrable envelopes uniformly on the compact class. Applying (6),

$$\partial_{c_{ij}}T(C)=2EH_{ij}=\frac{\kappa_{ij}(C_*)+\kappa_{ij}(C)}{2},\qquad V(C)=-\frac{1}{2}\{T(C)-T(I)\}. \tag{B.2}$$

The second identity follows by integrating over the feasible segment from I and using $V(I)=0$. It is an interpretation of the existing potential, not an independent discovery of variance-driven selection. It concerns conditional *active log likelihood*, followed by a true-observation average. It is distinct from nuisance log-weight variance, the variance of a stochastic gradient, and the variance of a predictive probability. A separate numerical check uses means $(-.6,.4)$, teacher correlation $.25$ and candidate correlations $(-.8,-.3,0,.4,.8)$. Direct conditional-variance quadrature at orders 32/48/64 agrees with the integrated covariance expression (64-point Legendre) to at most 5.85×10^{-14}; the identity is justified analytically above.

B.2. What the nearest asymptotics predict, and what they do not

Daudel–Roueff's v2 Proposition 1 proof, equation (208), has a subleading mean-gradient term $\sqrt{2\log K}\,\partial_\psi B$ in addition to $\partial_\psi E\Lambda$ (Daudel & Roueff, 2026). That mean-gradient formula takes particle count and B to infinity with $\log K=o(B^2)$. The relevant locator is the Gaussian log-weight random-field condition and Section D.4.1, equation (208), of Daudel–Roueff. Formally matching conditional log-weight means and variances in this model gives

$$\begin{aligned}B_C(y)^2&=nv+\mathrm{Var}_Z\,A_C(y,Z),\\ -\sqrt{2\log K}\,E_Y\partial_{c_{ij}}B_C(Y)&=-\frac{\sqrt{2\log K}}{2\sqrt{nv}}\partial_{c_{ij}}T(C)+O\left(\frac{\sqrt{\log K}}{n^{3/2}}\right).\end{aligned} \tag{B.3}$$

As $n\to\infty$ and $K\to\infty$ within the polynomial range, this predicts the coefficient in (8). It is a mechanism connection, not a direct application to the actual model: logistic log weights are not a Gaussian random field, and an ordinary CLT does not give relative pair-kernel accuracy near extremes or permit differentiation of an uncontrolled remainder. The precise Gaussian step is in Section D.4.1, proof of (208): normalized weights are independent of the centered derivatives of the standardized log-weight field, so the derivative contribution has zero expectation. Section D.3's Gaussian-field property supplies this independence; Section 3.4, after (49), explains that it is lost under merely approximate Gaussianity. We locate the condition by its content and these sections rather than its version-dependent label. Uniformity over the full correlation class and fixed $K\geq2$ also need to be established. Moreover, replacing a log weight by a moment-matched Gaussian changes its exact exponential integral in general: the latter is $\log Ee^{\Lambda}=E\Lambda+\mathrm{Var}(\Lambda)/2$. Thus the formal replacement need not preserve the statistical model.

Non-Gaussian log-weight sums already occur in Daudel et al., Theorem 6 (A2 and equations (37)–(38)), with $\log K=o(d^{1/3})$ (Daudel et al., 2023). The scalar expansion is not a uniform $o(w)$ parameter-difference or gradient theorem. Likewise, Bickel et al., Proposition 3.1, assumes the relative tail/density conditions of its Lemmas A.1–A.2 and $(\log K\log d)/d\to0$ (Bickel et al., 2008). On identifying their particle count with K and score standard deviation with $\sqrt{nv}$, their expected maximum-weight excess has the same large-K leading scale $\sqrt{2\log K}/\sqrt{nv}$ as $w_{n,K}$; that scale is not new. It does not directly provide the marked conditional pair estimate (A.13). Earlier collapse results (Bengtsson et al., 2008; Snyder et al., 2008) are therefore background, not novelty here. These comparisons do not rule out future extensions of those frameworks. We make no exponential-budget necessity claim for recovering this specific parameter choice.

B.3. Theorem-level comparison with nearby results

The distinction is uniform parameter-selection-scale analysis for the actual non-Gaussian model and its observation-risk consequence. The conditional-variance identity in B.1 connects the mechanism to established high-mismatch gradient theory. General Jensen bias, estimator-dependent learning, collapse, and correlation-matrix extreme-point geometry are established tools or phenomena. The following comparison concerns the stated analysis objects; it does not claim that earlier frameworks could never be extended to this model.

Result	Key assumptions / object	Established conclusion and distinction here
Monte Carlo objectives / FIVO (Maddison et al., 2017)	Positive unbiased likelihood estimators; expected logarithmic objectives, including particle filtering estimators.	General objective construction and lower-bound properties. The present question identifies the signed correlation field induced by an observation-equivalent nuisance representation.
AESMC, Claim 1 (Le et al., 2018)	Nonnegative unbiased normalizing-constant estimator under a proposal on an extended sampling space.	The expected-log gap is an extended-space KL to the estimator-induced target, not observation KL of the integrated candidate. It does not identify our uniform signed correlation field.
Daudel et al., Theorem 6 (Daudel et al., 2023)	High-dimensional VR-IWAE bounds and weight models, including non-Gaussian log-weight sums under specified assumptions.	A2 and equations (37)–(38), with $\log K = o(d^{1/3})$, yield scalar objective-gap asymptotics beyond an exactly log-normal model. Scalar flattening alone does not supply the full-correlation signed field and risk statement proved here.
Daudel and Roueff, v2 (Daudel & Roueff, 2026)	Gaussian log-weight random-field condition; derivative/expectation interchange; Section D.4.1 of Daudel–Roueff, Proposition 1, equation (208).	The standard-deviation derivative formally predicts our large-K coefficient via B.1. Actual non-Gaussian conditional pair control, uniformity over C and fixed K require additional analysis; an uncontrolled CLT remainder cannot be differentiated.
Particle-filter collapse (Bengtsson et al., 2008; Bickel et al., 2008; Snyder et al., 2008)	Bickel et al. Proposition 3.1: appendix relative tail/density conditions and $(\log K \log d)/d$ tending to zero.	Maximum-weight excess and extreme-value scale are established; this does not directly control marked overlaps conditional on active-score shifts.
Simulated likelihood (Lee, 1995)	Discrete-choice simulated maximum likelihood and logarithmic derivative bias.	Classical bias background; not a theorem about this growing-dimension uniform field.
Multiperiod probit simulations (Geweke et al., 1997)	GHK simulated likelihood for multinomial multiperiod probit models.	The published abstract reports downward bias in serial-correlation estimates. We do not infer a universal sign or a nuisance-growth mechanism from that report; the full experimental analysis was not rechecked here.

Result	Key assumptions / object	Established conclusion and distinction here
Extreme correlation matrices (Li & Tam, 1994)	Real correlation matrices and their extreme-point geometry.	The rank restriction is existing geometry. Our model-specific step places the minima of the particle-induced strictly concave potential in that extreme set, then transfers the separation to observation KL.
Latent-variable collapse (Lucas et al., 2019; Wang & Ziyin, 2022)	Analyzed linear latent-variable models and their likelihood/variational objectives.	Collapse and limits of variance learning predate this work. Our scale boundary uses independent nuisance collision, the constrained correlation class, and a Gaussian-logistic observation-KL limit. The uniform-weight Hessian is a specialization, not a separate novelty claim.
Present fixed-scale theorem	Fixed active dimension and Gaussian marginals; independent positive-compact nuisance scales; polynomial K; population regret $o(w)$.	Uniform first effective signed field, strictly concave matrix potential, floor-eigenvalue geometry and positive observation-risk bound. It does not establish a finite-optimizer limit.
Present free-active-scale boundary	Compact active means/scales, fixed unit nuisance scale, polynomial K; population regret $o(1)$.	Uniform leading-order convergence to a marginal objective and risk tending to teacher total correlation. Jointly free nuisance scales fall under A.9 instead; finite-program outcomes are separate evidence.

Appendix C. Finite selection, a population risk gap and local geometry

C.1. A fixed matched-weight program

Figure 1 of the main text displays this comparison. A positive truth-minus-candidate anchor gap favors the correlated candidate; the scalar Hessian determines whether the truth is a local minimum or maximum. Each panel uses 262,144 banks and its own prespecified 99% simultaneous audit family, giving at least 98% joint coverage under the numerical qualifications. Selection uses 16,384 outer banks and 65,536 labels; this single realization is distinct from the fresh-run probability bound.

Throughout this appendix, $m = 3$, the known mean is zero, $C_* = I$, $K = 100$, $C \in \mathcal{E}_{.1}^{(3)}$, and $\mathcal{R}(C) = D(Q_I \| Q_C)$. Write $F_n = F_{n,100}$. The score, after removing a candidate-independent nuisance constant, is

$$\begin{aligned} \phi_{n,y}(C; Z, a) &= -\log \sum_{k=1}^{100} a_{nk} \prod_{i=1}^{3} \sigma\big(y_i (C^{1/2} Z_k)_i\big), \\ \widehat{F}_n(C) &= B^{-1} \sum_{b=1}^{B} \sum_{y} q_y \phi_{n,y}(C; Z_b, a_b), \\ F_n(C) &= \mathbb{E}\, 8^{-1} \sum_{y} \phi_{n,y}(C; Z, a). \end{aligned} \tag{C.1}$$

The 65,536 labels and frequencies q_y are reused from Appendix H. There are $B = 16,384$ new shared selection banks, each containing 100 independent standard three-dimensional Gaussian vectors. For $n = 0$, $a_{0k} = 1/100$. For $n = 256$, $a_{256,k} = \exp(S_k)/\sum_l \exp(S_l)$, where S_k sums 256 independent standard-Gaussian log-sigmoid terms, independently of the active inputs. Only this weight input changes between branches.

The grid contains the 143 triples in $\{-.9, -.6, -.3, 0, .3, .6, .9\}^3$ satisfying $C \geq .1I$. Selection returns the lexicographically first point within 10^{-12} of the smallest binary64 score, with **no fallback**. Scores and outputs were fixed before evaluating risk. The anchor C_b has all off-diagonals .9. Data, active particles, grid, scoring function and selection rule are identical across branches; $K = 100$ was fixed before this run. This is a shared-bank finite search, not a general per-observation resampling and optimization procedure. In particular, K counts particles inside each logarithm, not the total learning budget.

Weight condition	Returned point	Numerical observation risk
$n = 0$	I	0
$n = 256$	C_b	.031103092779

Full 128-bit ball-arithmetic replay of the saved selection inputs gives an empirical continuous-region margin at least .00390670332106984 for $n = 256$: every C with $\mathcal{R}(C) \leq 10^{-6}$ scores above C_b. It treats saved binary64 inputs as exact mathematical inputs and includes root-residual allowances (Appendix H); it is deterministic, not a fresh-data failure probability. The risk value uses mutually checked Gaussian quadratures, not a certified quadrature remainder. The old $K = 2$ result, where both conditions fail, remains in Appendix H and is not a controlled comparison across budgets.

C.2. From a joint risk enclosure to a population gap

Put $\kappa(c) = \mathbb{E}[\tanh(X_1/2)\tanh(X_2/2)]$ for standard Gaussian marginals of correlation c. Central symmetry eliminates odd spin moments. Walsh inversion and orthogonality give exactly

$$\begin{gathered} Q_C(y) = \frac{1 + u_y}{8}, \qquad u_y = \sum_{i<j} \kappa(c_{ij}) y_i y_j, \\ \mathbb{E}_{Q_I} u = 0, \qquad \mathbb{E}_{Q_I} u^2 = \sum_{i<j} \kappa(c_{ij})^2, \qquad \mathcal{R}(C) = -\mathbb{E}_{Q_I} \log(1 + u). \end{gathered} \tag{C.2}$$

For $r_0 = 10^{-6}$ and $\mathcal{R}(C) \le r_0$, Pinsker applied to $\{y, -y\}$ gives $|u_y| \le a = \sqrt{8r_0} < 1$. For either sign of u, and by Price's identity (Price, 1958),

$$\begin{aligned} -\log(1+u) &= -u + u^2 \int_0^1 \frac{1-t}{(1+tu)^2}\, dt \ge -u + \frac{u^2}{2(1+a)^2}, \\ \kappa'(c) &= \tfrac{1}{4}\mathbb{E}[\operatorname{sech}^2(X_1/2)\operatorname{sech}^2(X_2/2)] \\ &\ge \tfrac{1}{4}\mathbb{E}e^{-(X_1^2+X_2^2)/4} \ge \beta := e^{-1/2}/4. \end{aligned} \tag{C.3}$$

The last inequalities use $\log\cosh v \le v^2/2$ and Jensen. Since $\kappa(0) = 0$, averaging (C.3) and using (C.2) proves

$$\mathcal{R}(C) \le r_0 \implies \|C - I\|_F^2 \le \frac{4(1+\sqrt{8r_0})^2 r_0}{\beta^2} =: d^2, \qquad d \simeq .013227076469306579. \tag{C.4}$$

For the curvature bound, set $S(t) = \sqrt{I + tD}$, $M_b = \max_k \|Z_{bk}\|_2$, and $\ell_k = \sum_i \log\sigma(y_i(S(t)Z_{bk})_i)$. With fully normalized weights π, $\phi'' = -\sum_k \pi_k \ell_k'' - \mathrm{Var}_\pi(\ell_k')$. Logistic derivative bounds and the square-root derivative bounds proved in Appendix H yield, whenever $\|D\|_F \le d < 1$,

$$\begin{aligned} |\phi''| &\le \left\{ \frac{(13/4)M_b^2}{4\lambda} + \frac{\sqrt{3}M_b}{4\lambda^{3/2}} \right\} \|D\|_F^2, \quad \lambda = 1 - d, \\ H(U;\lambda) &= \frac{(13/4)U}{4\lambda} + \frac{\sqrt{3U}}{4\lambda^{3/2}}, \quad b_K = \tfrac{8}{3}(\log K + 3\log 2) \ge \mathbb{E}M_b^2, \\ \inf_{\mathcal{R}(C)\le r_0} \widehat{F}_n(C) &\ge \widehat{F}_n(I) - d\|\nabla\widehat{F}_n(I)\|_2/\sqrt{2} - \tfrac{1}{2}H(U;1-d)d^2, \\ \inf_{\mathcal{R}(C)\le r_0} F_n(C) &\ge F_n(I) - c_d, \qquad c_d = \tfrac{1}{2}H(b_K;1-d)d^2. \end{aligned} \tag{C.5}$$

Here $U \ge B^{-1}\sum_b M_b^2$. The Gaussian bound follows from $\exp(tM_b^2) \le \sum_k \exp(t\|Z_{bk}\|^2)$ with $t = 3/8$. Jensen controls the first moment of M_b. Coordinate-sign symmetry makes the population gradient at I zero; the empirical gradient need not vanish. Taylor's theorem gives the last two lines. The maximum norm avoids a coarse linear-K curvature cost.

An independent audit uses $L = 262,144$ paired banks and true label weights $1/8$, which the learner does not use. Its prespecified contrasts are $g_n = F_n(I) - F_n(C_b)$ and $\Delta = g_{256} - g_0$. Set $A = \|\sqrt{C_b} - I\|_{\mathrm{op}}$, $T = 8$, and $M = 3AT$. A bank difference is set to zero if any active Gaussian coordinate exceeds T in magnitude. Single-branch truncated differences lie in $[-M, M]$; paired differences lie in $[-2M, 2M]$. Cauchy–Schwarz, the Gaussian moment bound and Mills' inequality give

$$\begin{aligned} t_T &= \sqrt{3}A\sqrt{b_K\, 2K(3)\varphi(T)/T}, \qquad \alpha_j = .01/3, \\ \mathrm{hw}_j &= \sqrt{\frac{2s_j^2\log(4/\alpha_j)}{L}} + \frac{7R_j\log(4/\alpha_j)}{3(L-1)} + t_j + \epsilon_j, \end{aligned} \tag{C.6}$$

Here s_j^2 is unbiased sample variance and R_j is range length. For one branch, $(R_j, t_j, \epsilon_j) = (32.821067231192,\ .000003077999,\ 10^{-9})$ to displayed precision; all three double for Δ, whose variance is measured from paired differences. The empirical-Bernstein inequality (Maurer & Pontil, 2009, Theorem 4) and a union bound give one qualified 99% family event. No bank was truncated. Thirty-two fixed banks per branch were checked with ball arithmetic; these diagnostics do not prove the numerical allowance at every population node. The coverage interpretation requires the stated iid Gaussian sampling model and numerical-error qualification. The all-node score enclosures in Appendix E concern the separate fresh-run moment audit in Appendix D, not these earlier anchor-gap banks.

Contrast	Mean	99% family lower	99% family upper
g_0	-.030106637632	-.032232312506	-.027980962758
g_{256}	.005905979917	.002342327475	.009469632359
Δ	.036012617550	.030377739156	.041647495943

Endpoints are rounded outward. The original prespecified enclosure used $d_{\mathrm{old}} = \sqrt{12r_0}/\beta$ and population cost .004361765095074056. Its region lower bound was $-.002019437619389396$: that gate failed and **six of seven original gates passed**. This record is unchanged.

Equation (C.4) is a **post hoc deterministic refinement**, using the same already-covered contrasts. It gives $c_d \simeq .0014471050001073832$ and, on that original event,

$$\begin{aligned} &\inf_{\mathcal{R}(C)\le r_0}\{F_{256}(C)-F_{256}(C_b)\}\ge \eta, \qquad \eta = .0008952224755772784 > 0,\\ &F_{256}(C) - \inf_{C'\in\mathcal{E}^{(3)}_{\cdot 1}} F_{256}(C') < \eta \implies \mathcal{R}(C) > 10^{-6}. \end{aligned} \tag{C.7}$$

The outward bound subtracts the new deterministic cost from the original g_{256} lower endpoint. No new random comparison or error budget is introduced. The implication follows because the infimum is at most $F_{256}(C_b)$. Continuity on the compact class, with integrable envelope $3\log 2 + 3M_b$, ensures a minimizer exists. Every minimizer has risk above 10^{-6}; neither risk .0311 for every minimizer nor global optimality of C_b follows. No claim certifies the high-dimensional asymptotic remainder or a general optimizer's regret.

The independent moment audit in Appendix D bounds the same population gap with its own joint event. Appendix D.5 reuses the deterministic radius and cost above to obtain a region margin greater than .00134 on the same event as the fresh-run risk bound. This later same-event consequence leaves the original audit table, failed gate and earlier margin unchanged.

C.3. An analytic positive-curvature control

Coordinate signs and permutations preserve either population objective. In the free coordinates (c_{12}, c_{13}, c_{23}), they imply exactly

$$\nabla F_{n,K}(I) = 0, \qquad \nabla^2 F_{n,K}(I) = h_{n,K} I_3. \tag{C.8}$$

We prove $h_{0,100} > .00410$. Let X_i be independent standard Gaussians, $p_i = \sigma(X_i)$, $W = p_1p_2p_3$, and define

$$\begin{aligned} &A(s) = \mathbb{E}e^{-sW}, \quad B(s) = \mathbb{E}[X_1X_2e^{-sW}],\\ &D(s) = \mathbb{E}[(X_1^2-1)(X_2^2-1)e^{-sW}],\\ &h_{0,K} = K\int_0^\infty A(s)^{K-2}\{A(s)D(s)+(K-1)B(s)^2\}\,\frac{ds}{s}. \end{aligned} \tag{C.9}$$

The identity follows by twice differentiating the Laplace representation of $-\log\sum_k W_k$; the one-particle Gaussian density scores are X_1X_2 and $(X_1^2-1)(X_2^2-1)$. Reflecting Gaussian coordinates absorbs any label and leaves this diagonal second derivative unchanged. For justification, write $S = \sum_k W_k$ and let J be its joint second density score. Then $\mathbb{E}J = \mathbb{E}(JS) = 0$, so $|\mathbb{E}Je^{-sS}| \le s^2\mathbb{E}(|J|S^2)/2$ near zero. For $T > 0$, the absolute tail integral is at most $\mathbb{E}[|J|\{E_1(T) - \log W_1\}] < \infty$, where $E_1(v) = \int_v^\infty e^{-u}du/u$. This uses $S \ge W_1$ and $-\log\sigma(x) \le \log 2 + |x|$. Gaussian density derivatives have locally uniform polynomial-Gaussian envelopes, also justifying twice-continuous differentiation.

Set $m_r = \mathbb{E}\sigma(X)^r$, $d_r = \mathbb{E}[(X^2-1)\sigma(X)^r]$. Symmetry gives $d_0 = d_1 = 0$. For $r \ge 2$, the even function $f_r = (p^r + (1-p)^r)/2$ increases strictly with $|X|$, hence $d_r > 0$. On $x \ge 0$,

$$\left[\frac{r+1}{r}f_r - f_{r+1}\right]' = \frac{r+1}{2}p'p(1-p)\{p^{r-2}-(1-p)^{r-2}\} \ge 0,$$
$$0 < d_{r+1} \le \frac{r+1}{r}d_r, \qquad D(s) = \sum_{r=2}^{\infty}\frac{(-s)^r}{r!}d_r^2 m_r \ge 0 \quad (0 \le s \le 4/3). \tag{C.10}$$

The covariance of increasing functions proves the first inequality for d_r. The absolutely convergent alternating series has consecutive magnitude ratio at most $s(r+1)/r^2 \le 3s/4$, proving its sign. No global nonnegativity of D is assumed.

Write $a_0 = 1/\sqrt{24} \le a = \mathbb{E}\sigma'(X)$, from $\operatorname{sech}^2(X/2) \ge e^{-X^2/4}$. Jensen gives $A(s) \ge e^{-s/8}$, and integration by parts gives

$$B(s) = s\mathbb{E}[\sigma'(X_1)\sigma'(X_2)p_3(sW-1)e^{-sW}],$$
$$|B(s)| \ge \tfrac{1}{2}a^2 s(1-s)e^{-s} \quad (0 \le s \le 1). \tag{C.11}$$

Integrate this positive contribution only over $[0, 1/8]$. For the remaining negative contribution take $T = 4/3$ and use $m_2 = 1/2 - a$, $e^{-u} \le 1 - u + u^2/2$, and $\|(X_1^2-1)(X_2^2-1)\|_2 = 2$. They give

$$P = \frac{100\cdot 99}{2304}(7/8)^2 e^{-57/32}/128,$$
$$A(T) \le q := 1 - T/8 + (T^2/2)(1/2 - a_0)^3,$$
$$\int_T^{\infty}|D(s)|\frac{ds}{s} \le 2\{e^{-T}/T + 3(\log 2 + 1)\},$$
$$N = 200q^{99}\{3(\log 2 + 1) + e^{-T}/T\}, \qquad h_{0,100} \ge P - N. \tag{C.12}$$

The tail bound follows from $E_1(TW) \le E_1(T) - \log W$, Cauchy–Schwarz and Minkowski. Equation (C.10) controls the rest of the D integral.

The strict numerical threshold uses rational arithmetic alone. Put $a_L = 204124/10^6$, $r = 57/32$, and

$$E_U = \sum_{j=0}^{12}\frac{r^j}{j!} + \frac{r^{13}/13!}{1 - r/14} > e^r, \qquad q_U = 1 - \tfrac{1}{6} + \tfrac{8}{9}(\tfrac{1}{2} - a_L)^3,$$
$$P_L = \frac{100\cdot 99}{2304}(7/8)^2/(128E_U), \quad N_U = 200q_U^{99}\{3(694/1000+1) + 1/5\},$$
$$24a_L^2 < 1, \quad \sum_{j=0}^{6}\frac{(694/1000)^j}{j!} > 2, \quad \sum_{j=0}^{5}\frac{(4/3)^j}{j!} > 15/4,$$
$$h_{0,100} \ge P_L - N_U > 41/10000 = .00410. \tag{C.13}$$

The exponential tail is bounded geometrically. The finite sums prove $\log 2 < .694$ and $e^{-4/3}/(4/3) < 1/5$; the last strict comparison is an exact rational inequality ($P_L - N_U \simeq .004101508638$). Thus I is a strict local minimum for $n = 0$, without Monte Carlo qualification. This does not establish its global optimality.

C.4. A bounded curvature audit and its scope

For $n = 256$ we use a new statistic and new banks. At I, fix label $(1,1,1)$ and let π_k normalize $a_k\prod_i \sigma(Z_{ki})$. Define $q_{ki} = \sigma(-Z_{ki})$, $r_{ki} = q_{ki}(2q_{ki}-1)$, $a_k^{ij} = q_{ki}q_{kj}$, $A_{ij} = \sum_k \pi_k a_k^{ij}$ and $B_{ij} = \sum_k \pi_k^2 a_k^{ij}$. Squaring the Price operator $\sum_k \partial_{Z_{ki}}\partial_{Z_{kj}}$ gives

$$
\begin{aligned}
H_{ij} = \sum_k \big[&-\pi_k(1-\pi_k)r_{ki}r_{kj} + 2\pi_k^2(1-\pi_k)(q_{ki}^2 r_{kj} + q_{kj}^2 r_{ki}) \\
&+\pi_k^2(1-4\pi_k)q_{ki}^2 q_{kj}^2\big] + A_{ij}^2 - 4A_{ij}B_{ij} + 6B_{ij}^2, \\
h_{n,K} = \mathbb{E}H_{ij}, \qquad &-33/8 \le H_{ij} \le 67/16.
\end{aligned}
\tag{C.14}
$$

The four displayed contributions lie respectively in $[-1, 1/8]$, $[-1/8, 1]$, $[-3, 1/16]$, and $[0, 3]$, using $-1/8 \le r \le 1$, $\sum \pi^2(1-\pi) \le 1/4$ and $0 \le B_{ij} \le A_{ij} \le 1$. Logistic derivatives and normalized weights are bounded; conditioning on nuisance weights justifies Price differentiation. At $K = 1$ the polynomial cancels exactly. This estimator and the square-root path Hessian have the same expectation, not necessarily the same bankwise value.

There are 262,144 new independent paired banks. The three pair kernels are averaged **within** each bank; they are not three independent observations. Apply (C.6) with zero tail allowance to $h_{0,100}$, $h_{256,100}$ and their difference, using range lengths $133/16$, $133/16$, $133/8$ and numerical allowances 10^{-9}, 10^{-9}, 2×10^{-9}. This is a separate $\alpha = .01$ family.

Curvature contrast	Mean	99% family lower	99% family upper
$h_{0,100}$	.028200713775	.027644213668	.028757213882
$h_{256,100}$	-.004265410162	-.004912712428	-.003618107895
$h_{256,100} - h_{0,100}$	-.032466123936	-.033641746822	-.031290501050

Endpoints are rounded outward. Manufactured small-K automatic differentiation checks test the polynomial; 32 fixed scientific banks give 192 ball-arithmetic checks with maximum discrepancy 8.35×10^{-15}. A subsequent deterministic computation encloses the exact saved-input n256 kernels at all 262,144 banks, including all three directions, and propagates their errors through the sample mean and standard deviation (Appendix E). It leaves the original table and family unchanged. Neither the original diagnostics nor this saved-input computation establishes a uniform Gaussian/log-sigmoid input-generation error bound, so the population sign retains that qualification.

On this event, (C.8) makes I a strict local maximum for $n = 256$. Every sufficiently small nonzero correlation perturbation decreases its finite population objective and increases exact observation risk; $\kappa'(c) \ge \beta$ ensures that only I has zero risk. The gradient at I is exactly zero, so an exact deterministic optimizer initialized there need not move. No optimizer was run in this curvature audit.

The two random families in this appendix have conservative joint coverage at least 98%, under their sampling and numerical conditions. Jointly including the mean-robustness family in Appendix F and the fresh-run moment family in Appendix D gives at least 96%; adding the earlier Appendix H family gives at least 95%. This covers those specified families, not every historical experiment. Independence between families is unnecessary. Deterministic refinements, the analytic positive-curvature proof and the saved-input arithmetic in Appendix E spend no additional probability budget.

C.5. Leading-term calibration at the retained finite instance

For the same $m = 3$, independent teacher, $n = 256$, $K = 100$ and C_b with all off-diagonals .9, deterministic coefficient quadratures in (A.4), (A.6) and (A.16) give

$$
\begin{aligned}
v(1) &\simeq .271514501801, \quad J_{100} \simeq 2.507593636442, \\
w_{256,100} &\simeq .300774255953, \quad V(C_b) \simeq -.052200051030.
\end{aligned}
$$

Here $V(I) = 0$. The leading objective difference is $w_{256,100}[V(I) - V(C_b)]$. Separately, formal differentiation of the limiting gradient field gives $h_{256,100}^{\text{formal}} = -w_{256,100}\kappa'(0)/4$, where $\kappa'(0) = \frac{1}{4}(E\,\operatorname{sech}^2(G/2))^2 \simeq .170768891292$.

Quantity	Leading or formal value	Retained finite-audit mean	Original qualified 99% family interval
g_{256}, objective gap	.015700431509	.005905979917	[.002342327475, .009469632359]
$h_{256,100}$, formal second-order comparison	$-.012840721555$	$-.004265410162$	[$-.004912712428$, $-.003618107895$]

Gap units are nats per complete observation; curvature units are nats per squared free correlation coordinate. The finite gap mean is 37.62% of its leading term, or the leading term is 2.658 times that mean. The formal curvature magnitude is 3.010 times the finite mean magnitude. Both comparisons lie outside their corresponding original intervals. The intervals belong to separate families; jointly their original conditional coverage is at least 98%. The finite gap and curvature entries are the retained audits in C.2 and C.4. These are comparisons with estimates, not certified asymptotic errors. In particular, the theorem supplies no small relative remainder at this n; the finite consequence is independently qualified. Uniform convergence of the normalized gradient cannot itself be differentiated to establish Hessian convergence, so the curvature row is a **formal second-order comparison**, not a prediction proved by Theorem 1. The leading gap concerns g_{256}, not the nuisance interaction $g_{256} - g_0$.

Deterministic Hermite40/64/96 with correlation Legendre32/48/64, a separately parameterized normal-weighted Legendre128 check, and full-line one-dimensional quadratures agree within 1.2×10^{-13} for the displayed coefficients. These rule sensitivities are not rigorous quadrature remainder bounds. This comparison adds no Monte Carlo data or training.

C.6. A fixed four-scale comparison: overlap and the marked learning field

This comparison keeps the independent three-variable teacher, $K = 100$, I/C_b and the free-coordinate convention of C.1. It uses 131,072 iid complete banks at $n = 64, 256, 1024, 4096$, with common active inputs and nuisance prefixes. The same banks are shared across scales, not independent training repetitions. Let $C(\rho)$ have all off-diagonals ρ, and let w_n denote the reference in (7). The frozen reference $m_H^{\mathrm{ref}} \simeq .0192712788530$ approximates $\kappa(.45)/4$. Define

$$\begin{aligned} g_n &= F_n(I) - F_n(C_b), \quad D_n = \partial_{c_{12}} F_n(C(.45)), \quad \omega_n = E \sum_k \gamma_{nk}, \\ \gamma_{nk} &= \pi_{nk}(1 - \pi_{nk}), \quad A_n = \frac{g_n}{w_n\{V(I) - V(C_b)\}}, \quad B_n = -\frac{D_n}{m_H^{\mathrm{ref}} w_n}. \end{aligned} \tag{C.15}$$

Primary target and bounded kernel. An independent uniform path coordinate is drawn once per complete bank and shared across n. Price differentiation, a Gaussian envelope on the compact positive-floor path, and Fubini give

$$\begin{aligned} g_n &= -\int_0^{.9} \sum_{i<j} \partial_{c_{ij}} F_n(C(\rho))\, d\rho = EX_n, \\ X_n &= .9\, \frac{1}{8} \sum_y \sum_k \gamma_{nk}(C(U)) \sum_{i<j} H_{k,ij}(C(U)), \qquad U \sim \mathrm{Uniform}(0, .9), \\ X_n &\in [-.675s, 1.35s], \quad D_{n,\mathrm{bank}} \in [-s/2, s/2], \quad \omega_{n,\mathrm{bank}} \in [0, s], \qquad s = 1 - 1/K = .99. \end{aligned} \tag{C.16}$$

The eight labels are enumerated with weight $1/8$. A free c_{12} changes both symmetric covariance entries, so its Price derivative has no extra factor $1/2$; the path derivative sums all three free coordinates. For the path support, the two equal-sign labels give a three-mark sum in $[0, 3]$, and the six mixed labels give $[-1, 1]$, yielding the narrower average support in (C.16). This is an unbiased estimator of the objective gap under the mathematical sampling model, not the same random variable as a realized square-root-path derivative.

Fixed-coordinate decomposition. At $C(.45)$, with the same fixed reference m_H^{ref} used in every bank,

$$\mathcal{C}_n = E\sum_k \gamma_{nk}(H_{k,12} - m_H^{\mathrm{ref}}), \qquad D_n = -m_H^{\mathrm{ref}}\omega_n - \mathcal{C}_n,$$
$$\frac{D_n}{w_n} + m_H^{\mathrm{ref}} = \underbrace{m_H^{\mathrm{ref}}\left(1 - \frac{\omega_n}{w_n}\right)}_{O_n} + \underbrace{\left(-\frac{\mathcal{C}_n}{w_n}\right)}_{J_n}, \tag{C.17}$$
$$\mathcal{C}_{n,\mathrm{bank}} \in [-(1/2 + m_H^{\mathrm{ref}})s, (1 - m_H^{\mathrm{ref}})s/2].$$

The bank forms average over all eight labels. The identity holds for any fixed m_H^{ref}; its interpretation as an asymptotic reference retains the reference's numerical qualification. The residuals below share one absolute normalized scale. Complete weights contain active as well as nuisance scores, so J_n is a marked-coupling residual, not an identification of every deeper source of non-Gaussian discrepancy.

n	A_n: mean [interval]	B_n: mean [interval]	ω_n/w_n: mean [interval]
64	-0.14273 [-0.16133, -0.12413]	-0.14859 [-0.17300, -0.12419]	0.99160 [0.98741, 0.99580]
256	0.33789 [0.30066, 0.37511]	0.32796 [0.27900, 0.37692]	1.02633 [1.01697, 1.03569]
1024	0.65135 [0.58246, 0.72023]	0.65173 [0.56171, 0.74175]	1.01815 [1.00267, 1.03363]
4096	0.82920 [0.70351, 0.95489]	0.83086 [0.66679, 0.99493]	1.00822 [0.98451, 1.03193]

n	O_n: mean [interval]	J_n: mean [interval]
64	0.00016179 [0.00008101, 0.00024257]	0.02197309 [0.02146427, 0.02248191]
256	-0.00050746 [-0.00068772, -0.00032720]	0.01345856 [0.01246015, 0.01445698]
1024	-0.00034978 [-0.00064800, -0.00005156]	0.00706136 [0.00528073, 0.00884199]
4096	-0.00015841 [-0.00061528, 0.00029846]	0.00341803 [0.00022369, 0.00661238]

At $n = 64$, the conditional intervals support $g_n < 0$ and $D_n > 0$, opposite to their respective leading signs although scalar overlap is close to its scale. The coupling term $J_{64} \simeq .02197309$ also exceeds m_H^{ref}, whereas $O_{64} \simeq .00016179$. At 64, 256 and 1024, the displayed joint intervals establish that J_n exceeds $|O_n|$; they do not establish this ordering at 4096. The conclusion concerns this coordinate and decomposition, not every point of the $I \to C_b$ path or a causal decomposition of training risk.

All targets and simultaneous intervals. The family contains five raw targets at each of four scales and six adjacent paired contrasts, for 26 targets. The raw targets are the path gap, fixed-coordinate gradient, overlap, coupling, and a secondary direct anchor-loss difference. With unbiased sample variance $\widehat{v}_j$ and prespecified range length R_j, the two-sided empirical-Bernstein radius is

$$\delta_j = .01/26, \qquad B = 131072,$$
$$\mathrm{hw}_j = \sqrt{\frac{2\widehat{v}_j \log(4/\delta_j)}{B}} + \frac{7R_j \log(4/\delta_j)}{3(B-1)} + t_j + \epsilon_j. \tag{C.18}$$

Here $\epsilon_j = 10^{-9}$ for each raw target; their affine transforms and contrasts scale this allowance. The first four raw supports are in (C.16)–(C.17), with zero tail allowance. For the direct gap, set the bank value to zero if any active $|Z| > 8$, as in C.2. Its support is $[-M, M]$, $M = 16.410533615595888$, with tail allowance $3.077999197742064 \times 10^{-6}$. No nuisance log-weight clipping is used. The direct-gap control does not replace the primary path estimator.

n	Target	Mean	Sample variance	Joint lower	Joint upper
64	g_n	-0.00448185566	0.0004559644	-0.00506565	-0.00389807
64	D_n	0.00172259379	0.0001016962	0.00143977	0.00200542
64	ω_n	0.596498369	0.03940728	0.59397700	0.59901974
64	$\mathcal{C}_n$	-0.0132178802	0.0001418427	-0.01352396	-0.01291180
64	g_n^{dir}	-0.00452243934	0.01260892	-0.01126389	0.00221901
256	g_n	0.00530496451	0.0004581111	0.00472058	0.00588935
256	D_n	-0.00190094564	0.0001032908	-0.00218471	-0.00161718
256	ω_n	0.308694352	0.04976738	0.30588104	0.31150766
256	$\mathcal{C}_n$	-0.0040479893	0.0001304913	-0.00434829	-0.00374769
256	g_n^{dir}	0.0055723925	0.04136146	-0.00225117	0.01339595
1024	g_n	0.00511321392	0.0003143611	0.00457247	0.00565396
1024	D_n	-0.00188881694	6.786048e-05	-0.00214970	-0.00162793
1024	ω_n	0.15311671	0.03318471	0.15078953	0.15544389
1024	$\mathcal{C}_n$	-0.00106193787	7.545536e-05	-0.00132973	-0.00079415
1024	g_n^{dir}	0.00613213357	0.06823211	-0.00237854	0.01464281
4096	g_n	0.00325468235	0.0001887566	0.00276135	0.00374801
4096	D_n	-0.00120397332	3.956504e-05	-0.00144172	-0.00096623
4096	ω_n	0.0758116501	0.01858578	0.07402902	0.07759428
4096	$\mathcal{C}_n$	-.000257014127	4.050285e-05	-0.00049721	-0.00001682
4096	g_n^{dir}	0.00365793433	0.08548928	-0.00522308	0.01253895

Means and variances are displayed approximations; interval endpoints are rounded outward. Adjacent differences are formed bankwise before estimating variance. If α_n is the appropriate affine scale, their variance is $\alpha_b^2\widehat{v}_b + \alpha_a^2\widehat{v}_a - 2\alpha_a\alpha_b\widehat{\mathrm{Cov}}(X_a, X_b)$. Their support is the difference of the two scaled supports, and their numerical allowances add. The covariance is not replaced by zero, nor is variance reduction assumed from pairing.

Quantity	Next minus previous n	Mean	Joint interval	Raw paired covariance	Paired variance
A	256-64	0.4806169	[0.43092, 0.53031]	-3.293773e-06	2.33423
A	1024-256	0.3134604	[0.21962, 0.40730]	1.566845e-05	6.705305
A	4096-1024	0.1778487	[0.00305, 0.35265]	9.030132e-06	16.76677
B	256-64	0.4765523	[0.41116, 0.54194]	5.843457e-07	3.813732
B	1024-256	0.3237732	[0.20000, 0.44755]	1.30637e-06	10.99817
B	4096-1024	0.1791244	[-0.05090, 0.40915]	6.744675e-07	26.60021

The complete 20-target and 26-target covariance matrices are retained in the machine-readable companion readout. The table gives all six paired covariances needed for the displayed comparisons. All three adjacent increases in A_n are resolved, with the last lower endpoint only about .00305. The last increase in B_n remains unresolved. At 4096 the two ratios' intervals do not establish inclusion in $[.9, 1.1]$; thus 10% relative accuracy is unconfirmed. The wider direct-gap intervals and all unresolved comparisons are retained, without outcome-based selection.

These are conditional 99% joint intervals under ideal iid complete-bank sampling, the stated numerical allowances and frozen coefficient references. They are not uniform native-arithmetic or exact coefficient certificates, and not an asymptotic remainder theorem. They spend one additional .01 family beyond the five earlier specified families in C.4; their joint

conditional coverage is at least 94% by a union bound, without requiring independence between families. The deterministic local certificates in I.8 spend no additional alpha.

The earlier C.5 gap estimate gives 37.62% of its leading term; this panel gives 33.79% at 256. Different banks and estimators are involved, and the new interval lies inside the earlier gap interval. Figure 2 uses only the new common panel; neither estimate is relabeled as the other's result. This comparison adds interpretation of the finite marked field, not a universal convergence rate, Hessian prediction or particle-budget rule.

In the displayed panel labels, m_H denotes the fixed numerical reference m_H^{ref} used in this section, not the exact expectation used from C.7 onward.

C.7. Exact label-conditioned kernel for the finite-scale correction

Sections C.7–C.14 establish the fixed-K correction and the finite-scale corollary in (11)–(12). They concern the same zero-mean model, but use deterministic model integrals rather than the conditional Monte Carlo intervals in C.6. First fix $K \geq 2$, finite m, a positive-definite correlation matrix C, and a label law $p_y = P(Y = y)$ independent of C. With $Z \sim N(0,1)$ and $X \sim N(0,C)$, put

$$\begin{aligned} L &= \log \sigma(Z), \quad \ell = EL, \quad v = \operatorname{Var} L, \\ \kappa_3 &= E(L-\ell)^3, \quad s = \sqrt{nv}, \\ A_y &= \sum_{i=1}^{m} \log \sigma(y_i X_i), \quad T_y = A_y - m\ell, \\ H_y &= y_1 y_2 \sigma(-y_1 X_1)\sigma(-y_2 X_2), \\ m_y &= E_X H_y, \quad c_y = E_X[(H_y - m_y)T_y], \\ m_H &= E_Y m_Y, \qquad \bar{c} = E_Y c_Y. \end{aligned} \tag{C.19}$$

The label is shared by every particle in a bank and is independent of the candidate particles. In the retained independent-teacher instance its law is uniform on eight labels. Let ξ be an independent centered standard Gumbel, $S_n = \sum_{j=1}^{n}(L_j - \ell)$, and let g, G be the density and CDF of $(S_n + \xi)/s$. All kernels below retain this Gumbel. For each fixed label define

$$\begin{aligned} f_y(z) &= E_X g(z - T_y/s), \quad F_y(z) = E_X G(z - T_y/s), \\ h_y(z) &= E_X[H_y g(z - T_y/s)], \quad q = K-2, \\ D_n &= -\frac{K(K-1)}{s} \sum_y p_y \int_{\mathbb{R}} h_y f_y F_y^q. \end{aligned} \tag{C.20}$$

The last identity follows from the ordered-pair Price formula in A.2 and the Gumbel maximum representation. A common translation by $m\ell/s$ inside each label integral exactly centers the active score. Conditional particle independence is used before the final label average; replacing the products by products of label-mixture densities would change the target. In particular, $\bar{c}$ is a same-particle, label-conditional mark–score covariance, not a covariance between particles or a training-noise statistic.

Write ϕ, Φ for the standard Gaussian density and CDF. If $M_K = \max_{k \leq K} Z_k$ for iid standard Gaussians, integration by parts gives

$$\begin{aligned} I_K &= \int \phi^2 \Phi^q, \quad T_K^{\mathrm{G}} = \int z\phi^2\Phi^q, \\ J_K &= EM_K = K(K-1)I_K, \\ B_K &= EM_K^2 - 1 = K(K-1)T_K^{\mathrm{G}}, \\ B_K &= \frac{K(K-1)(K-2)}{2} \int \phi^3 \Phi^{K-3} \quad (K \geq 3), \\ B_2 &= 0. \end{aligned} \tag{C.21}$$

Here T_y is the active centered score, whereas T_K^{G} and B_K are Gaussian order-statistic constants. The identities follow successively from $\phi' = -z\phi$, $z^2\phi = \phi - (z\phi)'$, and $(\phi^2)' = -2z\phi^2$; all boundary terms vanish.

C.8. Proof and interpretation of the signed second term

For each fixed K, m, C as above, there exist finite constants C_0, n_0 such that, for $n \geq n_0$, the remainder in

$$\begin{aligned} D_n &= -\frac{m_H J_K}{s} + \frac{B_K}{s^2}\left(\frac{m_H \kappa_3}{3v} - \bar{c}\right) + O_{K,m,C}(n^{-3/2}), \\ D_n^{(2)} &= -\frac{m_H J_K}{s} + \frac{b_2}{n}, \qquad b_2 = \frac{B_K}{v}\left(\frac{m_H \kappa_3}{3v} - \bar{c}\right). \end{aligned} \tag{C.22}$$

is bounded by $C_0 n^{-3/2}$. This is a fixed-K statement, with no uniformity in growing K or over a parameter class. Its existential constants do not themselves certify any of the four finite scales. The subsequent sections supply a separate explicit finite bound.

For completeness, the density argument supplies the norms needed by the nonlinear kernel. Put $\epsilon = n^{-1/2}$, $\alpha = \kappa_3/(6v^{3/2})$, $e = (z^3 - 3z)\phi$ and $\mathcal{P} = (1 - z^2)\phi$, so $\mathcal{P}' = e$. For $p = 1, \infty$ and sufficiently large n,

$$\begin{aligned} \|g - \phi - \epsilon\alpha e\|_p &= O(n^{-1}), \\ \|g' - \phi'\|_p = O(n^{-1/2}), &\qquad \|g''\|_p = O(1). \end{aligned} \tag{C.23}$$

To prove these bounds, the density of L on $x < 0$ is $p_L(x) = \phi(x - \log(1 - e^x))/(1 - e^x)$, extended by zero for $x \geq 0$. At $-\infty$ all fixed derivatives have Gaussian decay times a polynomial. At $0-$, writing $t = -\log(-x)$, each derivative is bounded by $\exp(-t^2/2 + O(t))$ times a polynomial and tends to zero. Thus this zero extension is Schwartz. Consequently $U = (L - \ell)/\sqrt{v}$ has all moments, its characteristic function ψ and its derivatives decay rapidly, and $\sup_{|u| \geq \delta} |\psi(u)| < 1$ for every $\delta > 0$. A nondegenerate density cannot have $|\psi(u)| = 1$ at a nonzero frequency. If p_n is the density of S_n/s, the local log expansion and its first two derivatives yield

$$\begin{aligned} n \log \psi(t/\sqrt{n}) &= -t^2/2 + \epsilon\alpha(it)^3 + O(t^4/n), \\ R_n(t) &= \psi(t/\sqrt{n})^n - e^{-t^2/2}(1 + \epsilon\alpha(it)^3), \\ \int_{\mathbb{R}} \left|\partial_t^j\{(it)^r R_n(t)\}\right| dt &= O(n^{-1}), \qquad 0 \leq r, j \leq 2. \end{aligned} \tag{C.24}$$

Here is the tail control behind the last line. On $|t| \leq n^{1/12}$, Taylor remainders and their first two derivatives are bounded by n^{-1} times a fixed polynomial times e^{-ct^2}, including the square of the cubic term. On $n^{1/12} < |t| \leq \delta\sqrt{n}$, choose fixed small δ so $|\psi(u)| \leq e^{-cu^2}$ and $\psi(u) \neq 0$; the differentiated products have at most polynomial factors in n, t, dominated by this Gaussian tail. Beyond $\delta\sqrt{n}$, retain the at most two differentiated factors and two undifferentiated factors; the remaining at least $n - 4$ factors contribute r_δ^{n-4} for some $r_\delta < 1$. The retained Schwartz factors have finite weighted integrals after rescaling, and all powers of n are dominated by that exponential. These three estimates prove (C.24), rather than leaving an uncontrolled Fourier tail.

Fourier inversion and two integrations by parts in frequency now imply $\sup_z (1 + z^2)|p_n^{(r)} - \phi^{(r)} - \epsilon\alpha e^{(r)}| = O(n^{-1})$. This gives both L^1 and L^∞ bounds. Since $E\xi = 0$, $E\xi^2 = \pi^2/6$, convolution with ξ/s gives $\|g - p_n\|_p \leq E\xi^2 \|p_n''\|_p/(2s^2)$, $\|g' - p_n'\|_p \leq E|\xi| \|p_n''\|_p/s$, and $\|g''\|_p \leq \|p_n''\|_p$, proving (C.23). The Gumbel therefore contributes no n^{-1} coefficient in (C.22), but remains in its remainder.

Gaussian active moments and $|H_y| \leq 1$ justify integral Taylor remainders. Centering $ET_y = 0$ gives, in both indicated density norms,

$$\begin{aligned} \|f_y - g\|_{1,\infty} + \|F_y - G\|_\infty &= O(s^{-2}), \\ h_y - m_y f_y &= -c_y g'/s + O_{1,\infty}(s^{-2}). \end{aligned} \tag{C.25}$$

For $\mathcal{I}(f) = \int f^2 F^q$, bounded densities and $L^1 + L^\infty$ errors control all products, with q fixed. Its linear variation at ϕ in direction e is

$$\begin{aligned} D\mathcal{I}_\phi[e] &= 2\int \phi e \Phi^q + q\int \phi^2 \mathcal{P}\Phi^{q-1} = -2T_K^{\mathrm{G}}, \\ \mathcal{I}(f_y) &= I_K - \frac{\kappa_3}{3vs} T_K^{\mathrm{G}} + O_K(n^{-1}), \\ \int (h_y - m_y f_y) f_y F_y^q &= \frac{c_y}{s} T_K^{\mathrm{G}} + O_K(n^{-1}). \end{aligned} \tag{C.26}$$

The first identity is obtained by integrating $[(z^2-1)\phi^2\Phi^q]'$; the last sign uses $\phi' = -z\phi$. The product remainder is $O_K(n^{-1})$ because each first-order density or CDF perturbation is $O(n^{-1/2})$. When $q = 0$, terms multiplied by q are omitted, with no negative CDF power. When $q = 1$, the CDF factor is linear and its second and higher variations vanish. Both cases follow from the same finite polynomial expansion; neither uses the divisions by $q - 1$ introduced later for $q = 98$. Substitution into (C.20), followed by (C.21), proves (C.22).

Finally, $E_X A_y = m\ell$ is independent of a free correlation coordinate. Price differentiation and the coordinate additivity of A_y show

$$\begin{aligned} \partial_{c_{12}} E_X T_y^2 = 2m_y, \qquad \partial_{c_{12}} E_X T_y^3 = 6c_y, \\ V_A = E_Y E_X T_Y^2, \quad T_A = E_Y E_X T_Y^3, \\ m_H = \tfrac{1}{2}\partial_{c_{12}} V_A, \qquad \bar{c} = \tfrac{1}{6}\partial_{c_{12}} T_A. \end{aligned} \tag{C.27}$$

The factor is one for the free off-diagonal Price operator: both symmetric covariance entries change together. Thus the second term combines nuisance skewness with the change of a label-conditional active third moment. For $K = 2$ it vanishes because $B_2 = 0$. It cannot be recovered from scalar overlap alone.

C.9. Five explicit finite remainder budgets

For the rest of the certificate fix $m = 3$, $C = C(\rho)$ with $\rho = 9/20$, the independent teacher, $K = 100$ and $q = 98$. Permutations and global reflection reduce the eight-label sum exactly to two classes. In (C.19)–(C.20), replace H_y by the mean of its three free-edge marks:

$$\begin{aligned} \widetilde{H}_y &= \tfrac{1}{3}(H_{12,y} + H_{13,y} + H_{23,y}), \\ p_+ &= 1/4\ (y = +++), \qquad p_0 = 3/4\ (y = ++-), \\ m_+ &= m_H + 1/4, \qquad m_0 = m_H - 1/12, \\ c_y &= E[(\widetilde{H}_y - m_y)T_y], \quad v_y = ET_y^2. \end{aligned} \tag{C.28}$$

For the mixed labels, a fixed edge is equal-sign twice and opposite-sign four times; after permutation to the representative this is precisely one equal-sign and two opposite-sign edges averaged. Hence the final derivative is still the free c_{12} derivative, not the three-edge radial derivative.

Set $\lambda = \kappa_3/(6vs)$. Use certified upper bounds for the following norms of the actual Gumbel-smoothed noise:

$$\begin{aligned} \|g^{(j)}\|_\infty \le L_j \quad (j = 0,1,2), \\ \|G - \Phi\|_\infty \le \delta_G, \quad \|g - \phi\|_\infty \le \delta_f, \\ \|g' - \phi'\|_\infty \le \delta_p, \quad \|g' - \phi'\|_1 \le V_{\mathrm{diff}}, \\ \|g - \phi\|_2^2 \le a_2^2, \quad \|g - \phi - \lambda e\|_\infty \le R_0, \\ \|G - \Phi - \lambda\mathcal{P}\|_\infty \le R_{\mathrm{CDF}}. \end{aligned} \tag{C.29}$$

$$
\begin{aligned}
M_2 &= \sum_y p_y E[|\widetilde{H}_y - m_y| T_y^2], \\
M_{mv} &= \sum_y p_y |m_y| v_y, \quad M_{cv} = \sum_y p_y |c_y| v_y, \\
v_{\max} &= \max_y v_y, \qquad \Delta = \delta_G + L_1 v_{\max}/(2s^2).
\end{aligned}
\tag{C.30}
$$

For an integer $r \geq 0$ and a real $d \geq 0$, define Gaussian comparison weights, retaining the extreme-value CDF factor:

$$
\begin{aligned}
W_1(r,d) &= \int_{\mathbb{R}} |\phi'| \min(\Phi + d, 1)^r, \\
W_{1,+}(r,d) &= \int_0^\infty |\phi'| \min(\Phi + d, 1)^r, \\
W_2(r,d) &= \int_{\mathbb{R}} |\phi''| \min(\Phi + d, 1)^r, \\
\mathcal{W}_r &= W_1(r,\Delta) + V_{\text{diff}}.
\end{aligned}
\tag{C.31}
$$

The five nonnegative budgets are

$$
\begin{aligned}
\mathsf{E}_1 &= \frac{K L_2 M_2}{2s^3}, \\
\mathsf{E}_2 &= \frac{K(K-1) M_{mv}}{2s^3} \left[\frac{L_2}{K-1} + L_1 \mathcal{W}_q \right], \\
\mathsf{E}_3 &= \frac{K(K-1) M_{cv}}{2s^4} \left[L_2 \mathcal{W}_q + q L_1 L_0 \mathcal{W}_{q-1} \right].
\end{aligned}
\tag{C.32}
$$

$$
\begin{aligned}
\mathsf{E}_4 &= \frac{K(K-1)|\bar{c}|}{s^2} \left[\frac{\delta_p}{K-1} + \delta_G W_2(q, \delta_G) \right], \\
\mathsf{E}_5 &= \frac{K(K-1)|m_H|}{s} \left\{ \frac{2R_0 + J_K R_{\text{CDF}}}{K-1} + a_2^2 + 2\delta_f \delta_G \right. \\
&\qquad \left. + \frac{q}{2} \delta_G^2 [W_{1,+}(q-1, \delta_G) + \delta_f] \right\}, \\
|D_n - D_n^{(2)}| &\leq \mathsf{E}_n := \sum_{j=1}^5 \mathsf{E}_j.
\end{aligned}
\tag{C.33}
$$

This implication holds for every $n \geq 1$ whenever its stated finite inputs hold; the noise construction below certifies those inputs at the four specified scales. The budgets E_j are distinct from the CDF correction function $\mathcal{P}$.

We give the propagation proof to make signs and constants explicit. Integral Taylor expansion gives

$$
\begin{aligned}
\|f_y - g\|_\infty &\leq \delta_{f,y} := L_2 v_y/(2s^2), \\
\|F_y - G\|_\infty &\leq \delta_{F,y} := L_1 v_y/(2s^2), \\
h_y - m_y f_y &= -c_y g'/s + e_y, \\
\|e_y\|_\infty &\leq L_2 E[|\widetilde{H}_y - m_y| T_y^2]/(2s^2).
\end{aligned}
\tag{C.34}
$$

Since $\int f_y F_y^q = 1/(K-1)$, the e_y contribution in (C.20) is bounded by E_1. Let $I_g = \int g^2 G^q$, $J_g = \int g' g G^q$ and $J_\phi = \int \phi' \phi \Phi^q = -T_K^{\text{G}}$. Two exact integrations by parts are

$$\begin{aligned}\mathcal{I}(f_y) - I_g &= \int (f_y - g) f_y F_y^q - \frac{1}{q+1}\int g'(F_y^{q+1} - G^{q+1}),\\ J_g - J_\phi &= \int (g' - \phi') g G^q - \frac{1}{q+1}\int \phi''(G^{q+1} - \Phi^{q+1}).\end{aligned} \tag{C.35}$$

The smooth bounded densities tend to zero at infinity; the derivative bounds make these boundary terms and integrals legitimate. Use $|u^a - v^a| \le a|u-v|\max(u,v)^{a-1}$ and $\max(F_y, G) \le \min(\Phi + \Delta, 1)$. Thus $\int |g'| \max(F_y, G)^r \le \mathcal{W}_r$, and

$$\begin{aligned}|\mathcal{I}(f_y) - I_g| &\le \frac{\delta_{f,y}}{K-1} + \delta_{F,y}\mathcal{W}_q,\\ \left|\int g' f_y F_y^q - J_g\right| &\le \delta_{f,y}\mathcal{W}_q + q\delta_{F,y}L_0\mathcal{W}_{q-1},\\ |J_g - J_\phi| &\le \frac{\delta_p}{K-1} + \delta_G W_2(q, \delta_G).\end{aligned} \tag{C.36}$$

Multiplying by the corresponding coefficients in (C.20) gives $\mathsf{E}_2, \mathsf{E}_3, \mathsf{E}_4$. Crucially, the common kernels are averaged over labels first: their coefficients are m_H and $\bar{c}$, retaining their signs, rather than averages of absolute values.

For E_5, put $d = g - \phi$, $D = G - \Phi$ and use the positive-density interpolation $f_t = \phi + td$, $F_t = \Phi + tD$ for $0 \le t \le 1$. The Edgeworth truncation is not assumed positive. The linear residual and the exact second variation satisfy

$$\begin{aligned}|D\mathcal{I}_\phi[d] - \lambda D\mathcal{I}_\phi[e]| &\le \frac{2R_0 + J_K R_{\mathrm{CDF}}}{K-1},\\ \frac{d^2}{dt^2}\mathcal{I}(f_t) &= 2\int d^2 F_t^q + 4q\int f_t d D F_t^{q-1}\\ &\quad + q(q-1)\int f_t^2 D^2 F_t^{q-2}.\end{aligned} \tag{C.37}$$

The first bound uses $\int \phi\Phi^q = 1/(K-1)$ and $q\int \phi^2\Phi^{q-1} = J_{K-1}/(K-1) \le J_K/(K-1)$. The first two terms of the second variation have absolute bounds $2a_2^2$ and $4\delta_f\delta_G$. For the last one,

$$\begin{aligned}\int f_t^2 F_t^{q-2} &= \int (f_t - \phi) f_t F_t^{q-2} - \frac{1}{q-1}\int \phi' F_t^{q-1}\\ &\le \frac{\delta_f + W_{1,+}(q-1, \delta_G)}{q-1},\\ |I_g - I_K - D\mathcal{I}_\phi[d]| &\le a_2^2 + 2\delta_f\delta_G\\ &\quad + \frac{q}{2}\delta_G^2[\delta_f + W_{1,+}(q-1, \delta_G)].\end{aligned} \tag{C.38}$$

On the negative half-line $\phi' > 0$, so its contribution with the minus sign is nonpositive and can be discarded. This justifies the positive-half-line weight; dropping the sign first would not. The last line retains the Taylor factor $\int_0^1 (1-t)dt = 1/2$. Combining (C.37)–(C.38), then (C.21) and (C.26), gives E_5 and closes (C.33). The divisions by $q-1$ are used only at the fixed $q = 98$.

C.10. Finite Fourier bounds for the actual noise

We use the convention $\widehat{f}(t) = \int e^{itz} f(z)\, dz$. Put $U = (L - \ell)/\sqrt{v}$, $\mu_3 = EU^3$, $\mu_4 = EU^4$, $\alpha = \mu_3/6$, $P_n(t) = \psi(t/\sqrt{n})^n$, and $Q_n(t) = e^{-t^2/2}(1 + \alpha(it)^3/\sqrt{n})$. The latter is the transform of $\phi + \lambda e$, not a probability density. The production split is $\delta = 1/4$, $a = 9/2$, with low, middle and high frequencies separated at $\delta\sqrt{n}, a\sqrt{n}$.

Low frequencies. Define

$$
\begin{aligned}
q_\delta &= \tfrac{1}{2} + |\mu_3|\delta/6 + \mu_4\delta^2/24, \quad z_\delta = \delta^2 q_\delta,\\
C_\delta &= \mu_4/24 + \frac{q_\delta^2}{2(1-z_\delta)},\\
D_\delta &= \frac{\mu_4/6 + (1+|\mu_3|\delta/2)q_\delta}{1-z_\delta},\\
c &= \tfrac{1}{2} - C_\delta\delta^2, \quad c_{\mathrm{mod}} = \tfrac{1}{2}\{1-(2\mu_4+6)\delta^2/24\}.
\end{aligned}
\tag{C.39}
$$

The certified fourth moment is $\mu_4 \simeq 4.53778313038698$; the saved intervals give $z_\delta < .034973$, $c > .4780439$, $c_{\mathrm{mod}} > .4803703$. Taylor expansion through degree three, $|\log(1+w) - w| \le |w|^2/[2(1-|w|)]$, and ψ'/ψ imply $\log\psi(u) = -u^2/2 + \alpha(iu)^3 + r(u)$, with $|r(u)| \le C_\delta|u|^4$ and $|r'(u)| \le D_\delta|u|^3$. No fifth moment is needed for these finite bounds. For $0 \le t \le \delta\sqrt{n}$,

$$
\begin{aligned}
|P_n - Q_n| &\le A_0(t) := e^{-ct^2}\frac{C_\delta t^4}{n} + e^{-t^2/2}\frac{\alpha^2 t^6}{2n},\\
|P_n' - Q_n'| &\le A_1(t) := tA_0(t) + e^{-t^2/2}\frac{3\alpha^2 t^5}{n}\\
&\qquad + e^{-ct^2}\left(\frac{3|\alpha|C_\delta t^6}{n^{3/2}} + \frac{D_\delta t^3}{n}\right),\\
|P_n(t)| &\le e^{-\max(c,c_{\mathrm{mod}})t^2}.
\end{aligned}
\tag{C.40}
$$

The cubic phase is purely imaginary, so $|e^{ib} - 1 - ib| \le b^2/2$ retains it without an $e^{|b|}$ loss. The additional modulus bound follows by applying the fourth-order cosine bound to $U - U'$, whose fourth moment is $2\mu_4 + 6$. Positive coefficients are rounded upward and decay rates downward.

Middle and high frequencies. At the 273 frequencies $u_j = 1/4 + j/64$, $j = 0, \dots, 272$, certified complex finite sums give upper bounds b_j for $|\psi(u_j)|$. Since $|\psi'| \le E|U| \le 1$, each of the 272 cells has the valid cap

$$
\begin{aligned}
\rho_j &= \min\{1, \max(b_j, b_{j+1}) + 1/128\} < 1,\\
|\psi(u)| &\le \min(1, B/|u|), \qquad B = \frac{2\sqrt{v}(1+e^{1/2})}{\sqrt{2\pi}} < 1.102.
\end{aligned}
\tag{C.41}
$$

For the high-frequency bound, $p_L(\log\sigma(x)) = \phi(x)(1+e^x)$ and its log derivative in x is $-x+\sigma(x)$, strictly decreasing with opposite endpoint signs. The density is unimodal and vanishes at both ends. Hence $\|p_L'\|_1 = 2\sup p_L \le 2(1+e^{1/2})/\sqrt{2\pi}$; scaling and integration by parts give (C.41). This supplies an explicit tail constant. On a middle cell $|P_n| \le \rho_j^n$ and $|P_n'| \le \sqrt{n}\rho_j^{n-1}$. For high-frequency moments,

$$
\begin{aligned}
\int_{a\sqrt{n}}^\infty t^p (B\sqrt{n}/t)^k dt &= \frac{n^{(p+1)/2}B^k a^{p-k+1}}{k-p-1}, \quad k > p+1,\\
I_p(c;t_0) &:= \int_{t_0}^\infty t^p e^{-ct^2} dt,\\
I_0(c;t_0) &= \frac{\sqrt{\pi}}{2\sqrt{c}}\operatorname{erfc}(\sqrt{c}\,t_0), \quad I_1(c;t_0) = \frac{e^{-ct_0^2}}{2c},\\
I_p(c;t_0) &= \frac{t_0^{p-1}e^{-ct_0^2} + (p-1)I_{p-2}(c;t_0)}{2c} \quad (p \ge 2).
\end{aligned}
\tag{C.42}
$$

Middle-cell polynomial integrals are exact endpoint power differences. Formula (C.42) handles powers $n, n-1, 2n, 2n-2$ and all Gaussian polynomial tails. Low-frequency positive envelopes may be extended to infinity conservatively; their squares include every cross term. Gaussian comparison tails start at $t_0 = \delta\sqrt{n}$ and are counted once, not once per middle cell. No Fourier numerical quadrature is required.

Gumbel and norm conversion. For $\chi_n(t) = Ee^{it\xi/s}$, centering and $E\xi^2 = \pi^2/6$ imply

$$\begin{aligned}
&|\chi_n| \le 1, \quad |\chi_n - 1| \le b(t) := \pi^2 t^2/(12nv),\\
&|\chi_n'| \le d(t) := \pi^2|t|/(6nv),\\
&R_g := \chi_n P_n - Q_n,\\
&|R_g| \le A_g := A_0 + b|Q_n|,\\
&|R_g'| \le A_g' := A_1 + d\, e^{-\max(c,c_{\mathrm{mod}})t^2} + b|Q_n'| \quad (0 \le t \le \delta\sqrt{n}).
\end{aligned} \tag{C.43}$$

Here A_g' denotes a bound for $|R_g'|$, not a derivative of the envelope A_g. On the remaining frequencies use $|R_g| \le M + |Q_n|$ and $|R_g'| \le M_1 + dM + |Q_n'|$, where M, M_1 are the preceding caps for $|P_n|, |P_n'|$. The centered Gumbel is therefore present in every finite error norm.

Let $r_j = g^{(j)} - (\phi + \lambda e)^{(j)}$. Inversion, CDF inversion and Parseval give the following sufficient bounds:

$$\begin{aligned}
\|r_j\|_\infty &\le \pi^{-1} \int_0^\infty t^j A_g(t)dt \quad (j = 0, 1, 2),\\
\|G - \Phi - \lambda\mathcal{P}\|_\infty &\le \pi^{-1} \int_0^\infty A_g(t)/t\, dt,\\
\|r_0\|_2^2 &\le \pi^{-1} \int_0^\infty A_g(t)^2 dt,\\
\|r_1\|_1 &\le \left[\int_0^\infty \{t^2 A_g^2 + (A_g + tA_g')^2\}\, dt\right]^{1/2},\\
\|g^{(j)}\|_\infty &\le L_{j,\mathrm{direct}} := \pi^{-1} \int_0^\infty t^j M(t)dt.
\end{aligned} \tag{C.44}$$

For the L^1 line, Cauchy–Schwarz with $(1 + z^2)^{-1}$ and Parseval gives $\|r_1\|_1^2 \le \frac{1}{2}(\|tR_g\|_2^2 + \|(tR_g)'\|_2^2)$. Symmetry turns this into the displayed positive-frequency integral, with no additional π factor. The CDF integral is finite at zero: the Gumbel error starts at t^2 and the remaining error at t^4. On middle and high frequencies, the production bounds take the Minkowski sum of the CF and Gaussian vector L^2 caps before squaring; this preserves the same normalization. For direct density caps use $|\chi_n| \le 1$.

Write R_1, R_{02}, R_{11} for the resulting upper bounds on $\|r_1\|_\infty, \|r_0\|_2, \|r_1\|_1$. The inputs in (C.29) are recovered, including the first Edgeworth term, by

$$\begin{aligned}
\delta_f &= |\lambda|\|e\|_\infty + R_0, \quad \delta_p = |\lambda|\|e'\|_\infty + R_1,\\
\delta_G &= \min(1, |\lambda|\phi(0) + R_{\mathrm{CDF}}),\\
V_{\mathrm{diff}} &= |\lambda|\|e'\|_1 + R_{11}, \quad a_2^2 = (|\lambda|\|e\|_2 + R_{02})^2,\\
L_0 &= \min(L_{0,\mathrm{direct}}, \phi(0) + \delta_f),\\
L_1 &= \min(L_{1,\mathrm{direct}}, \phi(1) + \delta_p), \qquad L_2 = L_{2,\mathrm{direct}}.
\end{aligned} \tag{C.45}$$

All comparisons use upper endpoints. The required Hermite norms are closed scalar expressions. With $a_* = \sqrt{3 - \sqrt{6}}$, $b_* = \sqrt{3 + \sqrt{6}}$ and probabilists' Hermite polynomials He_j,

$$\begin{aligned}
\|e\|_\infty &= \max_{z \in \{a_*, b_*\}} |\,\mathrm{He}_3(z)\phi(z)|,\\
\|e'\|_\infty &= \max_{z \in \{0, \sqrt{5-\sqrt{10}}, \sqrt{5+\sqrt{10}}\}} |\,\mathrm{He}_4(z)\phi(z)|,\\
\|e'\|_1 &= 4\sqrt{6}[a_*\phi(a_*) + b_*\phi(b_*)], \qquad \|e\|_2^2 = \frac{15}{16\sqrt{\pi}}.
\end{aligned} \tag{C.46}$$

These follow from the zeros of $e' = -\operatorname{He}_4 \phi$, $e'' = \operatorname{He}_5 \phi$, total variation between extrema, and Gaussian even moments.

Certification of the finite CF table and fourth moment. The original Gaussian variable uses a full-weight trapezoidal sum with $h = 1/16$, $x_j = jh$, $|j| \le 192$, cutoff $R = 12$, strip half-width $\eta = 1$ and 192-bit ball arithmetic. Set $d_0 = E \log\cosh(Z/2)$, so $L - \ell = x/2 - \log\cosh(x/2) + d_0$. On $|\operatorname{Im} x| \le \eta$, $|\operatorname{Im} \log \sigma(x)| \le \eta$. The strip integral caps and resulting errors are

$$\begin{aligned} M_\psi(u) &= \exp(\eta^2/2 + |u|\eta/\sqrt{v_-}), \\ \varepsilon_\psi(u) &= \frac{2M_\psi(u)}{e^{2\pi\eta/h} - 1} + \frac{2\phi(R)}{R}, \\ B_4^* &= d_{0,+} + \eta + \log\sec(\eta/2), \\ M_4 &= e^{\eta^2/2} E(|Z| + B_4^*)^4, \\ \varepsilon_4 &= \frac{2M_4}{e^{2\pi\eta/h} - 1} + \sum_{k=0}^{4} \binom{4}{k} d_{0,+}^{4-k} P_k(R). \end{aligned} \tag{C.47}$$

The fourth-moment strip bound uses $|L - \ell| \le |\operatorname{Re} x| + B_4^*$. Its expectation is expanded with absolute Gaussian moments $1, \sqrt{2/\pi}, 1, 2\sqrt{2/\pi}, 3$. The infinite trapezoidal alias bound is the strip-boundary integral divided by $e^{2\pi\eta/h} - 1$, summed over both boundaries. For the omitted real grid tail, monotonicity of $x^k\phi(x)$ for $x \ge R \ge 2$, $k \le 4$, gives

$$\begin{aligned} P_0(R) &= 2\phi(R)/R, \quad P_1(R) = 2\phi(R), \\ P_2(R) &= 2\phi(R)(R + 1/R), \\ P_3(R) &= 2\phi(R)(R^2 + 2), \\ P_4(R) &= 2\phi(R)(R^3 + 3R + 3/R). \end{aligned} \tag{C.48}$$

The raw fourth moment receives ε_4 before division by the saved interval for v^2. Each complex sum receives the complex-modulus error ε_ψ, and its true modulus cap is $|\text{finite sum}| + \varepsilon_\psi$. All input dyadic intervals and arithmetic radii are retained. The 385 shared Gaussian nodes produce 273 complex CF sums and one real fourth-moment sum: 547 real scalar sums and 105,105 complex exponentials. These are model-noise calculations, not direct evaluations of D_n.

C.11. Active moment caps without new active integrals

All active inputs follow from the previously certified scalars $V_A, V_H, s_0, m_H, \bar{c}$. Here $V_H = E_Y \operatorname{Var}_X H_{12,Y}$ and $s_0 = E \operatorname{sech}^2(Z/2)$. Write $d(x) = \log\cosh(x/2)$, $\mathcal{D} = \sum_{i=1}^3 [d(X_i) - Ed(Z)]$, $S_y = y^T X/2$, and $a_y = Cy/2$. Then $T_y = S_y - \mathcal{D}$; odd/even symmetry gives $ES_y\mathcal{D} = 0$. Put $\lambda_C = 1 + 2\rho$, $\kappa = 4m_H$ and $t_2 = 1 - s_0$.

$$\begin{aligned} V_d &= E\mathcal{D}^2 = V_A - 3/4, \\ \sigma_+^2 &= 3\lambda_C/4, \quad \sigma_0^2 = (3 - 2\rho)/4, \\ v_+ &= \sigma_+^2 + V_d, \qquad v_0 = \sigma_0^2 + V_d. \end{aligned} \tag{C.49}$$

For the class covariances, Price differentiation along the three-edge ρ path is used only as a moment identity. Twice integrating by parts gives $ES_y^2\mathcal{D} = (s_0/4)\|a_y\|^2$, hence

$$\begin{aligned} ET_y^3 &= -\tfrac{3s_0}{4}\|a_y\|^2 - E\mathcal{D}^3, \\ \frac{d}{d\rho} ET_y^3 &= 6\sum_{i<j} E[T_y H_{ij,y}] = 18c_y, \\ \|a_+\|^2 - \|a_0\|^2 &= 4\rho + 2\rho^2, \\ c_+ = \bar{c} - \frac{s_0(1+\rho)}{8}, &\qquad c_0 = \bar{c} + \frac{s_0(1+\rho)}{24}. \end{aligned} \tag{C.50}$$

The factor 18 is the cubic mixed-derivative factor 6 times three edges. Dividing by it and using $\bar{c} = (c_+ + 3c_0)/4$ proves the last line; no radial derivative is substituted for D_n.

For the fourth moment, write $X = C^{1/2}Z_3$. Since $|d'| \le 1/2$, $\|\nabla_{Z_3}\mathcal{D}\|^2 \le 3\lambda_C/4$. Gaussian Poincare applied to $\mathcal{D}^2$ yields

$$E\mathcal{D}^4 - V_d^2 \le 4E[\mathcal{D}^2\|\nabla_{Z_3}\mathcal{D}\|^2] \le 3\lambda_C V_d, \qquad Q_4 := V_d^2 + 3\lambda_C V_d. \tag{C.51}$$

One may obtain the Poincare inequality here from $\operatorname{Var} u = 2\int_0^\infty E\|\nabla P_t u\|^2 dt$, $\nabla P_t u = e^{-t}P_t\nabla u$ and Jensen. The functions and their gradients are square-integrable by Gaussian moments.

The potentially signed mixed term requires more than an absolute-value bound. Twice integrating by parts gives

$$\begin{aligned} ES_y^2\mathcal{D}^2 &= \sigma_y^2 V_d + 2E(a_y\cdot\nabla\mathcal{D})^2 + 2E[\mathcal{D}(a_y\cdot\nabla)^2\mathcal{D}], \\ E[\mathcal{D}(a_y\cdot\nabla)^2\mathcal{D}] &\le 0. \end{aligned} \tag{C.52}$$

To prove the second line, condition on $X_i = x$. Each $E[d(X_j) \mid X_i = x]$ is even and convex in x, hence nondecreasing in $|x|$, whereas $d''(x) = \frac{1}{4}\operatorname{sech}^2(x/2)$ is nonincreasing in $|x|$. For independent copies R, R' of $|X_i|$, the covariance identity $2\operatorname{Cov}(a(R), b(R)) = E[(a(R) - a(R'))(b(R) - b(R'))] \le 0$ proves $\operatorname{Cov}(\sum_j d(X_j), d''(X_i)) \le 0$. The Hessian of $\mathcal{D}$ is diagonal, so summing with coefficients $(a_y)_i^2 \ge 0$ proves the required sign.

$$\begin{aligned} Q_y &:= E(a_y\cdot\nabla\mathcal{D})^2 = \tfrac{1}{4}\{(t_2-\kappa)\|a_y\|^2 + \kappa(\mathbf{1}^T a_y)^2\}, \\ ET_y^4 &\le 3(\sigma_y^2)^2 + 6\sigma_y^2 V_d + 12Q_y + Q_4, \\ \overline{T}_4 &:= \frac{27+36\rho^2}{16} + \left(\frac{9}{2} + 3\lambda_C\right)V_d + V_d^2 \\ &\qquad + \frac{9}{4}(1+2\rho^2)t_2 + \frac{9}{2}\rho(2+\rho)\kappa, \\ \sum_y p_y ET_y^4 &\le \overline{T}_4. \end{aligned} \tag{C.53}$$

Odd powers of S_y vanish under global reflection. The label averages used in the last two lines are $E_y\sigma_y^2 = 3/4$, $E_y(\sigma_y^2)^2 = (9+12\rho^2)/16$, $E_y\|a_y\|^2 = (3+6\rho^2)/4$, and $E_y(\mathbf{1}^T a_y)^2 = 3\lambda_C^2/4$. Jensen for the three-edge average and label-permutation symmetry imply $\sum_y p_y \operatorname{Var}(\widetilde{H}_y) \le V_H$. Cauchy–Schwarz on (y, X) now gives

$$M_2 \le \sqrt{V_H\overline{T}_4} < .307468813. \tag{C.54}$$

This is an upper bound for a remainder moment, not a measured value of M_2. Together with (C.28), (C.49) and (C.50), it supplies every active input in (C.30), without new active Gaussian integration.

C.12. Gaussian weight enclosures

The four weights required for each scale are $W_1(98, \Delta)$, $W_1(97, \Delta)$, $W_2(98, \delta_G)$ and $W_{1,+}(97, \delta_G)$. Their computation uses only ϕ, Φ on the 3073-node grid $[-12, 12]$ with spacing $1/128$, at 192 bits. For a cell $[a, b]$, the weight $w(z) = \min(\Phi(z) + d, 1)^r$ is nondecreasing. Use $w_- = \min(\underline{\Phi}(a) + d, 1)^r$ and $w_+ = \min(\overline{\Phi}(b) + d, 1)^r$, clipping the CDF enclosures to $[0, 1]$. The grid includes $-1, 0, 1$, so no cell crosses an unaccounted derivative sign change.

$$\begin{aligned}
&\int_a^b |\phi'| = |\phi(b) - \phi(a)|,\\
&\int_a^b |\phi''| = |-b\phi(b) + a\phi(a)|,\\
&M_- w_- \le \int_a^b w(z)|\phi^{(j)}(z)|dz \le M_+ w_+,\\
&\int_{|z|>R} |\phi'| = 2\phi(R), \quad \int_R^\infty |\phi'| = \phi(R),\\
&\int_{|z|>R} |\phi''| = 2R\phi(R) \qquad (R \ge 2).
\end{aligned} \tag{C.55}$$

The first two lines apply on cells not crossing 0 and $\{-1, 1\}$, respectively; $[M_-, M_+]$ is the interval for the relevant derivative mass. Summing lower and upper products, and adding the displayed tail only to the upper sum, encloses each weight. The propagation uses upper endpoints and an upper value of d. Thus all 16 full-line weight bounds follow from saved Gaussian values and monotonicity, with no quadrature remainder inferred from grid agreement.

C.13. Model constants and complete four-scale synthesis

For clarity, the strict constants are model integrals, not fitted gradient coefficients. A low-dimensional evaluation of (C.19) uses $t(x) = \tanh(x/2)$, $d(x) = \log\cosh(x/2)$ and $X_i = \sqrt{\rho}U_0 + \sqrt{1-\rho}V_i$, with independent standard Gaussians. Define $\tau(u) = E_V t(X)$, $\beta(u) = E_V d(X)$, $\gamma(u) = E_V[t(X)d(X)]$, $\nu(u) = E_V t(X)^2$ at $X = \sqrt{\rho}u + \sqrt{1-\rho}V$. Label averaging and Gaussian integration by parts give

$$\begin{aligned}
d_0 &= Ed(Z), \quad s_0 = 1 - Et(Z)^2, \quad v = \tfrac{1}{4} + \operatorname{Var} d(Z),\\
\kappa_3 &= -3s_0/16 - E[(d(Z) - d_0)^3], \quad \kappa = E\tau(U_0)^2,\\
m_H &= \kappa/4,\\
\bar{c} &= -\rho s_0/8 - \{2E[\tau\gamma] + E[\tau^2\beta] - 3d_0\kappa\}/4,\\
V_H &= \{2Et(Z)^2 + E\nu(U_0)^2 - \kappa^2\}/16,\\
V_A &= 3/4 + 3(v - 1/4) + 6\{E\beta(U_0)^2 - d_0^2\}.
\end{aligned} \tag{C.56}$$

Here each outer expectation is over U_0. The third active coordinate contributes $E[\tau^2\beta] - d_0\kappa$ and is not discarded. These expressions evaluate free-coordinate moments at the equicorrelation point; they do not differentiate the three-edge path.

Their certified evaluation used the full-weight Gaussian trapezoid with spacing $1/16$, cutoff 10, strip half-width $3/4$ and 192 bits. The bounds $|t(x+iy)| \le 1$ for $|y| \le \pi/2$ and $|d(x+iy)| \le |x|/2 + \log\sec(|y|/2) + |y|/2$ give the same alias formula as (C.47) with explicit polynomial Gaussian strip and tail envelopes. Each inner conditional integral is enlarged by its own alias and tail bounds before products are formed; the outer sums then receive their own errors. The Gaussian order-statistic integrals in (C.21) are treated separately, using $|\Phi(x+i\eta)| \le 1 + \eta e^{\eta^2/2}/\sqrt{2\pi}$. The full envelopes and exact saved intervals are provided with the companion constant calculation; no ordinary floating-point rule difference is used as a strict error bound.

Constant	Approximate value
v	.271514501800558731839
κ_3	$-$.16208683226381804873
m_H	.01927127885302918387252
$\bar{c}$	$-$.0547240726452943675562
J_{100}	2.50759363644168
B_{100}	5.47243065910866
V_H	.02360133572724289470720
V_A	.84051923403528451804

$$
\begin{aligned}
b_2^{\text{nuisance}} &= B_{100} m_H \kappa_3/(3v^2) \simeq -.0772914085493864,\\
b_2^{\text{mark}} &= -B_{100}\bar{c}/v \simeq 1.102974945903173,\\
1.02568353735378 < b_2 &< 1.02568353735380.
\end{aligned}
\tag{C.57}
$$

The large positive correction is primarily the signed mark–score term. This is a mechanism in the analytic expansion, not a causal attribution of training outcomes. The table decimals are for reading; the strict coefficient interval and the final outward bounds below use the full saved dyadic inputs.

n	$D_n^{(2)}$, approximately	Strict upper bound on E_n	Outward enclosure of D_n	Sign
64	+.00443369615	99.704837	[$-$99.70041, 99.70928]	Unresolved
256	$-$.00178972824	.35027010	[$-$.35205983, .34848037]	Unresolved
1024	$-$.00189650820	.011695409	[$-$.013591917, .009798901]	Unresolved
4096	$-$.00119866512	.000846727778	[$-$.002045393, $-$.0003519373]	Strictly negative

All five budget expressions are retained below as approximate readouts; none is an actual-error lower bound.

n	E_1	E_2	E_3	E_4	E_5
64	.135264699	12.7311527	82.8304548	.445542322	3.56242253
256	.0114049517	.0790174116	.223704548	.00979174301	.0263514419
1024	.00140485780	.00341057177	.00422225078	.00071844314	.00193928492
4096	.00017560722	.00026772948	.00014413379	.00006758914	.00019166814

The valid but loose 64-scale bound is not an explanation of its old positive derivative estimate. Its second-order center also lies outside the old C.6 interval. At 256 and 1024, a negative center does not settle the actual sign. At 1024, even setting $\mathsf{E}_2 = \mathsf{E}_3 = 0$ while retaining the other budgets leaves $\mathsf{E}_1 + \mathsf{E}_4 + \mathsf{E}_5 \simeq .00406258585509 > |D_{1024}^{(2)}|$; E_5 alone already exceeds that window. Thus reducing only those two budgets cannot make this fixed recipe pass. This says nothing about a lower bound on the true error or the impossibility of another method.

At the retained 4096-scale point the full finite synthesis proves

$$
\mathsf{E}_{4096} < .000846727778, \qquad -.002045393 < D_{4096} < -.0003519373 < 0. \tag{C.58}
$$

There is also a direct certified connection to the original leading field. Let $D_n^{(0)} = -m_H J_{100}/\sqrt{nv}$. Then

$$
\begin{aligned}
D^{(0)}_{4096} &\simeq -.00144907613957, \qquad b_2/4096 \simeq .000250411019862,\\
|D_{4096} - D^{(0)}_{4096}| &\le |b_2|/4096 + \mathsf{E}_{4096}\\
&< .001097138798 < |D^{(0)}_{4096}|.
\end{aligned}
\tag{C.59}
$$

The first-order relative upper budget is about 75.7%, and the second-order remainder budget is about 70.6% of its center magnitude. This is a sign-level finite applicability witness, not a 10% approximation guarantee. The assertion is at $n = 4096$, not a threshold or a statement for all larger n; it does not establish a full-matrix finite remainder, a parameter region, or an optimizer outcome.

The old four-scale panel was visible before the correction was derived. Model constants were not regressed on its gradients, but the comparison remains informed retrospective analysis. The new deterministic enclosures are not confidence intervals, spend no alpha, and do not upgrade the old ideal-iid or native-arithmetic qualifications. The former numerical reference for J_{100} is preserved in C.6 with its own provenance.

The strictness of (C.58)–(C.59) comes from analytic envelopes and saved ball arithmetic, followed by exact endpoint composition. Model-constant certification used 1296 scalar sums and 103041 nested nodes; the later noise and weight computations used the counts in C.10 and C.12. Separate ordinary Gaussian-Hermite and Gaussian-Legendre expressions check agreement, including all 273 CF values, but are not second strict certificates. Independent rational reconstruction checks the 16 weight enclosures, 20 budget expressions and four final intervals from saved endpoints; it does not re-prove all analytic envelopes. No direct D_n integral, new random bank or optimization is used. The companion records contain the full finite sums, error envelopes and arithmetic receipts. The alternative quantile-window interface is not part of this full-line certificate: its central window was not computed, and its separate tail budget is not added here.

C.14. A local observation-risk consequence at the certified finite point

For three centered Gaussian logits, global reflection eliminates all odd spin moments. Conditional independence and expansion of the three Bernoulli factors therefore give the first identity below. The teacher is uniform on eight labels, and $\kappa(c) = E_c[\tanh(X_1/2)\tanh(X_2/2)]$. Each two-variable marginal depends only on its own correlation, so a free c_{12} change leaves $\kappa(c_{13}), \kappa(c_{23})$ unchanged:

$$
\begin{aligned}
Q_C(y) &= \tfrac{1}{8}\left(1 + \sum_{i<j} \kappa(c_{ij}) y_i y_j\right),\\
\partial_{c_{12}} \mathcal{R}(C)|_{C=C(\rho)} &= -\frac{\kappa'(\rho)}{8}\left(\frac{2}{1+3\kappa(\rho)} - \frac{2}{1-\kappa(\rho)}\right)\\
&= \frac{\kappa'(\rho)\kappa(\rho)}{(1-\kappa(\rho))(1+3\kappa(\rho))} > 0, \qquad \rho = 9/20.
\end{aligned}
\tag{C.60}
$$

Among the two all-equal labels, $y_1 y_2$ sums to 2 and the sum of the three pair products is 3. Among the other six labels these values are -2 and -1, respectively; differentiating uniform-teacher cross entropy gives (C.60), with KL differing only by a constant. Equation (5) gives $\kappa'(c) > 0$; since $\kappa(0) = 0$ and $|\tanh| < 1$, $0 < \kappa(9/20) < 1$. This is an analytic positivity argument, not a new numerical integral for κ'. Changing all three edges along ρ would multiply this derivative by 3.

For $A = e_1 e_2^\top + e_2 e_1^\top$, set $C_h = C(9/20) + hA$. Here $\|A\|_{\mathrm{op}} = 1$ and $\lambda_{\min}(C(9/20)) = 11/20$, so $|h| < 9/20$ keeps C_h inside the paper's $\delta = .1$ correlation class. Both Gaussian and fair observation marginals stay fixed. This bound ensures feasibility only. Combining differentiability, (C.58) and (C.60), there exists $\varepsilon > 0$ such that every $0 < h < \varepsilon$ is feasible and satisfies $F^*_{4096,100}(C_h) < F^*_{4096,100}(C(9/20))$ and $\mathcal{R}(C_h) > \mathcal{R}(C(9/20))$. No explicit monotonicity radius or finite risk increase is supplied. The conclusion is one-sided, local and coordinate-specific; it does not assert a stationary point or an optimizer outcome.

C.15. Complete finite-program statements

This instance tests matched-run selection, low-risk population near-minimizers, curvature at truth and fresh-run risk. Its separate comparisons and audits do not require a finite remainder bound for Theorem 1.

Fix $m = 3$, $\mu = 0$, $C_* = I$, $\delta = .1$, $K = 100$. Two programs share 65,536 labels, active particles, 143 feasible candidates and lexicographic tolerance 10^{-12}. Replacing uniform weights by weights from 256 independent nuisance factors changes one matched realization's selection from I (KL zero) to equicorrelation C_b with off-diagonals .9, spectrum (.1,.1,2.8) and KL .03110309. Selection does not use risk. Scores average 16,384 shared outer banks over empirical label frequencies; 100 is the inner particle count.

Write $F_n = F_{n,100}$ after removing candidate-independent nuisance constants. For the continuous low-risk set $\mathcal{R}(C) \le r_0 = 10^{-6}$, Price–Pinsker and matrix-square-root derivative bounds give $\widehat{F}(C) \ge \widehat{F}(I) - L_r$. Hence an output no worse than C_b obeys

$$\widehat{F}(I) - \widehat{F}(C_b) > L_r \quad \Longrightarrow \quad \mathcal{R}(\widehat{C}) > r_0.$$

Interval replay gives a saved-input empirical margin .003906703321, excluding the continuous low-risk set, not only the grid point I.

Appendix C retains the failed original population-region gate. A deterministic refinement gives population cost $c_d <$ .001447106; subtracting it from the separate-bank audit bound on $F_{256}(I) - F_{256}(C_b)$ leaves a margin above .00134 on the same 99% event (Appendix D.5). With $\eta = .00134$ and the declared sampling/input qualifications,

$$\inf_{\mathcal{R}(C)\le 10^{-6}} F_{256}(C) > F_{256}(C_b) + \eta,$$
$$F_{256}(\widehat{C}) - \inf_{C} F_{256}(C) < \eta \implies \mathcal{R}(\widehat{C}) > 10^{-6}.$$

This neither proves global optimality of C_b nor gives its point risk .03110309 to every minimizer.

Under the declared ideal iid sampling and input numerical qualifications, a separate $L = 262{,}144$-bank moment audit gives, with at least 99% audit confidence, the following bound for a **fresh run** of the mathematical nuisance program with 143 candidates, $K = 100$, $N = 65{,}536$, $B = 16{,}384$ and the same tolerance:

$$\Pr_{\text{fresh}}\{\mathcal{R}(\widehat{C}) \ge r_{\text{grid}}\} \ge .7035,$$
$$r_{\text{grid}} = \frac{0.09\,\exp(-1)}{32} > .00103466.$$

Here 70.35% bounds a probability, not an observed frequency; 99% is confidence in that bound. Every non-I grid point has a correlation of magnitude at least .3, giving r_{grid} by Price–Pinsker, while a positive empirical anchor gap excludes I. Appendix D combines Cantelli with shared-bank moment and variance bounds; the NB pairs are dependent. The population-region and fresh-run statements share an audit event but assert different conclusions. Neither covers continuous optimizers, finer grids or uniform-branch recovery. Native floating-point execution additionally needs a future-run error guarantee beyond the fresh-run bound's sampling/input qualifications.

Appendix D. Fresh-run risk for a fixed finite search

D.1. The mathematical program and its risk separation

Fix the centered three-variable Gaussian-logistic model, independent teacher $C_* = I$, unit marginal variances, $K = 100$ and $n = 256$. The candidate set $\mathcal{C}_{\mathrm{grid}}$ consists of the 143 triples in $\{-.9, -.6, -.3, 0, .3, .6, .9\}^3$ whose unit-diagonal correlation matrix satisfies $C \succeq .1I$. The truth I and the equicorrelation anchor C_b, with off-diagonals .9, both belong to this set. All square roots below are positive symmetric roots.

A fresh run draws $N = 65,536$ iid active labels from the uniform distribution on $\{-1, 1\}^3$, and independently draws $B = 16,384$ iid banks. Bank W_b contains K standard three-dimensional active Gaussian vectors and, for each particle, 256 independent standard Gaussian nuisance coordinates. The same banks are shared across every observation and candidate. Define

$$\begin{aligned} S_k &= \sum_{j=1}^{256} \log \sigma(G_{kj}), \qquad a_k(W) = \frac{e^{S_k}}{\sum_{l=1}^{K} e^{S_l}}, \\ \phi(C; y, W) &= -\log \sum_{k=1}^{K} a_k(W) \prod_{j=1}^{3} \sigma\Big(y_j (C^{1/2} Z_k)_j\Big), \\ \widehat{F}(C) &= \frac{1}{NB} \sum_{\ell=1}^{N} \sum_{b=1}^{B} \phi(C; Y_\ell, W_b), \\ d(y, w) &= \phi(I; y, w) - \phi(C_b; y, w), \quad g = \mathbb{E} d(Y, W), \quad \widehat{g} = \frac{1}{NB} \sum_{\ell, b} d(Y_\ell, W_b). \end{aligned} \tag{D.1}$$

The mathematical program returns the lexicographically first candidate within $\mathrm{tol} = 10^{-12}$ of the minimum mathematical score. It uses no observation-risk comparison or fallback. This definition distinguishes mathematical evaluation from its floating-point implementation.

Every nonidentity grid point has some $|c_{ij}| \geq .3$. For the pair-spin moment κ, Appendix C proves $\kappa(0) = 0$ and $\kappa'(c) \geq \beta = e^{-1/2}/4$. A bounded pair-spin statistic gives total variation at least $|\kappa(c_{ij})|/2$, and Pinsker in the true-to-candidate KL direction gives

$$\begin{aligned} &\mathcal{R}(C) \geq \tfrac{1}{2}\beta^2 (.3)^2 = r_{\mathrm{grid}} := \frac{0.09 \exp(-1)}{32} > .00103466 \quad (C \in \mathcal{C}_{\mathrm{grid}} \setminus \{I\}), \\ &\widehat{g} > \mathrm{tol} \implies \widehat{F}(I) > \min_{C \in \mathcal{C}_{\mathrm{grid}}} \widehat{F}(C) + \mathrm{tol} \implies \mathcal{R}(\widehat{C}) \geq r_{\mathrm{grid}}. \end{aligned} \tag{D.2}$$

The first implication uses $\min_C \widehat{F}(C) \leq \widehat{F}(C_b)$. It does not require identifying the winner among the other 142 points. The exact risk constant is used throughout; its upward-rounded approximation .00103466093 is not a certified lower threshold.

D.2. Shared-bank variance and Gaussian tails

Let $m(y) = \mathbb{E}_W d(y, W)$ and $q(w) = \mathbb{E}_Y d(Y, w)$. The crossed-average decomposition is

$$\begin{aligned} d(y, w) &= g + a(y) + b(w) + r(y, w), \qquad a(y) = m(y) - g, \quad b(w) = q(w) - g, \\ \mathbb{E}[r(Y, W) \mid Y] &= \mathbb{E}[r(Y, W) \mid W] = 0, \\ \mathrm{Var}(\widehat{g}) &= \frac{\sigma_Y^2}{N} + \frac{\sigma_W^2}{B} + \frac{\sigma_r^2}{NB}, \\ \sigma_Y^2 &= \mathbb{E} a(Y)^2, \qquad \sigma_W^2 = \mathbb{E} b(W)^2, \qquad \sigma_r^2 = \mathbb{E} r(Y, W)^2. \end{aligned} \tag{D.3}$$

Conditional centering makes the residual covariance vanish whenever either index differs. The three terms are also mutually orthogonal. This proves (D.3) without treating the NB crossed terms as independent observations.

Coordinate permutations preserve C_b and its root; global sign reflection preserves the Gaussian bank law. Consequently $m(y)$ has only two values: m_{same} for the two all-equal labels, and m_{split} for the remaining six. Hence

$$g = \tfrac{1}{4}m_{\mathrm{same}} + \tfrac{3}{4}m_{\mathrm{split}}, \qquad \Delta = m_{\mathrm{same}} - m_{\mathrm{split}}, \qquad \sigma_Y^2 = \tfrac{3}{16}\Delta^2. \tag{D.4}$$

This is a symmetry of conditional expectations. The eight realized differences in one bank need not take only two values, and the program does not symmetrize its inputs.

The function $\log\sigma$ is 1-Lipschitz, as is log-sum-exp in the maximum norm. The nuisance weights are identical at the two anchors, so they disappear from the following envelope. The Gaussian exponential-moment argument in Appendix C, using $t = 3/8$, gives

$$\begin{aligned} A = \|\sqrt{C_b} - I\|_{\mathrm{op}} = \max(1-\sqrt{.1}, \sqrt{2.8}-1), \qquad M_Z = \max_{k\le K}\|Z_k\|_2, \\ |d(y,W)| \le \sqrt{3}\,AM_Z, \qquad \mathbb{E}M_Z^2 \le b_K := \tfrac{8}{3}(\log K + 3\log 2), \\ \sigma_r^2 \le \mathbb{E}d(Y,W)^2 \le 3A^2 b_K. \end{aligned} \tag{D.5}$$

The last bound follows from orthogonality in (D.3). It controls the residual term directly, without an additional random audit.

Set $D = 3K = 300$, $T = 6$ and $E_T = \{\max_{k,j}|Z_{kj}| \le T\}$. Define a bank's class contrast $u(w)$ by the mean of its two same-sign differences minus the mean of its six split-sign differences. Then $\mathbb{E}u(W) = \Delta$. The three truncated audit variables and their ranges are

$$\begin{aligned} X = q(W)\mathbf{1}_{E_T}, \qquad D_T = u(W)\mathbf{1}_{E_T}, \qquad M = 3AT, \\ X \in [-M, M], \qquad X^2 \in [0, M^2], \qquad D_T \in [-2M, 2M]. \end{aligned} \tag{D.6}$$

Write φ and $\overline{\Phi}$ for the standard Gaussian density and upper tail. The coordinate union bound and Mills' inequality control the tail probability. To control the second moment, bound M_Z^2 by the sum of all D squared coordinates and the tail-event indicator by the sum of coordinate exceedance indicators. Diagonal terms use $\mathbb{E}[Z^2\mathbf{1}_{\{|Z|>T\}}] = 2(T\varphi(T) + \overline{\Phi}(T))$; off-diagonal terms use independence. No disjointness of exceedance events is assumed. Thus

$$\begin{aligned} \mathbb{P}(E_T^c) \le p_T := 2D\varphi(T)/T, \\ \mathbb{E}[M_Z^2\mathbf{1}_{E_T^c}] \le 2D\{T\varphi(T) + D\overline{\Phi}(T)\} \le Q_T := 2D\varphi(T)(T + D/T), \\ \tau_2 := 3A^2Q_T, \qquad \tau_1 := \sqrt{\tau_2 p_T}, \\ |g - \mathbb{E}X| \le \tau_1, \qquad |\Delta - \mathbb{E}D_T| \le 2\tau_1, \qquad \sigma_W^2 \le \mathbb{E}q(W)^2 \le \mathbb{E}X^2 + \tau_2. \end{aligned} \tag{D.7}$$

Cauchy–Schwarz proves the mean-tail bounds. Numerically, $\tau_1 \simeq 1.31901920513 \times 10^{-5}$ and $\tau_2 \simeq 2.86347137776 \times 10^{-4}$; the actual calculation encloses these constants outward.

D.3. A three-component moment audit on separate banks

This is an author computation using a separately generated bank sample, not a third-party audit. The audit uses $L = 262,144$ new banks, with all eight labels and both anchors evaluated within each bank. These are separate from the future program's N labels and B banks. Earlier independent banks were used only for development and choosing the truncation design; none enter the new confidence statistics. The formal run fixed $T = 6$, the three targets and the error allocations before generating its fresh banks. Reproducible generation used PCG64DXSM with seed sequence $[202610060501, j]$, with separate active and nuisance streams $j = 1, 2$. The probability interpretation assumes the specified iid Gaussian model; reproducibility does not establish that assumption.

For each saved binary64 scalar sequence $\widetilde{V}_1, \ldots, \widetilde{V}_L$, choose a common dyadic denominator and accumulate integers exactly:

$$\widetilde{V}_b = 2^{-e} z_b, \quad S_1 = \sum_b z_b, \quad S_2 = \sum_b z_b^2, \qquad \widetilde{\overline{V}} = \frac{S_1}{L2^e}, \quad \widetilde{s}_V^2 = \frac{LS_2 - S_1^2}{L(L-1)2^{2e}}. \tag{D.8}$$

The exponents are 56, 95 and 68 for X, X^2 and D_T, respectively. High-precision ball arithmetic then encloses the standard deviations, confidence endpoints and final probability. The displayed sample moments are descriptive rounded values, not bounds on true variances.

Audit variable	Sample mean	Unbiased sample variance
X	.00562444984697731	.04106054244035259
X^2	.04109202024289478	.011140404592328061
D_T	.23838858527527407	.05025621128454873

No bank exceeded the truncation threshold. Allocate $\alpha_j = .01/3 = 1/300$ to each component: a one-sided lower mean bound for X, a one-sided upper mean bound for X^2, and a two-sided mean interval for D_T. For range length R_j, the empirical-Bernstein inequality (Maurer & Pontil, 2009, Theorem 4), with numerical sample errors propagated through both mean and standard deviation, uses

$$\begin{aligned} \epsilon_X &= 10^{-9} + 10^{-12}, \qquad \epsilon_D = 2 \cdot 10^{-9} + 10^{-12}, \\ \epsilon_2 &= (2M + \epsilon_X)\epsilon_X + 10^{-12}, \\ B_j &= \sqrt{\frac{2\ell_j}{L}} \left(\widetilde{s}_j + \sqrt{\frac{L}{L-1}} \epsilon_j \right) + \frac{7R_j\ell_j}{3(L-1)}, \\ (R_X, R_2, R_D) &= (2M, M^2, 4M), \qquad (\ell_X, \ell_2, \ell_D) = (\log 600, \log 600, \log 1200). \end{aligned} \tag{D.9}$$

Here $|\widetilde{V}_b - V_b| \le \epsilon_j$ is the relevant numerical condition. The centered Euclidean norm proves $|s_j - \widetilde{s}_j| \le \sqrt{L/(L-1)}\epsilon_j$. Thus (D.9) uses the exact mathematical ranges, inflates the sample standard deviation, and also allows ϵ_j at the mean endpoint. Appendix E gives the complete arithmetic method and remaining input qualifications. In particular, all 2,097,152 saved bank-label differences were subsequently enclosed, with maximum arithmetic error below 3.331×10^{-14} for exact saved Z/S; the original 10^{-9} allowance was retained. That deterministic replay introduces no new confidence family and does not certify input generation or all future runs.

On the simultaneous audit event, which has probability at least .99 under those conditions, define outward endpoints by

$$\begin{aligned} g_L &= \widetilde{\overline{X}} - \epsilon_X - B_X - \tau_1, \\ U_2 &= \widetilde{\overline{X^2}} + \epsilon_2 + B_2 + \tau_2, \\ \Delta_- &= \widetilde{\overline{D_T}} - \epsilon_D - B_D - 2\tau_1, \qquad \Delta_+ = \widetilde{\overline{D_T}} + \epsilon_D + B_D + 2\tau_1, \\ U_\Delta &= \max(|\Delta_-|, |\Delta_+|), \\ g \ge g_L, \quad \sigma_W^2 &\le U_2, \quad |\Delta| \le U_\Delta, \qquad \mathrm{Var}(\widehat{g}) \le V_U := \frac{3U_\Delta^2}{16N} + \frac{U_2}{B} + \frac{3A^2 b_K}{NB}. \end{aligned} \tag{D.10}$$

The bank contribution uses a population second-moment upper bound, not an unqualified replacement of true variance by sample variance. The following values display the certified endpoints conservatively; inference uses the full outward endpoints.

Quantity	Conservative display
g_L	.00279405023657203
U_2	.05074113097360336
U_Δ	.24317070176659291

Quantity	Conservative display
V_U	$3.289456534426161 \times 10^{-6}$
p_L	.70355036

The three contributions to V_U are shown below, rounded upward. They are contributions to this variance **upper bound**, not an empirical variance decomposition or a causal attribution.

Bound contribution	Upper display
Data: $3U_\Delta^2/(16N)$	$1.691779809884757 \times 10^{-7}$
Banks: U_2/B	$3.096992857275596 \times 10^{-6}$
Residual: $3A^2b_K/(NB)$	$2.328569616208936 \times 10^{-8}$

D.4. The probability statement and its limits

Since $g_L > \mathrm{tol}$, one-sided Cantelli applied to the fresh-run crossed average, followed by (D.2), gives on the same audit event

$$\begin{aligned}\mathbb{P}_{\mathrm{fresh}}\{\mathcal{R}(\widehat{C}) \geq r_{\mathrm{grid}}\} &\geq \mathbb{P}_{\mathrm{fresh}}\{\widehat{g} > \mathrm{tol}\} \\ &\geq \frac{(g_L - \mathrm{tol})^2}{V_U + (g_L - \mathrm{tol})^2} \geq p_L > .7035.\end{aligned} \tag{D.11}$$

For completeness, Cantelli bounds $\mathbb{P}\{\widehat{g} \leq \mathrm{tol}\}$ by $V/(V + (g - \mathrm{tol})^2)$, which increases with V and decreases with $g > \mathrm{tol}$. The recorded lower endpoint is approximately .703550362875; the table and main text use downward conservative displays.

The 99% statement is audit confidence in a lower bound on fresh-run risk probability. The 70.35% value is a lower bound, not an observed failure frequency. The audit neither repeatedly ran the 143-point search nor drew fresh training labels; it established the moments needed for this sufficient-condition argument. Nonnegativity of KL also gives the deterministic consequence

$$\mathbb{E}_{\mathrm{fresh}}\mathcal{R}(\widehat{C}) \geq .7035\, r_{\mathrm{grid}} > .00072788. \tag{D.12}$$

It uses the same audit event and spends no additional error budget.

This result concerns the $n = 256$ program alone. It does not establish that at least 70.35% of paired fresh runs change from a correct $n = 0$ output to an incorrect $n = 256$ output. Nor does it assert that every bad output is C_b or has risk .0311. Finer grids may contain additional low-risk points and invalidate the single-anchor shortcut. The continuous-region result of Appendix C and the present coarse-grid probability result provide different guarantees. Neither is a guarantee for Adam or arbitrary continuous optimization.

A floating-point program requires a further condition. Suppose that, on an event of fresh-run probability at least $1 - \delta_{\mathrm{alg}}$, all candidate scores are finite, the program returns a feasible grid point according to its stated tolerance rule, and $|\widehat{g}_{\mathrm{fp}} - \widehat{g}| \leq \epsilon_{\mathrm{alg}}$. With $t_{\mathrm{alg}} = \mathrm{tol} + \epsilon_{\mathrm{alg}} < g_L$, a union bound gives

$$\mathbb{P}_{\mathrm{fresh}}\{\mathcal{R}(\widehat{C}_{\mathrm{fp}}) \geq r_{\mathrm{grid}}\} \geq \left[\frac{(g_L - t_{\mathrm{alg}})^2}{V_U + (g_L - t_{\mathrm{alg}})^2} - \delta_{\mathrm{alg}}\right]_+, \qquad [x]_+ = \max(x, 0). \tag{D.13}$$

The event on the left counts only valid grid outputs meeting the risk threshold; unsuccessful executions are not counted. For a uniform valid-output and error guarantee ($\delta_{\mathrm{alg}} = 0$), $\epsilon_{\mathrm{alg}} \leq .00022$ would retain a lower bound exceeding $.66824 > 2/3$. This is a conditional sensitivity statement: such a uniform guarantee has not been established. Saved-input arithmetic enclosures do not certify Gaussian generation, accumulated nuisance transforms, the ideal iid sampling law, or this future-program condition.

This audit is a new 99% family. Combining it with the anchor-gap, centered-curvature and known-mean families of Appendices C and F gives conservative joint coverage at least 96%; including the older finite-selection family in Appendix H gives at least 95%. These bounds concern the named families, not all historical experiments in the paper. Independence between families is unnecessary. The full saved-input arithmetic refinements and (D.12) add no statistical comparisons.

D.5. A continuous-region consequence of the same audit event

The quantity g in (D.1) is exactly $F_{256}(I) - F_{256}(C_b)$ from Appendix C. Both definitions average the same normalized-weight loss over uniform three-bit teacher labels and the same Gaussian bank law, with known zero means, unit marginal variances, $K = 100$, $n = 256$ and the equicorrelation anchor .9. A candidate-independent nuisance normalization cancels from the point difference and from population regret. The cutoff $T = 6$ truncates the audit statistic, not the population target: the mean-tail correction in (D.10) already makes g_L a lower bound for the untruncated g.

The radius and population cost in (C.4)–(C.5), $c_d = H(b_{100}; 1-d)d^2/2$, are deterministic. They require no additional random event. Applying that regional inequality on the same 99% joint event as the present moment audit, with its input, numerical and iid-model qualifications, gives

$$\begin{aligned} &\inf_{\substack{C \in \mathcal{E}^{(3)}_{\cdot 1} \\ \mathcal{R}(C) \le 10^{-6}}} \{F_{256}(C) - F_{256}(C_b)\} \ge g_L - c_d > .00134, \\ &F_{256}(C) - \inf_{C' \in \mathcal{E}^{(3)}_{\cdot 1}} F_{256}(C') < .00134 \implies \mathcal{R}(C) > 10^{-6}. \end{aligned} \tag{D.14}$$

Outward scalar arithmetic verifies the strict margin using the existing lower endpoint for g and upper endpoint for c_d. The second line follows because the population infimum is no larger than $F_{256}(C_b)$. Thus the audit event supporting the fresh-run probability statement also supports the continuous-region implication, both restated in Section C.15, with conservative regret threshold $.00134$.

This is a deterministic consequence of the same event, with no additional statistical comparison or alpha allocation. It supplies no fresh-run failure probability for a continuous optimizer. The earlier margin near $.0008952$ and the original six-of-seven gate record in Appendix C remain unchanged. The joint coverage statements for the four or five named families remain 96% and 95%, respectively; this corollary does not change their scope.

Appendix E. Arithmetic enclosures for saved audit inputs

E.1. Three distinct levels of guarantee

The finite-population results use an ideal iid Gaussian sampling model. We distinguish this probability assumption from two numerical questions. First, **saved-input arithmetic** treats each stored binary64 active Gaussian coordinate Z_{ki} and nuisance log-weight sum S_k as an exact real number, then encloses the corresponding mathematical sigmoid, weights, curvature or score. Second, **input transformation error** concerns how these saved numbers approximate intended inputs, including the generation and accumulation of 256 log-sigmoid terms into S_k. Complete arithmetic enclosures address the first level; they do not set errors in the second level to zero. The iid Gaussian law remains an explicit statistical model assumption; equivalence of a pseudorandom implementation to that law is not asserted.

Two deterministic passes now cover every saved centered $n = 256$ curvature kernel of Appendix C and every saved bank–label anchor difference in the new probability audit of Appendix D. They give numerical support for those specified mathematical quantities. They do not guarantee the arithmetic error of every future native floating-point execution. Their statistical interpretation retains the stated input and sampling qualifications.

E.2. Uniform curvature sensitivity

For one pair (i, j), put $q_{ki} = \sigma(-Z_{ki})$, $r_{ki} = q_{ki}(2q_{ki} - 1)$, and let $\pi_k \propto e^{S_k} \prod_i \sigma(Z_{ki})$ denote exactly normalized weights for the centered all-positive-label curvature. The diagonal Price kernel in (C.14) is

$$\begin{aligned} H_{ij}(\pi, q) = & -\sum_k \pi_k(1-\pi_k) r_{ki} r_{kj} \\ & + 2\sum_k \pi_k^2(1-\pi_k)(q_{ki}^2 r_{kj} + q_{kj}^2 r_{ki}) \\ & + \sum_k \pi_k^2(1-4\pi_k) q_{ki}^2 q_{kj}^2 + P(A, B), \\ A = \sum_k \pi_k q_{ki} q_{kj}, \quad & B = \sum_k \pi_k^2 q_{ki} q_{kj}, \quad P(A,B) = A^2 - 4AB + 6B^2. \end{aligned} \tag{E.1}$$

On the probability simplex and $0 \le q \le 1$, we have $0 \le B \le A \le 1$, $|r| \le 1$ and $|r'| \le 3$. The following constants bound the derivative contribution of each term, using the ℓ_1 norm for weight changes and the maximum norm over all logistic inputs:

Term in (E.1)	Weight constant	Logistic-input constant
First local sum	1	6
Second local sum	4	5
Third local sum	10	12
$P(A, B)$	14	12

For the weight column, use $|1 - 2\pi| \le 1$, $|2\pi - 3\pi^2| \le 1$ and $|2\pi - 12\pi^2| \le 10$. For the input column, the two-coordinate derivative sums of $r_i r_j$, $q_i^2 r_j + q_j^2 r_i$ and $q_i^2 q_j^2$ are at most $6, 10, 4$. Their weight sums are bounded by $1, 1/2, 3$, respectively: in particular, $2\sum \pi_k^2(1-\pi_k) \le 1/2$ and $\sum \pi_k^2 |1 - 4\pi_k| \le 3$.

The global polynomial includes the dependence of B on squared weights. Writing $a_k = q_{ki} q_{kj}$, its derivatives and the resulting bound are

$$\begin{aligned} \partial_{\pi_k} P &= a_k\{(2 - 8\pi_k)A + (-4 + 24\pi_k)B\}, \\ \partial_{q_{ki}} P &= \pi_k q_{kj}\{(2 - 4\pi_k)A + (-4 + 12\pi_k)B\}, \\ |H_{ij}(\widetilde{\pi}, \widetilde{q}) - H_{ij}(\pi, q)| &\le 29\|\widetilde{\pi} - \pi\|_1 + 35\|\widetilde{q} - q\|_\infty. \end{aligned} \tag{E.2}$$

For $A > 0$, divide each bracket by A and put $t = B/A \in [0, 1]$. The first bilinear expression has corner values

$2, -6, -2, 14$, so its magnitude is at most 14. The second has corner values $2, -2, -2, 6$; summing $\pi_k q_{kj}$ over particles gives at most 6 for each of two coordinates. The case $A = 0$ follows directly. Integrating these derivative bounds along the simplex and cube proves (E.2). Averaging the three pair kernels within one bank preserves the constants 29 and 35; it neither multiplies them by three nor creates independent observations.

For log weights ℓ and perturbation e, the softmax Jacobian yields an expectation of $|e - Ee|$. If $m = \min e_k$ and $M = \max e_k$, then $\mathrm{Var}(e) \le (M - Ee)(Ee - m) \le (M - m)^2/4$. Integrating along $\ell + te$ gives

$$\begin{aligned} \|\operatorname{softmax}(\ell + e) - \operatorname{softmax}(\ell)\|_1 &\le \int_0^1 E_{\operatorname{softmax}(\ell+te)}|e - Ee|\, dt \\ &\le \tfrac{1}{2}\operatorname{osc}(e) = \inf_c \|e - c\mathbf{1}\|_\infty. \end{aligned} \tag{E.3}$$

A common log-weight shift therefore cancels. These bounds require weights in the exact simplex; rounded stored weights need not satisfy that premise. The interval implementation encloses exact normalization itself rather than assuming that the rounded weights are exactly normalized.

E.3. Propagation through sample statistics

Let v_b be an exact target bank kernel, t_b its stored floating reference, and $|v_b - t_b| \le \epsilon_b$ for every one of L banks. With the centering projection $J = I_L - \mathbf{1}\mathbf{1}^T/L$, the sample standard deviation based on unbiased sample variance, $s(t) = \|Jt\|_2/\sqrt{L-1}$, satisfies

$$\begin{aligned} |\bar{v} - \bar{t}| \le \bar{\epsilon}, \qquad |s(v) - s(t)| &\le \frac{\|J(v-t)\|_2}{\sqrt{L-1}} \le d_s := \sqrt{\frac{\sum_b \epsilon_b^2}{L-1}}, \\ |s(v)^2 - s(t)^2| &\le 2s(t)d_s + d_s^2. \end{aligned} \tag{E.4}$$

Thus numerical error affects both the mean and the standard deviation entering the empirical Bernstein bound. Let $\bar{t}_U, s_U, \bar{\epsilon}_U, d_{s,U}$ be reliable upper enclosures for these quantities and R the exact target kernel's range length. On the corresponding empirical Bernstein event,

$$h \le \bar{t}_U + \bar{\epsilon}_U + \sqrt{\frac{2\log(4/\alpha_j)}{L}}(s_U + d_{s,U}) + \frac{7R\log(4/\alpha_j)}{3(L-1)}. \tag{E.5}$$

The lower endpoint is obtained symmetrically. For centered curvature, $L = 262144$, $R = 133/16$ and $\alpha_j = .01/3$, giving $\log 1200$. The range remains that of the exact target because its sample statistics have been enclosed; this is not an unchanged variance calculation with an allowance added only to the mean. If every bank error is at most ϵ, the extra mean-and-standard-deviation contribution is at most $\{1 + \sqrt{2\log 1200/(L-1)}\}\epsilon$, approximately 1.007355ϵ. Reliable aggregation separately covers its own rounding.

For the score audit, with its fixed truncation indicators, a per-label arithmetic error at most η gives error at most η in the eight-label mean and 2η in the difference of the two label-class means. If the exact truncated mean satisfies $|X| \le M$, its squared summary has error at most $2M\eta + \eta^2$. These bounds enter (E.4) for the scalar summaries; the original numerical allowances in Appendix D were retained after the full score pass.

E.4. Outward arithmetic for the full panels

The implementation assumes correctly rounded, separately executed IEEE binary64 addition, subtraction, multiplication and division, gradual underflow, and reliable nextafter and binary exponent scaling. Each endpoint operation is expanded outward by one representable step. Reductions use an explicit pairwise tree. Manufactured probes exercise ordinary values, subnormals and rejected inputs; they do not prove these platform assumptions. Invalid intervals, nonfinite inputs or lost positive denominators cause failure rather than sample deletion.

For exponentials, an Arb-verified binary64 enclosure for $\log 2$ is fixed once. An integer k is used only if the entire reduced interval $r = x - k\log 2$ lies in $[-1, 1]$. Interval Horner evaluation gives

$$e^x = 2^k e^r, \qquad e^r \in \left[\sum_{j=0}^{20} \frac{r^j}{j!} - \frac{3}{21!},\ \sum_{j=0}^{20} \frac{r^j}{j!} + \frac{3}{21!}\right], \quad |r| \le 1,$$
$$x \le -800 \quad \Longrightarrow \quad 0 < e^x < 2^{-1074}. \tag{E.6}$$

The Taylor remainder uses $e < 3$. The last bound follows from $\log 2 < .7$ and $800 > 1074(.7)$, allowing a rigorous subnormal enclosure. Power-of-two scaling receives an outward rounding step. From $u = e^{-|z|}$, the two sigmoid signs are evaluated as $u/(1+u)$ and $1/(1+u)$, avoiding subtraction from one. The curvature implementation directly encloses all terms of (E.1).

For positive logarithms, exact binary normalization writes $x = 2^e m$ with $1 \le m < 2$. Set $u = (m-1)/(m+1) \in [0, 1/3)$ and use

$$\log x = e \log 2 + 2 \sum_{j=0}^{19} \frac{u^{2j+1}}{2j+1} + R_{20},$$
$$0 \le R_{20} \le \frac{2u^{41}}{41(1-u^2)} \le \frac{9}{164\, 3^{41}}. \tag{E.7}$$

The remainder bounds the positive tail geometrically. Monotonicity permits endpoint evaluation for positive interval inputs. No ordinary library exp/log output is used as the exact target at these nodes. A small number of constants and scalar aggregation log/square-root operations are enclosed with Arb.

For the score audit, the equicorrelation anchor uses its exact closed-form square root, not a rounded eigendecomposition. With $T_C = \sqrt{C}$ and $S_{\max} = \max_k S_k$,

$$C_b = \tfrac{1}{10} I + \tfrac{9}{10} \mathbf{1}\mathbf{1}^T, \qquad T_{C_b} = aI + b\mathbf{1}\mathbf{1}^T,$$
$$a = \sqrt{1/10}, \qquad b = \{\sqrt{14/5} - a\}/3,$$
$$M_C(y; Z, S) = \sum_k e^{S_k - S_{\max}} \prod_{i=1}^{3} \sigma\{y_i (T_C Z_k)_i\},$$
$$d_y(Z, S) = \log M_{C_b}(y; Z, S) - \log M_I(y; Z, S). \tag{E.8}$$

The constants a, b are enclosed using 192-bit Arb. The common nuisance normalization and any common weight shift cancel exactly in the score difference. All eight labels are evaluated, with the stored score difference used only for error comparison.

E.5. Coverage, outcomes and interpretation

The arithmetic methods, manufactured checks, source versions and saved-input identities were fixed before each full pass. These were post hoc deterministic additions to already completed audits; original statistical contracts and results were retained. The underlying curvature banks used seed 202610060301 and the new probability-audit banks used seed 202610060501, with PCG64DXSM SeedSequence roles 1/2 for active/nuisance normals and blocks of 256 banks. The arithmetic passes drew no new randomness. Indexed lower/upper bounds, original references, error envelopes and content hashes permit complete local replay and link each pass to its inputs.

Arithmetic target	Complete saved coverage	Largest error envelope, rounded upward
Centered $n = 256$ curvature	262144 banks; 786432 pair kernels	4.382×10^{-13} per bank average
New probability-audit anchor differences	262144 banks; 2097152 bank–label differences	3.331×10^{-14} per difference

The unrounded recorded maxima are approximately $4.3818074169088566 \times 10^{-13}$ and $3.33066907387547 \times 10^{-14}$. For

curvature, the average error bound is at most 2.268×10^{-14} and the standard-deviation perturbation bound at most 3.630×10^{-14}. Propagating them through (E.5), with all aggregation errors enclosed, gives the outward conditional interval

$$h_{256,100} \in [-.004912711427809,\ -.003618108895746]. \tag{E.9}$$

This remains conditional on the declared input and sampling qualifications. The score pass satisfies the original 10^{-9} per-label arithmetic allowance at every saved point; it changes neither the original exact scalar statistics nor the probability lower bound in Appendix D. The two full passes took 126.438 and 339.672 seconds, respectively.

Independent Arb calculations at 32 fixed banks checked 96 $n = 256$ curvature entries at 192 bits and 256 label differences at 256 bits; all reference balls lay inside the full-pass intervals. These are auxiliary implementation checks. Complete coverage comes from applying the outward algorithm to every saved input, not from extrapolating the sampled checks. The earlier matched-weight point-gap audit retains its own 32-bank diagnostics; the 18-component asymmetric-mean derivative audit in Appendix F is also not covered by these two passes.

Both passes are deterministic computations on existing samples and spend no additional alpha. They neither prove a deterministic expectation sign nor establish a uniform error bound for the native algorithm on future data and banks.

For clarity, the input-error interface for the two-anchor score is

$$|d_y(\widetilde{Z}, \widetilde{S}) - d_y(Z, S)| \le \operatorname{osc}(\widetilde{S} - S) + 3(1 + \sqrt{2.8})\|\widetilde{Z} - Z\|_\infty. \tag{E.10}$$

Each candidate's log mixture changes by a value between the smallest and largest nuisance log-weight errors, giving the oscillation term. The matrix T_{C_b} has nonnegative entries and row sum $\sqrt{2.8}$, while $\log \sigma$ and log-sum-exp have the required unit Lipschitz bounds. This proves (E.10) for all labels and banks. It is a conditional bridge, not a numerical bound on unknown input errors. Future native-program conditions on ϵ_{alg} are stated separately in Appendix D; no such condition follows merely from complete saved-input coverage.

Appendix F. Known means and observation-equivalent scales

F.1. Matched means and smoothness

Fix finite K, three unit Gaussian marginal variances and a known mean μ shared by teacher and candidate. Only $\theta = (c_{12}, c_{13}, c_{23})$ is fitted, with $C(\theta)$ the unit-diagonal correlation matrix and teacher $C_* = I$. Write $p_i = E\sigma(\mu_i + X)$ for $X \sim N(0,1)$, $q_i(+1) = p_i$, $q_i(-1) = 1 - p_i$. For normalized nonnegative weights a_k summing to one, independent of active Gaussians and labels, with law independent of (θ, μ), define

$$\begin{aligned} Q_{I,\mu}(y) &= \prod_i q_i(y_i), \\ \Phi_y(X, \mu, a) &= -\log \sum_{k=1}^{K} a_k \prod_{i=1}^{3} \sigma\{y_i(\mu_i + X_{ki})\}, \\ F_n(\theta, \mu) &= \sum_y Q_{I,\mu}(y) E_{a,X} \Phi_y(X, \mu, a), \quad X_k \overset{\text{iid}}{\sim} N(0, C(\theta)), \\ G_n &= \nabla_\theta F_n, \qquad Q_{C,\mu}(y) = E_{X \sim N(0,C)} \prod_i \sigma\{y_i(\mu_i + X_i)\}, \\ \mathcal{R}(\theta, \mu) &= D(Q_{I,\mu} \| Q_{C(\theta),\mu}). \end{aligned} \tag{F.1}$$

Here $n = 0$ means $a_k = 1/K$; $n = 256$ uses the independent nuisance weights. Normalization removes only a parameter-independent additive term. No inverse weight moments are required: uniformly over weights,

$$0 \le \Phi_y(X, \mu, a) \le 3 \log 2 + \|\mu\|_1 + \sum_i \max_k |X_{ki}|. \tag{F.2}$$

On compact positive-definite covariance neighborhoods, Gaussian density derivatives have polynomial-times-Gaussian envelopes. Every positive-order derivative of $\log \sigma$ is bounded; derivatives of the log sum are finite combinations of these derivatives and normalized posterior weights. These bounds permit differentiation under expectation to every fixed order, including derivatives of the teacher probabilities. Thus F_n is jointly C^∞. This concerns known matched means, not means fitted jointly or estimated with error.

For $D = \operatorname{diag}(d_1, d_2, d_3)$, $d_i \in \{-1, 1\}$, let $S_D = \operatorname{diag}(d_1 d_2, d_1 d_3, d_2 d_3)$. Gaussian sign changes and label relabeling give

$$F_n(S_D\theta, D\mu) = F_n(\theta, \mu), \qquad G_n(S_D\theta, D\mu) = S_D G_n(\theta, \mu). \tag{F.3}$$

At truth, $G_{n,ij}(0, \mu)$ is odd separately in μ_i, μ_j and even in the remaining mean. Two applications of smooth division, followed by coordinate permutation symmetry, yield

$$G_{n,ij}(0, \mu) = \mu_i \mu_j b_{n,ij}(\mu) = b_n \mu_i \mu_j + O(\|\mu\|^4). \tag{F.4}$$

Each smooth $b_{n,ij}$ is even in every mean coordinate and shares $b_{n,ij}(0) = b_n$. At most one nonzero mean coordinate therefore leaves I exactly stationary for any finite mean magnitude. Its extremum type need only persist locally for small means. General matched means need not leave I stationary; Appendix G supplies a counterexample, without resolving the corresponding coefficient at $K = 100$.

F.2. Nearby stationary branches and exact risk

At $\mu = 0$, Appendix C gives $G_n(0, 0) = 0$ and $D_\theta G_n(0, 0) = h_n I_3$: at $K = 100$, $h_0 > .00410$ analytically, while $h_{256} < 0$ on the centered curvature event with its numerical qualification. Whenever $h_n \neq 0$, the implicit function theorem gives a unique smooth stationary branch in a neighborhood of truth. Hessian continuity preserves its strict local minimum or maximum type. Sign equivariance and local uniqueness imply

$$\theta_{n,ij}(\mu) = \mu_i\mu_j t_{n,ij}(\mu) = -\frac{b_n}{h_n}\mu_i\mu_j + O(\|\mu\|^4),$$
$$\theta_n(\mu) = O(\|\mu\|^2). \tag{F.5}$$

The neighborhood can lie within $C \succ .1I$. It is local, need not be uniform in K, and does not certify that a specified moderate mean belongs to it.

Set $v_i = E\sigma'(\mu_i + X) > 0$. Price's identity at I gives the probability derivative $y_iy_jv_iv_j\prod_{\ell\neq i,j} q_\ell(y_\ell)$. Under the teacher product law, $E[Y_i/q_i(Y_i)] = 0$ and $E[\{Y_i/q_i(Y_i)\}^2] = 1/[p_i(1-p_i)]$. Consequently the exact observation-risk Hessian is diagonal Fisher information:

$$\nabla^2_\theta\mathcal{R}(0,\mu) = \mathrm{diag}_{i<j}\, r_{ij}(\mu), \qquad r_{ij}(\mu) = \frac{v_i^2v_j^2}{p_i(1-p_i)p_j(1-p_j)} > 0,$$
$$\mathcal{R}(\theta,\mu) = \frac{1}{2}\sum_{i<j} r_{ij}(\mu)\theta_{ij}^2 + O(\|\theta\|^3),$$
$$\mathcal{R}(\theta_n(\mu),\mu) = \frac{r_*}{2}\left(\frac{b_n}{h_n}\right)^2\sum_{i<j}\mu_i^2\mu_j^2 + O(\|\mu\|^6) = O(\|\mu\|^4),$$
$$r_* = 16\{E\sigma'(X)\}^4. \tag{F.6}$$

The coefficient b_n has not been proved nonzero. The risk order is an upper bound, not $\Theta(\|\mu\|^4)$; risk is exactly zero on the mean axes. More generally, every feasible $C \neq I$ has strictly positive risk: the pair probability $E[\sigma(\mu_i + X_i)\sigma(\mu_j + X_j)]$ has strictly positive correlation derivative $E[\sigma'(\mu_i + X_i)\sigma'(\mu_j + X_j)]$, so a nonzero correlation changes an observed pair marginal.

F.3. Full derivative audit at an asymmetric mean

The fixed point $\mu_* = (-.2, .1, .3)$, $K = 100$, uses 262,144 fresh iid banks, with active Gaussians shared between the two weight conditions. No mean, particle count or bank count was selected from these results. Teacher probabilities are approximately $(.458758831114, .520651702363, .561706803459)$, enclosed by 128-bit Acb integration on $[-12, 12]$ with Mills tail bounds and checked against GH64/96. Each bank averages **all eight labels with their teacher probabilities**; the single all-positive-label shortcut is unavailable here.

For fixed y, put $x_{ki} = \mu_i + Z_{ki}$ and let π_k be proportional to $a_k\prod_i \sigma(y_ix_{ki})$. Define

$$q_{ki} = y_i\sigma(-y_ix_{ki}), \quad r_{ki} = \sigma(-y_ix_{ki})\{2\sigma(-y_ix_{ki}) - 1\},$$
$$A_p = \sum_k \pi_k q_{ki}q_{kj}, \quad B_p = \sum_k \pi_k^2 q_{ki}q_{kj}, \quad p = (i,j). \tag{F.7}$$

The Price operator $L_p = \sum_k \partial_{x_{ki}}\partial_{x_{kj}}$ has no extra factor $1/2$. It gives the gradient kernel below and, for distinct pairs $p = (a,b)$ and $q = (a,c)$, the off-diagonal Hessian kernel:

$$\mathsf{G}_p = -\sum_k \pi_k(1-\pi_k)q_{ki}q_{kj},$$
$$\mathsf{H}_{pq} = \sum_k q_{kb}q_{kc}\left[-\pi_k(1-\pi_k)(1-2\pi_k)r_{ka} + 3\pi_k^2(1-2\pi_k)q_{ka}^2\right] + A_pA_q - 2(B_pA_q + A_pB_q) + 6B_pB_q. \tag{F.8}$$

The diagonal kernel is (C.14), with these signed q_{ki}. Formula (F.8) follows by applying the fourth-order derivative partitions of a negative log sum, then combining equal- and distinct-particle terms. Since $|q_{ki}| \le 1$, $r_{ki} \in [-1/8, 1]$, the gradient lies in $[-1, 1]$ and the diagonal kernel in $[-33/8, 67/16]$. For each pair, all $q_{ki}q_{kj}$ have the same label sign, so $B_p = sA_p$

with $s \in [0,1]$. The last three terms in (F.8) have magnitude at most 3, since $|1-2s-2t+6st| \le 3$; the local sum has magnitude at most 4. Thus $\mathsf{H}_{pq} \in [-7,7]$. Label averaging preserves these bounds.

The two arms' three gradients and six unique Hessian entries form one 18-component family with $\alpha = .01$. For unbiased bank variance s_j^2, range length L_j and $N = 262144$, its empirical Bernstein intervals use

$$\widehat{m}_j \pm \left\{ \sqrt{\frac{2s_j^2 \log 7200}{N}} + \frac{7L_j \log 7200}{3(N-1)} + \varepsilon_j \right\}, \qquad L_j \in \{2, 133/16, 14\}. \tag{F.9}$$

Here $\varepsilon_j = 10^{-9} + M_j\delta_Q$, where $M_j \in \{1, 67/16, 7\}$ bounds the absolute kernel and $\delta_Q \le 5.56 \times 10^{-16}$ bounds teacher-probability rounding. Banks, rather than labels, coordinates or arms, are the independent units. All 18 intervals follow, with endpoints rounded outward to 12 decimal places; Hessian indices follow $(12, 13, 23)$.

Component	$n = 0$ interval	$n = 256$ interval
g_{12}	[-0.000184565302, 0.000179529158]	[-0.000253339714, 0.000229858472]
g_{13}	[-0.000195341459, 0.000168423016]	[-0.000270877759, 0.000210500179]
g_{23}	[-0.000177779284, 0.000186080091]	[-0.000235312039, 0.000246075882]
$H_{12,12}$	[0.027349272457, 0.028693509363]	[-0.004963286860, -0.003481915149]
$H_{13,13}$	[0.027054242599, 0.028398354055]	[-0.004906087758, -0.003426349185]
$H_{23,23}$	[0.027163138163, 0.028507306983]	[-0.004915420006, -0.003435231666]
$H_{12,13}$	[-0.001116578916, 0.001113107665]	[-0.001128737702, 0.001143888725]
$H_{12,23}$	[-0.001109989535, 0.001119719329]	[-0.001123088450, 0.001149345972]
$H_{13,23}$	[-0.001113738311, 0.001115930316]	[-0.001136064635, 0.001136714384]

Gershgorin bounds from the full, unrounded simultaneous intervals give

$$\lambda_{\min}(H_0) \ge .02482173336837, \qquad \lambda_{\max}(H_{256}) \le -.00114574607762. \tag{F.10}$$

All six gradient intervals contain zero; this establishes neither zero nor nonzero gradients, and does not identify I as an extremum at μ_*. On 32 prespecified banks, 576 derivative components were recomputed with 128-bit Arb; the largest discrepancy was 2.19×10^{-16}. These checks support the explicit allowance, not a global floating-point proof for every Gaussian, log-sigmoid or weight evaluation. The population conclusion retains that numerical qualification.

F.4. Risk conflict and scale continuity

Negative definiteness at truth implies $\nabla_\theta^2 F_{256} \preceq -\lambda I_3$ in some neighborhood, with $\lambda > 0$. Writing $g = G_{256}(0,\mu)$ gives

$$\begin{aligned} F_{256}(\theta,\mu) - F_{256}(0,\mu) &\le g^\top\theta - \frac{\lambda}{2}\|\theta\|^2, \\ F_{256}(\theta,\mu) + F_{256}(-\theta,\mu) - 2F_{256}(0,\mu) &\le -\lambda\|\theta\|^2 < 0 \quad (\theta \ne 0). \end{aligned} \tag{F.11}$$

Thus every sufficiently short unoriented line through I has at least one side that lowers the finite objective; both nonzero sides increase exact risk. Every direction descends only when $g = 0$. This applies at μ_* on its own event, and to sufficiently small means on the centered event. It makes no optimizer claim.

At $K = 100$, keep all 256 nuisance observations and replace their logits by τG_j, $0 \le \tau \le 1$. Each remains an independent fair bit, so

$$\begin{aligned} Q_{C,\mu,\tau}(y,u) &= Q_{C,\mu}(y)2^{-256}, \\ a_k(\tau) &= \frac{\prod_{j=1}^{256}\sigma(\tau G_{kj})}{\sum_\ell \prod_{j=1}^{256}\sigma(\tau G_{\ell j})}. \end{aligned} \tag{F.12}$$

At $\tau = 0$ the weights are uniform, while the observations remain present. Pointwise continuity and weight-independent envelopes make the correlation Hessian continuous in τ. At $\mu = 0$ it is $h(\tau)I_3$, with $h(0) > .00410$ and $h(1) < 0$ on the centered event. At least one interior zero-curvature scale follows. For sufficiently small general means, or for μ_* on its own event, endpoint Hessians are positive and negative definite, so their continuous eigenvalues must cross zero. With nonstationary truth this is a Hessian-signature change, not a stationary bifurcation. No scale scan, location, uniqueness, monotonicity or global stationary-branch continuation is established.

The matched-weight population audit, centered curvature audit and asymmetric-mean audit each have a 99% family event. Their joint coverage is conservatively at least 97%; including the older $K = 2$ finite-selection family gives at least 96%, without requiring independence. Deterministic risk, branch and scale consequences use the same relevant events and consume no additional probability budget. These probability statements preserve each audit's numerical qualification.

Appendix G. Matched means need not preserve finite-particle stationarity

G.1 A two-particle counterexample

Consider three logistic observations, unit Gaussian marginal variances, independent teacher coordinates, and known matched means $\mu_t = (t, t, 0)$. Write $\sigma(x) = (1 + e^{-x})^{-1}$ and let P_t be the teacher label distribution on $\{-1, 1\}^3$. With two independent candidate particles $X_k \sim N(\mu_t, C)$ and no nuisance coordinates, define

$$\begin{aligned} F_{0,2}(C; \mu_t) &= E_{Y \sim P_t} E_X \left[-\log \left\{ \frac{1}{2} \sum_{k=1}^{2} \prod_{i=1}^{3} \sigma(Y_i X_{ki}) \right\} \right], \\ g(t) &= \partial_{c_{12}} F_{0,2}(C; \mu_t)|_{C=I}\,. \end{aligned} \tag{G.1}$$

The teacher distribution remains fixed when differentiating the candidate correlation. The other two candidate correlations are zero.

Proposition G.1. There is a constant $\Gamma > 0$ such that

$$\lim_{t \to +\infty} e^{2t} g(t) = -e\Gamma < 0. \tag{G.2}$$

Consequently, there exists a finite T such that $g(t) < 0$ for every real $t > T$. Moreover, g is the real restriction of a nonzero entire function, is even, and satisfies $g(0) = 0$. Thus there exists $\epsilon_0 > 0$ such that $g(t) \neq 0$ whenever $0 < |t| < \epsilon_0$. Neither threshold is quantified. The proposition does not determine the first nonzero Taylor order or the sign near zero.

G.2 Four label contributions and their limits

Let $\varepsilon = e^{-t}$, $R_{ki} = e^{-Z_{ki}}$ for $i = 1, 2$, and $B_k = \sigma(Z_{k3})$, where all Z_{ki} are independent standard Gaussians. Put $a = ER_{ki} = e^{1/2}$, $p_t = E\sigma(t + Z)$, and $q_t = 1 - p_t$. Dominated convergence gives $p_t \to 1$ and $q_t/\varepsilon \to a$. For the third label, reflection of Z_{k3} leaves the law of B_k unchanged; its two probabilities sum to one, so there is no additional factor of two.

For $s = (s_1, s_2) \in \{+, -\}^2$ and $\delta_i = 1_{\{s_i = -\}}$, define

$$\begin{aligned} \pi_k^s(\varepsilon) &= \frac{B_k R_{k1}^{\delta_1} R_{k2}^{\delta_2} / [(1 + \varepsilon R_{k1})(1 + \varepsilon R_{k2})]}{\sum_{\ell=1}^{2} B_\ell R_{\ell 1}^{\delta_1} R_{\ell 2}^{\delta_2} / [(1 + \varepsilon R_{\ell 1})(1 + \varepsilon R_{\ell 2})]}, \\ d_{ki}^{+} &= \frac{\varepsilon R_{ki}}{1 + \varepsilon R_{ki}}, \qquad d_{ki}^{-} = -\frac{1}{1 + \varepsilon R_{ki}}. \end{aligned} \tag{G.3}$$

The common factor $\varepsilon^{\delta_1 + \delta_2}$ cancels in the weights. Price's identity (Price, 1958), applied to the mixed derivative of the negative log particle average, gives

$$\begin{aligned} G^s &= -\sum_{k=1}^{2} \pi_k^s (1 - \pi_k^s) d_{k1}^{s_1} d_{k2}^{s_2}, \\ g(t) &= p_t^2 EG^{++} + p_t q_t (EG^{+-} + EG^{-+}) + q_t^2 EG^{--}. \end{aligned} \tag{G.4}$$

For $0 < \varepsilon \leq 1$, valid dominating bounds are

$$\begin{aligned} &\varepsilon^{-2} |G^{++}| \leq \sum_k R_{k1} R_{k2}, \quad \varepsilon^{-1} |G^{+-}| \leq \sum_k R_{k1}, \\ &\varepsilon^{-1} |G^{-+}| \leq \sum_k R_{k2}, \qquad\qquad |G^{--}| \leq 1. \end{aligned} \tag{G.5}$$

These lognormal envelopes are integrable. Since $B_k > 0$ almost surely, the weights converge pointwise to those proportional to $B_k R_{k1}^{\delta_1} R_{k2}^{\delta_2}$; no inverse moment of B is required. For independent standard Gaussians U_1, U_2, also independent of B_1, B_2, introduce the collision rate

$$T_k(v) = B_k e^{\sqrt{v}U_k}, \qquad c(v) = E\sum_{k=1}^{2}\left(\frac{T_k(v)}{T_1(v)+T_2(v)}\right)^2, \qquad v \geq 0. \tag{G.6}$$

Multiplication by one or two independent R factors gives $v = 1$ or $v = 2$. After applying (G.5), independence of each unused R coordinate gives

$$\begin{aligned} \varepsilon^{-2}EG^{++} &\longrightarrow -a^2[1-c(0)], \\ \varepsilon^{-1}EG^{+-},\ \varepsilon^{-1}EG^{-+} &\longrightarrow a[1-c(1)], \\ EG^{--} &\longrightarrow -[1-c(2)]. \end{aligned} \tag{G.7}$$

In particular, the teacher tail probabilities in (G.4) are essential:

$$\begin{aligned} \lim_{t\to\infty} e^{2t}g(t) &= -a^2[1-c(0)] + 2a^2[1-c(1)] - a^2[1-c(2)] \\ &= e\{c(0) - 2c(1) + c(2)\}. \end{aligned} \tag{G.8}$$

G.3 A strictly signed Fourier representation

Set $L = \log B$ and $\phi_L(\omega) = Ee^{i\omega L}$. The two-particle log odds satisfy

$$\begin{aligned} D(v) &= L_1 - L_2 + \sqrt{v}(U_1 - U_2), \\ Ee^{i\omega D(v)} &= |\phi_L(\omega)|^2 e^{-v\omega^2}, \\ c(v) &= 1 - \tfrac{1}{2}E\,\mathrm{sech}^2(D(v)/2). \end{aligned} \tag{G.9}$$

The nonnegative squared modulus follows from the independent, identically distributed L_1, L_2; L itself need not be symmetric. With the convention $\widehat{f}(\omega) = \int_{\mathbb{R}} f(x)e^{-i\omega x}\,dx$, substitution $u = e^x$ for $f(x) = \mathrm{sech}^2(x/2)$ yields

$$\widehat{f}(\omega) = 4\int_0^\infty \frac{u^{-i\omega}}{(1+u)^2}\,du = 4\Gamma(1-i\omega)\Gamma(1+i\omega) = \frac{4\pi\omega}{\sinh(\pi\omega)}. \tag{G.10}$$

Here the Beta integral has parameters with real part one; the Gamma recurrence and reflection formulas apply (National Institute of Standards and Technology, 2026, Sections 5.12 and 5.5). At zero the continuous value is four. Since $\widehat{f}$ is integrable and $|\phi_L| \leq 1$, Fourier inversion and expectation can be interchanged. Hence

$$\begin{aligned} c(v) &= 1 - \int_{\mathbb{R}} \frac{\omega}{\sinh(\pi\omega)}|\phi_L(\omega)|^2 e^{-v\omega^2}\,d\omega, \\ c(0) - 2c(1) + c(2) &= -\Gamma, \\ \Gamma &= \int_{\mathbb{R}} \frac{\omega}{\sinh(\pi\omega)}|\phi_L(\omega)|^2(1-e^{-\omega^2})^2\,d\omega > 0. \end{aligned} \tag{G.11}$$

Strict positivity is analytic: $\omega/\sinh(\pi\omega) > 0$ (with value $1/\pi$ at zero), and continuity with $\phi_L(0) = 1$ makes its squared modulus positive on a neighborhood of zero. The remaining factor is positive away from zero. Equations (G.8) and (G.11) prove (G.2), without numerical integration or a finite-t certificate.

G.4 Entire continuation with an explicit envelope

For real $x \in \mathbb{R}^6$ and $y \in \{-1,1\}^3$, keep the bounded Price kernel fixed:

$$\begin{aligned} \pi_k^y(x) &= \frac{\prod_{i=1}^3 \sigma(y_i x_{ki})}{\sum_{\ell=1}^2 \prod_{i=1}^3 \sigma(y_i x_{\ell i})}, \\ H_y(x) &= -\sum_{k=1}^2 \pi_k^y(x)(1-\pi_k^y(x)) y_1 \sigma(-y_1 x_{k1}) y_2 \sigma(-y_2 x_{k2}), \\ |H_y(x)| &\le 1. \end{aligned} \tag{G.12}$$

Move the mean into the Gaussian density. For $z = a + ib$ with $|z| \le R$, the elementary bound $(x-a)^2 \ge x^2/2 - a^2$ gives

$$\left| e^{-(x-z)^2/2} \right| = e^{-(x-a)^2/2 + b^2/2} \le e^{R^2/2} e^{-x^2/4}. \tag{G.13}$$

There are four shifted coordinates and two unshifted ones. Consequently the complexified density and its kernel integral obey

$$\begin{aligned} \rho_z(x) &= (2\pi)^{-3} \exp\left[-\frac{1}{2} \sum_{k=1}^2 \{(x_{k1}-z)^2 + (x_{k2}-z)^2 + x_{k3}^2\} \right], \\ |H_y(x)\rho_z(x)| &\le (2\pi)^{-3} e^{2R^2} e^{-\|x\|^2/4}, \\ h_y(z) &= \int_{\mathbb{R}^6} H_y(x)\rho_z(x)\,dx. \end{aligned} \tag{G.14}$$

The displayed bound is integrable and uniform on every compact complex disk. Since the integrand is entire in z, dominated holomorphic integration makes each h_y entire. The same one-dimensional argument makes $p(z) = \int_{\mathbb{R}} \sigma(x)\varphi(x-z)\,dx$ entire, where φ is the standard Gaussian density continued in its argument. Each teacher weight is $w_y(z) = \frac{1}{2}\prod_{i=1}^2 p(z)^{1\{y_i=1\}}[1-p(z)]^{1\{y_i=-1\}}$. The finite sum $\sum_y w_y(z)h_y(z)$ is therefore an entire extension of g. This construction evaluates logistic functions only at real x; it does not assume that a complex log-sigmoid is entire.

Simultaneously reflecting the first two coordinates preserves c_{12} and changes t to $-t$, giving $g(-t) = g(t)$. Single-coordinate reflection at zero gives $g(0) = 0$. Equation (G.2) excludes the identically zero function. Isolated zeros and evenness now imply, for some integer $r \ge 1$ and real $A \ne 0$,

$$g(t) = At^{2r} + O(t^{2r+2}), \qquad 0 < |t| < \epsilon_0 \implies g(t) \ne 0. \tag{G.15}$$

This refutes universal stationarity under known matched means. It neither proves a nonzero quadratic coefficient b_2 nor transfers to $K = 100$. In particular, it does not resolve the six gradient intervals containing zero at the asymmetric mean used in Appendix F, establish nonstationarity for every finite K, or negate the exact symmetry result when at most one mean coordinate is nonzero.

Appendix H. Earlier finite selection and its negative control

H.1. Selection conditions and the role of a region certificate

For a neighborhood U of $\mathcal{M}$, define $\Delta_U = \inf_{C\notin U} V(C) - \min_C V(C) > 0$, and let $e = \sup_C |F(C) - F(I) - wV(C)|$, $\zeta = \operatorname{osc}_C(\widehat{F} - F)$ and $\eta = \widehat{F}(\widehat{C}) - \inf_C \widehat{F}(C)$. Comparing with any minimizer of V gives the familiar sufficient condition

$$2e + \zeta + \eta < w\Delta_U \implies \widehat{C} \in U. \tag{H.1}$$

Indeed an output outside U has empirical excess over that minimizer at least $w\Delta_U - 2e - \zeta$, contradicting the bound η. Writing this condition alone gives no finite numerical certificate. We instead bound the empirical criterion on a whole low-risk region directly, and separately audit a finite-population comparison. We do not estimate global e, ζ or optimizer regret from random restarts or from a grid minimum.

H.2. A simultaneous lower bound on the continuous low-risk set

Here fix $m = 3$, $\mu = 0$, $C_* = I$, $\delta = .1$ and $K = 2$. Let q_y be any empirical active-label frequencies, and let $Z_{bk} \in \mathbb{R}^3$ and positive nuisance weights a_{bk} be any fixed finite bank inputs. The symmetric root $S(C) = C^{1/2}$ couples candidates. Removing only a candidate-independent additive constant gives

$$\widehat{F}(C) = \frac{1}{B}\sum_{b=1}^{B}\sum_{y} q_y \phi_y(S(C)Z_b), \qquad \phi_y(X) = -\log\sum_{k=1}^{K} a_{bk}\prod_{i=1}^{m}\sigma(y_i X_{ki}), \qquad a_{bk} = \frac{e^{S^{\mathrm{nuis}}_{bk}}}{\sum_\ell e^{S^{\mathrm{nuis}}_{b\ell}}}. \tag{H.2}$$

The nuisance weights remain random likelihood factors; they are not analytically removed as in FACT. For $r_0 = 10^{-6}$, let $\mathcal{G}_{r_0} = \{C \in \mathcal{E}^{(3)}_{.1} : \mathcal{R}(C) \le r_0\}$. Price, $\log\cosh u \le u^2/2$ and Jensen imply $\kappa'(c) \ge \beta = e^{-1/2}/4$. Pinsker applied to $y_i y_j$ gives $|\kappa(c_{ij})| \le \sqrt{2r_0}$, since the true pair moment is zero. Therefore

$$C \in \mathcal{G}_{r_0} \implies \|C - I\|_F \le d := \frac{\sqrt{12r_0}}{\beta} = .02284535206697714\ldots, \qquad \lambda := 1 - d > 0. \tag{H.3}$$

For $D = C - I$, every matrix on $I + tD$ has minimum eigenvalue at least λ. Differentiating the root identity gives $SS' + S'S = D$ and $SS'' + S''S = -2(S')^2$. The Sylvester inverse bound yields

$$\|S'\|_F \le \frac{\|D\|_F}{2\sqrt{\lambda}}, \qquad \|S''\|_F \le \frac{\|D\|_F^2}{4\lambda^{3/2}}, \qquad \|\nabla_X \phi_y\|_2 \le \sqrt{m}, \quad \|\nabla^2_X \phi_y\|_{\mathrm{op}} \le m + \tfrac{1}{4}. \tag{H.4}$$

The last bound uses the weighted emission Hessians (norm at most $1/4$) and the covariance of the log-sum-exp gradients (norm at most m). If $a_b^2 = \sum_k \|Z_{bk}\|^2$ and $A_2 = B^{-1}\sum_b a_b^2$, the chain rule followed by $B^{-1}\sum_b a_b \le \sqrt{A_2}$ bounds the second directional derivative. Writing g for the three free-coordinate derivatives at I, Taylor's formula gives

$$H(A_2) = \frac{(m + 1/4)A_2}{4\lambda} + \frac{\sqrt{mA_2}}{4\lambda^{3/2}}, \qquad L_r = \frac{\|g\|_2 d}{\sqrt{2}} + \frac{H(A_2)d^2}{2}, \qquad \inf_{C\in\mathcal{G}_{r_0}} \widehat{F}(C) \ge \widehat{F}(I) - L_r. \tag{H.5}$$

The factor $1/\sqrt{2}$ converts free off-diagonal coordinates to Frobenius norm. This bound holds simultaneously for every matrix in the continuous low-risk set, regardless of whether it is on the search grid. Consequently,

$$\widehat{g} := \widehat{F}(I) - \widehat{F}(C_b) - L_r > 0, \qquad \widehat{F}(\widehat{C}) \le \widehat{F}(C_b) \quad \Longrightarrow \quad \mathcal{R}(\widehat{C}) > r_0. \tag{H.6}$$

H.3. A separate finite-population comparison

The true uniform label distribution and the Gaussian prior make $F(C)$ invariant under simultaneous sign conjugation of C. Thus its free-coordinate gradient at I is zero, for either nuisance condition. Taking expectations in (H.4), with $Ea_b^2 = Km = 6$ and $Ea_b \le \sqrt{6}$, gives the same bound with $A_2 = 6$:

$$\inf_{C\in\mathcal{G}_{r_0}} F(C)-F(C_b)\ge g-\frac{H(6)d^2}{2},\qquad g=F(I)-F(C_b),\qquad \frac{H(6)d^2}{2}=.00158844598226789635\ldots. \tag{H.7}$$

The independent audit estimates only this prespecified point difference. It averages the eight active patterns exactly using true weights $1/8$; these weights never enter the learner. If $D_b=h_b(I)-h_b(C_b)$ and $A=\|C_b^{1/2}-I\|_{\mathrm{op}}=\max(1-\sqrt{.1},\sqrt{2.8}-1)$, Lipschitz continuity gives $|D_b|\le\sqrt{3}A\max_k\|Z_{bk}\|$. Set $T=8$, $E_b=\{\max_{k,i}|Z_{bki}|\le T\}$, $D_b^t=D_b1_{E_b}$ and $M=3AT$. Gaussian Mills and Cauchy bounds yield

$$|D_b^t|\le M=16.410533615595888\ldots,\qquad |ED_b-ED_b^t|\le t_T:=\sqrt{3}A\sqrt{6\cdot 12\varphi(T)/T}<2.526\times10^{-7}. \tag{H.8}$$

For $L=2097152$ independent banks, the scaled two-sided empirical Bernstein inequality (Maurer & Pontil, 2009, Theorem 4) gives the half-width below. Here s^2 is the unbiased sample variance of D_b^t, $\alpha_j=.005$, and the union over two conditions has total error at most .01 under the iid sampling model and the stated numerical allowance:

$$b_L=\sqrt{\frac{2s^2\log(4/\alpha_j)}{L}}+\frac{7(2M)\log(4/\alpha_j)}{3(L-1)}+t_T+10^{-9},\qquad g\in[\overline{D}^t-b_L,\overline{D}^t+b_L]. \tag{H.9}$$

H.4. Fixed program, inputs and complete outcomes

The single training dataset has 65,536 iid three-bit observations from the independent teacher. Nuisance labels are integrated by Gaussian sign symmetry; the n random nuisance likelihood factors remain in every bank. Each correlation coordinate takes $\{-9,-6,-3,0,3,6,9\}/10$. All principal minors of $10(C-.1I)$ are checked with integers, leaving 143 feasible candidates. The lexicographic first candidate within 10^{-12} of the smallest double score is a proposal. An interval comparison either confirms its score is no worse than the fixed reference $(.9,.9,.9)$ or returns that reference. Risk and spectrum are computed only after outputs are locked. Neither the true correlation direction nor a risk oracle selects among candidates.

PCG64DXSM SeedSequence uses `[202610060101,role]`: roles 0/1/2 are training labels, active Gaussian banks and nuisance Gaussian factors; 3/4 are independent audit active/nuisance Gaussians. Banks are generated in blocks of 4096 in [bank, particle, coordinate] order. Both conditions share labels and active priors; $n=0$ sets nuisance weights to $1/2$, while $n=256$ uses the same active inputs with 256 independent nuisance factors. There are 65,536 training banks and 2,097,152 independent audit banks. No seed, dimension, particle count, risk threshold or family changed after execution. Shared inputs count once: 4,325,376 prior vectors, each with three active and 256 nuisance Gaussian coordinates, and 65,536 observed active patterns. No neural training or gradient-based optimizer updates are performed.

The realized bounds, in nats per observation, are:

n	Empirical point gap	Linear cost	Quadratic cost	Empirical region lower
0	.003594355617	.000015488961	.001585398366	.001993468290
256	.004741770692	.000019724039	.001585398366	.003136648287

The independent point audit gives:

n	Population point mean	99% family interval	Population region lower
0	.002478573642	[.001864904388, .003092242896]	.000276458406
256	.003266233361	[.002280561286, .004251905435]	.000692115304

Displayed values are rounded; all positivity decisions use the unrounded enclosures. Both proposals already equal C_b; no fallback or comparison-based change occurs. Both output spectra are $(.1,.1,2.8)$ and their numerical observation KL is

.031103092779, whereas the true independent reference has zero risk. The analytic certificate proves only $\mathcal{R} > 10^{-6}$, not a sharp lower bound of .0311. In particular, the $n = 0$ control also excludes the low-risk region. The study resolves finite wrong selection for these specified programs, but does not isolate nuisance addition as necessary, determine the full global minimizing set, or establish failure for typical stochastic optimizers.

H.5. Numerical verification and retained failure

Manufactured inputs compare double scores and free-coordinate derivatives to 128-bit Arb ball arithmetic; score/gradient discrepancies are below 3×10^{-17} and central finite-difference gradient error below 2.4×10^{-11}. Every saved training bank is replayed with ball arithmetic at I, C_b and the returned point, including the gradient at I. Binary64 bank inputs are treated as exact numbers for this realized-program certificate. For a positive definite rounded symmetric root S_0, a ball residual bound gives $\|S_0 - C^{1/2}\|_F \le \|S_0^2 - C\|_F/\sqrt{\delta}$; the resulting score allowance is at most $\sqrt{mA_2}$ times that bound. Positivity of S_0 is checked by Sylvester principal minors.

The population audit stores every paired bank difference and recomputes its mean and variance. Thirty-two prespecified evenly spaced banks per condition are checked with Arb. These checks qualify the explicit 10^{-9} numerical allowance in (H.9), but do not prove a global floating-point error bound or constitute interval quadrature at every population node. Thus the population coverage statement retains that numerical qualification. The deterministic empirical region certificate does not rely on this sampled population check.

After selection, observation probabilities use the exact centered three-spin representation $Q_C(y) = \{1 + \sum_{i<j} \kappa(c_{ij}) y_i y_j\}/8$. Odd spin moments vanish by symmetry. Pair Gaussian quadrature orders 32/48/64 are checked against independent three-dimensional orders 32/48. Maximum probability disagreement is below 3.21×10^{-13}, risk change below 3.23×10^{-13} and mass error below 1.12×10^{-16}. These are numerical qualifications, not certified Gaussian integration remainders; the region proof uses Price and Pinsker, not a quadrature error claim.

After completing data generation, grid proposals, population differences and the full interval selection certificate, the run failed on a NumPy Boolean in the variance-summary serialization. A separately frozen continuation corrected scalar types and completed only the unsaved diagnostics on the same banks and locked selections. Original errors, saved inputs/outputs and recovery costs remain; no data, prior particles or learning output were added. Computation used NumPy/SciPy double precision and python-flint 0.8.0 (Arb, 128-bit) for ball replay. Full raw nuisance Gaussian coordinates are regenerable from the fixed RNG; their sufficient log-sums, active banks, labels and per-bank differences are retained locally. Hash records are not a replacement for those local arrays.

Appendix I. Joint learning and local stability without a designated removable block

The main argument certifies a local minimum with every bias and loading free, then shows that a dependence change preserving candidate logit marginals lowers risk from that same point. Read the model, full derivatives and Laplace representation in I.1–I.3, followed by the root certificate in I.8 and the angular-path proof in I.10. The explicit teacher-family extension in I.11 proves the quantitative angular-persistence corollary; the qualitative continuation in I.9 concerns the original radial risk direction. Sections I.4–I.7 give the numerical discovery context and its limits, including the empirical fits and K11/K12 comparisons, which retain their numerical status.

I.1. Model, targets, and the weak-loading interface

This appendix studies a finite-dimensional model distinct from the nuisance-growth setting of Theorems 1–2. Let $m = 8$, $Z \sim N(0, I_2)$, and allow every bias and loading to vary:

$$\begin{aligned} A_y(z; b, W) &= \prod_{j=1}^{m} \sigma(b_j + W_j z)^{y_j} [1 - \sigma(b_j + W_j z)]^{1-y_j}, \qquad y \in \{0,1\}^m, \\ p_\theta(y) &= E_Z A_y(Z; \theta), \qquad F_K^q(\theta) = \sum_y q_y E\left[-\log\left\{ K^{-1} \sum_{k=1}^{K} A_y(Z_k; \theta) \right\} \right], \\ F_\infty^q(\theta) &= -\sum_y q_y \log p_\theta(y), \qquad R(\theta) = D_{\mathrm{KL}}(q_* \| p_\theta). \end{aligned} \tag{I.1}$$

The target q is fixed during candidate differentiation: either empirical pattern frequencies or the teacher distribution q_*. The teacher has $b_* = 0$ and rows $W_{*,j} = .8(\cos\phi_{*,j}, \sin\phi_{*,j})$, with $\phi_* = (-.7, -.5, -.3, -.1, .1, .3, .5, .7)$. All rows are nonzero and every off-diagonal loading inner product is positive; Gaussian logistic monotonicity also gives nonzero pairwise observation dependence. There is no designated independent observation block to delete. Bernoulli latent-variable modelling is established (Tipping, 1998); the contribution here concerns a particular objective geometry, not a new model or generic latent-collapse phenomenon (Burda et al., 2016; Lucas et al., 2019; Urban & Bauer, 2021).

Since $\mathrm{rank}(WW^\top) \le 2$, this class is not the positive-spectral-floor full-correlation class of Theorems 1–2. Neither their spectral conclusion nor their nuisance-growth rate is transferred to this model.

For the weak-loading calculation only, use spins $Y_j \in \{-1, 1\}$, zero biases, $W_j^\top = \varepsilon v_j(\phi)$, and $v_j = s_j(\cos\phi_j, \sin\phi_j)^\top$ with fixed positive s_j. Let $c = (v_i^\top v_j)_{i<j}$ and let the teacher have the same ε, s and angle ϕ_*. For fixed m, K, uniformly with two angle derivatives on a compact angle neighbourhood,

$$\begin{aligned} F_{K,\varepsilon}(\phi) - F_{K,\varepsilon}(\phi_*) &= \varepsilon^4 \{ a_K(\|c\|^2 - \|c_*\|^2) - b_K c_*^\top (c - c_*) \} + O(\varepsilon^6), \\ a_K &= \frac{(K-1)(K-2)}{32K^2}, \qquad b_K = \frac{K-1}{16K}. \end{aligned} \tag{I.2}$$

For a smooth path through the teacher, write $\dot c = dc/dt|_0$. The corresponding risk and directional derivatives are

$$\begin{aligned} R_\varepsilon(\phi) &= \frac{\varepsilon^4}{32} \|c - c_*\|^2 + O(\varepsilon^6), \\ F'_{K,\varepsilon}(0) &= -\frac{K-1}{8K^2} \varepsilon^4 c_*^\top \dot c + O(\varepsilon^6), \\ R'_\varepsilon(0) &= 0, \qquad R''_\varepsilon(0) = \frac{\varepsilon^4}{16} \|\dot c\|^2 + O(\varepsilon^6). \end{aligned} \tag{I.3}$$

Here is a coefficient derivation. Put $x_j = v_j^\top Z$ and expand $2^m A_y = 1 + \varepsilon h_1 + \varepsilon^2 h_2 + \varepsilon^3 h_3 + \cdots$:

$$
\begin{aligned}
h_1 &= \tfrac{1}{2}\sum_j y_j x_j, \qquad h_2 = \tfrac{1}{4}\sum_{i<j} y_i y_j x_i x_j,\\
h_3 &= \tfrac{1}{8}\sum_{i<j<\ell} y_i y_j y_\ell x_i x_j x_\ell - \tfrac{1}{24}\sum_j y_j x_j^3,\\
[-\log(1+\textstyle\sum_{r\ge1}\varepsilon^r \bar h_r)]_{\varepsilon^4} &= -\bar h_4 + \bar h_1\bar h_3 + \tfrac{1}{2}\bar h_2^2 - \bar h_1^2\bar h_2 + \tfrac{1}{4}\bar h_1^4.
\end{aligned}
\tag{I.4}
$$

Bars average the K particles. Uniform-spin orthogonality and Gaussian fourth moments give the following coefficients of each c_{ij}^2; terms depending only on the fixed row norms cancel in differences.

Term in (I.4)	Coefficient of c_{ij}^2
$-E\bar h_4$	0
$E\bar h_1\bar h_3$	0
$E\bar h_2^2/2$	$(K+1)/(32K)$
$-E\bar h_1^2\bar h_2$	$-(K+1)/(8K^2)$
$E\bar h_1^4/4$	$3/(16K^2)$

In particular,

$$
\begin{aligned}
E(\overline{x_i x_j})^2 &= (1+K^{-1})c_{ij}^2 + K^{-1}s_i^2 s_j^2,\\
E(\bar x_i \bar x_j \overline{x_i x_j}) &= K^{-2}\{s_i^2 s_j^2 + (K+1)c_{ij}^2\},\\
E_Z[-\bar h_2 + \bar h_1^2/2] &= \frac{\sum_j s_j^2}{8K} - \frac{K-1}{4K}\sum_{i<j} y_i y_j c_{ij}.
\end{aligned}
\tag{I.5}
$$

The teacher moment $E(Y_iY_j) = \varepsilon^2 c_{*,ij}/4 + O(\varepsilon^4)$ supplies the b_K term. Expanding the finite-alphabet KL and using the same spin orthogonality gives (I.3). At $K = 2$, the exact two-particle log-cosh identity independently yields cancellation of the two $\|c\|^2$ coefficients, confirming $a_2 = 0$ and $b_2 = 1/32$.

The remainder is not a pointwise uniform Taylor bound on unbounded Gaussian inputs. Real derivatives of log-sigmoid are bounded; derivatives of log-sum-exp are weighted cumulants with weights in $[0, 1]$. Through the required sixth loading derivative and two angle derivatives, a polynomial in $\max_k \|Z_k\|$ dominates on compact parameter sets. Gaussian moments justify differentiation under expectation. Simultaneously reversing all latent particles makes the objective even in ε. These facts give the stated remainder; no explicit sixth-order constant, growing-K uniformity, or threshold for $\varepsilon = .8$ is supplied.

For $K > 2$, completing the square in (I.2) gives, up to candidate-angle-independent terms, $\varepsilon^4 a_K\|c - \{K/(K-2)\}c_*\|^2 + O(\varepsilon^6)$: this fixed-row-length weak-loading approximation favours amplified pairwise inner products. For the stated teacher angles and $s_j = 1$, let L_* be the Laplacian with off-diagonal edge weights $\cos(\phi_{*,i} - \phi_{*,j})$. At the collinear candidate, the leading angle Hessian on $\mathbf{1}^\perp$ is $\varepsilon^4\{b_K L_* - 2a_K(8)I\}$. Its minimum changes sign formally at $K = 16/(8 - \lambda_{\min}(L_*|_{\mathbf{1}^\perp})) \simeq 4.32690638$, obtained from one fixed eigendecomposition, without a K scan. This is a real-valued boundary of the leading coefficient for fixed equal row lengths and sufficiently small common loading, not a quantitative prediction at loading .8. The qualified 11/12 separation in I.6 instead concerns the complete finite-loading target with biases and row lengths refit at each K. The weak expansion gives preference intuition but does not quantitatively predict that refitted transition; neither number identifies a universal critical particle count.

For $s_j = 1$ and $\phi(t) = (1-t)\phi_*$, every $c_{*,ij} > 0$ and $\dot c_{ij} = (\phi_{*,i} - \phi_{*,j})\sin(\phi_{*,i} - \phi_{*,j}) > 0$. Thus sufficiently weak loadings give finite-objective descent and local risk increase. $K = 1$ is exactly angle invariant. The finite-ε check retains both unsuccessful precision cases:

ε	$F_2(.05) - F_2(0)$	$R(.05)$	$F_2'(0)$	GH16/20 derivative difference	Qualification
.125	$-2.46194897\,10^{-6}$	$1.53201221\,10^{-7}$	$-5.01126089\,10^{-5}$	$9.92045\,10^{-18}$	Passed

ε	$F_2(.05) - F_2(0)$	$R(.05)$	$F_2'(0)$	GH16/20 derivative difference	Qualification
.25	$-3.78093733\,10^{-5}$	$2.20782999\,10^{-6}$	$-7.68065055\,10^{-4}$	$1.01103\,10^{-12}$	Passed
.5	$-4.60084001\,10^{-4}$	$2.51342565\,10^{-5}$	$-.00932588949$	$1.40079\,10^{-7}$	Failed
.8	$-.00187535140$	$1.00148195\,10^{-4}$	$-.0379671263$	$2.04217\,10^{-5}$	Failed

These use complete four-dimensional $K = 2$ banks at GH orders 8/12/16/20; derivative and paired-loss qualification requires successive 16/20 differences at most 10^{-7}. The separate two-dimensional teacher-PMF check passed at .8, permitting learning, without qualifying its four-dimensional angular calculation.

I.2. Full derivatives and the conditional local-type criterion

Proposition 3 (rank-one Hessian structure). Fix q and K, allowing the integrated objective, and let $W = (a, 0)$ with $a_j > 0$. In coordinates (b, u, v), write H_{ba} for the bias/first-column block, $H_\perp$ for the second-column block, $g_a = \partial_a F_K^q$ and $D_a = \mathrm{diag}(a)$. Then

$$\begin{aligned} H_{\mathrm{full}} &= \mathrm{diag}(H_{ba}, H_\perp), \\ H_\phi &= D_a H_\perp D_a - \mathrm{diag}(a \odot g_a), \\ H_\perp a &= g_a, \qquad H_\phi \mathbf{1} = 0. \end{aligned}$$

Return to bits. At $W_j = (a_j, 0)$, with $a_j > 0$, write

$$\begin{aligned} &p_k = \sigma(b + a z_k), \quad r_k = y - p_k, \quad D_k = \mathrm{diag}\{p_{kj}(1 - p_{kj})\}, \quad \pi_k = \frac{A_y(z_k)}{\sum_\ell A_y(z_\ell)}, \\ &G_k = \begin{pmatrix} r_k \\ z_k r_k \end{pmatrix}, \qquad B_k = \begin{pmatrix} D_k & z_k D_k \\ z_k D_k & z_k^2 D_k \end{pmatrix}. \end{aligned} \tag{I.6}$$

Expectations below include $y \sim q$ and iid particles. Differentiating log-sum-exp in the coordinate order (b, a) gives

$$\begin{aligned} g_{ba} &= -E \sum_k \pi_k G_k, \\ H_{ba} &= E \sum_k \pi_k B_k - E \sum_k \pi_k (1 - \pi_k) G_k G_k^\top \\ &\quad + E \sum_{k \neq \ell} \pi_k \pi_\ell G_k G_\ell^\top. \end{aligned} \tag{I.7}$$

The last term is essential: the negative Hessian is the weighted conditional negative Hessian minus the particle-score covariance. Cross-particle outer products are formed before averaging over labels.

In full coordinates (b, u, v), where $W_j = (u_j, v_j)$, reflection of the second Gaussian coordinate gives $F(b, u, v) = F(b, u, -v)$. At $(u, v) = (a, 0)$,

$$\begin{aligned} g_v = 0, \qquad H_{(b,u),v} = 0, \qquad H_{\mathrm{full}} = \mathrm{diag}(H_{ba}, H_\perp), \\ H_\perp = E\left[\sum_k \pi_k D_k - \sum_k \pi_k (1 - \pi_k) r_k r_k^\top \right]. \end{aligned} \tag{I.8}$$

Independent, centred second particle coordinates remove cross-particle terms only from the transverse block. A norm-preserving angle path also has second variation $\ddot{W}_{j1} = -a_j \dot{\phi}_j^2$, so

$$H_\phi = \mathrm{diag}(a) H_\perp \mathrm{diag}(a) - \mathrm{diag}(a \odot g_a), \qquad H_\perp a = g_a, \qquad H_\phi \mathbf{1} = 0. \tag{I.9}$$

For completeness, Gaussian integration by parts applies to

$$\begin{aligned}\partial_{z_k}(\pi_k r_{ki}) &= \pi_k(1-\pi_k)(a^\top r_k)r_{ki} - \pi_k a_i (D_k)_{ii},\\ E[z_k \pi_k r_{ki}] &= E[\partial_{z_k}(\pi_k r_{ki})].\end{aligned} \tag{I.10}$$

This proves the Ward relation in (I.9). The equivalent signed Laplacian is

$$\begin{aligned} w_{ij} &= a_i a_j E\sum_k \pi_k(1-\pi_k) r_{ki} r_{kj}, \qquad i \neq j,\\ (H_\phi)_{ij} &= -w_{ij}, \qquad (H_\phi)_{ii} = \sum_{j\neq i} w_{ij},\\ d^\top H_\phi d &= \sum_{i<j} w_{ij}(d_i - d_j)^2. \end{aligned} \tag{I.11}$$

Weights may be negative; a structural zero row sum alone does not validate quadrature. At $K = 1$, $H_\phi = 0$ exactly.

At a **genuine** stationary point $g_b = g_a = 0$, let V have orthonormal columns spanning $a^\perp$. The sufficient condition is

$$\lambda_{\min}(H_{ba}) > 0, \qquad \lambda_{\min}(V^\top H_\perp V) > 0. \tag{I.12}$$

The slice $a^\top v = 0$ is transverse to the common-rotation orbit. By (I.8), (I.12) makes its Hessian positive definite, yielding a strict local minimum modulo rotation; for $m = 8$ there are 23 positive nonrotation directions. A negative transverse direction gives local descent. Discrete reflection adds no continuous zero mode. These conditional statements do not turn a small floating-point gradient into a certified stationary point. Section I.8 supplies the additional error bounds and root inclusion for the population K8 problem; the empirical fits and the K11/K12 continuation retain their numerical status. Compact-neighbourhood Gaussian polynomial domination justifies the derivatives, and positivity of each integrated label probability gives the analogous local argument for F^q_∞.

I.3. One-particle Laplace computation and paired differences

For each label choose any $c_y > 0$, put $x = A_y/c_y$, $u = e^v$, and define

$$\Psi_y(u) = Ee^{-ux}, \qquad B_{0,y}(u) = E[xe^{-ux}], \qquad M_{G,y}(u) = E[xe^{-ux}G]. \tag{I.13}$$

The identities $S^{-1} = \int_0^\infty e^{-uS}du$ and $S^{-2} = \int_0^\infty ue^{-uS}du$, with particle exchangeability, reduce the bank expectation to a single latent integral and one auxiliary integral. For discrete normal nodes (z_ℓ, ω_ℓ), replace the inner expectations by weighted sums:

$$\begin{aligned} d_{y\ell} &= \omega_\ell x_\ell K \int_{\mathbb{R}} ue^{-ux_\ell}\Psi_y(u)^{K-1}\,dv,\\ c_{y\ell} &= \omega_\ell x_\ell K(K-1)\int_{\mathbb{R}} u^2 e^{-ux_\ell} B_{0,y}(u)\Psi_y(u)^{K-2}\,dv. \end{aligned} \tag{I.14}$$

The labelwise derivatives are then

$$\begin{aligned} g_{ba,y} &= -\sum_\ell d_{y\ell} G_\ell,\\ H_{ba,y} &= \sum_\ell d_{y\ell} B_\ell - \sum_\ell c_{y\ell} G_\ell G_\ell^\top\\ &\quad + K(K-1)\int_{\mathbb{R}} u^2 M_{G,y}(u) M_{G,y}(u)^\top \Psi_y(u)^{K-2}\,dv. \end{aligned} \tag{I.15}$$

Average these quantities with q_y afterwards. Signed-term Fubini is justified by $\sum_k \pi_k(1-\pi_k) \le 1 - 1/K$ and integrable Gaussian score moments. At $K = 1$, $g = -EG$ and $H = EB$. For integrated likelihood, with posterior $\pi_y(dz) \propto A_y(z)\varphi(z)dz$,

$$\begin{aligned} g_{ba,\infty,y} &= -E_{\pi_y} G, \\ H_{ba,\infty,y} &= E_{\pi_y} B - E_{\pi_y} GG^\top + (E_{\pi_y} G)(E_{\pi_y} G)^\top, \\ H_{\perp,\infty,y} &= E_{\pi_y} D - E_{\pi_y} rr^\top. \end{aligned} \tag{I.16}$$

Reflection and the Ward relations still apply after target averaging. At an integrated model's own target, its collinear angle Hessian vanishes; strict positivity is not imposed on this check. For a numerical normal measure ν of raw mass M, retain its weights and the finite-bank mass M^K:

$$\begin{aligned} F^{(\nu)}_{K,y} &= -M^K \log c_y + \int_{\mathbb{R}} \{\Psi^{(\nu)}_y(e^v)^K - M^K e^{-Ke^v}\}\, dv, \\ \sum_\ell d_{y\ell} &= M^K, \qquad 0 \le \sum_\ell c_{y\ell} \le M^K(1 - 1/K). \end{aligned} \tag{I.17}$$

The equalities refer to the untruncated auxiliary integral; their numerical residuals are checked. Integrated posterior weights instead have mass one. No normal or target weight is silently renormalized.

For matched noncollinear paths with an angle direction $d^{(12)}_{\mathrm{ang}} \in \mathbb{R}^8$, the common scale is fixed at the baseline:

$$W_j(t) = a_j(\cos(td^{(12)}_{\mathrm{ang},j}), \sin(td^{(12)}_{\mathrm{ang},j})), \qquad c_y = \frac{\int A_{y,0}\, d\nu}{M}, \qquad \Psi^{(\nu)}_{y,t}(u) = \int e^{-uA_{y,t}/c_y}\, d\nu. \tag{I.18}$$

The primary changes are evaluated directly as

$$\begin{aligned} \Delta F^{(\nu)}_K(t) &= \sum_y q_y \int_{\mathbb{R}} \{\Psi^{(\nu)}_{y,t}(e^v)^K - \Psi^{(\nu)}_{y,0}(e^v)^K\}\, dv, \\ \Delta R^{(\nu)}(t) &= -\sum_y q_y \{\log p^{(\nu)}_t(y) - \log p^{(\nu)}_0(y)\}. \end{aligned} \tag{I.19}$$

Common constants cancel by the Frullani identity. Only the inner integral becomes two-dimensional, not $2K$-dimensional. Exponent differences use sign-preserving `expm1`; absolute-loss subtraction is an interface check. Small-u evaluation uses the original mass with `log1p`/`expm1`; very small sums use log-sum-exp. Under the true Gaussian law, fixed row norms preserve every univariate observation marginal, and reflection makes both changes even in t. Local smoothness gives

$$F_K(t) - F_K(0) = \tfrac{1}{2}t^2(d^{(12)}_{\mathrm{ang}})^\top H_\phi d^{(12)}_{\mathrm{ang}} + O(t^4), \qquad R(t) - R(0) = \tfrac{1}{2}t^2(d^{(12)}_{\mathrm{ang}})^\top H_{\phi,\infty} d^{(12)}_{\mathrm{ang}} + O(t^4). \tag{I.20}$$

The computed finite differences below are not generated from (I.20).

I.4. Full learning and the mixed fixed-marginal control

Four exploratory streams each contain 4096 complete observations from the .8 teacher. All methods share data and initialization: Jeffreys-smoothed marginal logits and iid $N(0, .25^2)$ loading entries. Biases and loading entries are constrained to $[-2, 2]$. Projected Adam takes 1200 updates, learning rates .02 for 800 steps and .005 thereafter, $(\beta_1, \beta_2) = (.9, .999)$, and stabilizer 10^{-8}; the output is LAST. Each step shares 128 fresh two-dimensional prior vectors between $K = 8$ and 64, arranged as 16 or 2 banks. Each bank takes its own log-mean before averaging; pooling banks would optimize a different objective. GH20 supplies the integrated empirical comparator. Equal particle counts do not equalize gradient noise or total cost.

REF starts at GH-Adam LAST and uses GH40 L-BFGS-B, at most 300 iterations/450 calls, gradient tolerance 10^{-7} and function tolerance 10^{-12}. All four already met the gradient criterion: one call, zero further iterations. IND fits

independent Bernoulli margins with Jeffreys smoothing. The teacher enters only locked-output population evaluation, not training. Data/prior streams use PCG64DXSM with SeedSequence $[202610070100, r, \mathrm{role}]$; roles 0–4 specify latent draws, emission uniforms, initialization, training priors and independent evaluation.

Stream	K8 KL	K64 KL	REF KL	IND KL
0	.010714173	.000882779	.001013446	.106851719
1	.013518917	.005017770	.004563076	.108354984
2	.008767421	.003466091	.002591933	.107382320
3	.011357337	.002199853	.002356989	.107318617
Mean	.011089462	.002891623	.002631361	.107476910

K64 has lower risk than K8 in every stream, but beats REF in two streams: more accurate integration is not uniformly best in finite data. K8 unit-row Gram matrices have second eigenvalues $1.20\,10^{-5}$–$2.30\,10^{-4}$, versus 1.3403–1.5988 for REF and 1.0785–1.4567 for K64. This is near-one-factor **loading structure**, not rank-one observation covariance.

For each stream, 4096 new 64-particle banks evaluate fixed models jointly; F_8 uses the first eight particles. Entries below are K8 minus REF, with conditional MCSE in parentheses, in nats per eight-bit observation.

Stream	Empirical F_8 difference	Population F_8 difference
0	$-$.017111839 (.000420685)	$-$.019030398 (.000421690)
1	$-$.017389742 (.000861898)	$-$.017994448 (.000870445)
2	$-$.016952623 (.000519001)	$-$.019887290 (.000506188)
3	$-$.017387109 (.000383751)	$-$.018698806 (.000387336)

Both empirical integrated NLL and population KL prefer REF; F_{64} evaluation also prefers REF in all streams and both targets. These are fixed-output ordering comparisons, not global-optimality or training-population confidence claims.

The staged profile control fixes REF biases and row norms but replaces unit directions by K8 or K64 directions, producing H8/H64 without teacher alignment or refitting. A separately frozen bank family $[202610070300, r, 0]$ uses the same evaluation budget. All candidates remain feasible, and the maximum marginal discrepancy is $6.67\,10^{-16}$.

Stream	H8$-$REF KL	Empirical F_8 difference (MCSE)	Population F_8 difference (MCSE)
0	+.015649258	$-$.000384622 (.000474800)	$-$.001914842 (.000475939)
1	+.015691309	$-$.000283583 (.000921181)	+.000805674 (.000943760)
2	+.015529662	$-$.002605471 (.000575264)	$-$.000903313 (.000572604)
3	+.016110010	+.001209140 (.000399319)	$-$.000149413 (.000400167)

Thus H8 worsens risk throughout, but its F_8 ordering is mixed and some differences are comparable to MCSE. F_{64} consistently penalizes H8. H64$-$REF KL is $(-.000009563, +.000896783, +.000587691, +.000160217)$; its mean KL is .00304014, versus H8's .01837642. Neither this improvement control nor later curvature results erase the mixed H8 comparison. No banks were added to obtain a preferred sign.

I.5. Row-length controls and full stationary signatures

Canonical collinear anchors use positive row norms, not the original first loading column. Each stream combines K8/REF biases with K8/REF row norms; at fixed K8 bias, row norms also interpolate at $\lambda = .25, .5, .75$. At K8 biases and row norms, the minimum eigenvalues of the angle Hessian projected onto $\mathbf{1}^{\perp}$ are:

Stream	Empirical F_8	Population F_8	Population F_{64}	Risk
0	+.002854452	+.003744359	−.004021564	−.005337262
1	+.004466493	+.002889122	−.006012170	−.007533965
2	+.004168796	+.002341562	−.008817385	−.010706974
3	+.003203522	+.003711700	−.003994969	−.005288578

Changing only to REF row norms makes empirical F_8 minima range from $-.01958$ to $-.01101$, and population minima from $-.02006$ to $-.01332$. Changing only biases does not change endpoint types in this panel. Interpolation gives numerical sign brackets $(0, .25)$ or $(.25, .5)$, without uniqueness or monotonicity. These anchors have nonzero bias/radial gradients; angle stability alone is not a full stationary-point claim.

To check that distinction, four empirical F_8 problems E0–E3 and one population problem P optimize all (b, a), with $b \in [-2, 2]^8$, $a \in [.05, 2]^8$. They start from the corresponding K8 canonical anchor; P uses stream 0. P is an oracle diagnostic, not a fifth independent dataset.

Endpoint	$\|g_{ba}\|_\infty$	$\lambda_{\min}(H_{ba})$	$\lambda_{\min}(V^\top H_\perp V)$	KL
E0	$9.93946\,10^{-11}$	.05029121	.01062942	.01110942
E1	$4.53156\,10^{-11}$	.05191026	.01891921	.01373345
E2	$5.02400\,10^{-14}$	.05465116	.01576894	.00908874
E3	$2.96443\,10^{-12}$	.05410205	.01389994	.01229271
P	$2.93578\,10^{-12}$	.05396628	.01485717	.01043357

All pass the numerical full-local-minimum signature. The displayed values remain readouts at the original approximate endpoints. The population P objective additionally has the nearby exact local minimum certified in I.8; this certification does not extend to E0–E3. At P, the integrated-risk gradient has norm .02456458. Along its fixed negative unit (b, a) direction, all five prespecified finite steps pass their qualification:

Step	ΔF_8	ΔR
−.02	$+1.49974827\,10^{-5}$	$+5.09524051\,10^{-4}$
−.01	$+3.74871128\,10^{-6}$	$+2.50202674\,10^{-4}$
0	0	0
+.01	$+3.74715392\,10^{-6}$	$-2.41091767\,10^{-4}$
+.02	$+1.49850334\,10^{-5}$	$-4.73081829\,10^{-4}$

The positive .02 step reduces KL from .01043357299 to .00996049116 while increasing F_8. At the same five endpoints, population F_{64} and risk angle blocks have negative directions; they were not optimized to stationarity there.

I.6. Refitting each K and the common-direction comparison

Each population problem is then refit, starting at the preceding feasible LAST, for the complete coarse sequence $9, 10, 12, 16, 24, 32, 64, 128, \infty$. Only afterwards is the first adjacent qualified positive-to-negative finite pair refined, with at most six integer midpoints and interpolated starts. One refinement at 11 yields the retained pair $[11, 12]$. The table uses Cartesian transverse eigenvalues, not angle-direction curvatures.

K	$\|g_{ba}\|_\infty$	$\lambda_{\min}(H_{ba})$	Transverse minimum	Negative modes	KL
8, reused	$2.936\,10^{-12}$	.0539663	+.01485717	0	.010433573
9	$4.910\,10^{-14}$	.0530561	+.00959400	0	.009461482
10	$5.902\,10^{-11}$	.0524036	+.00530421	0	.008842072

K	$\|g_{ba}\|_\infty$	$\lambda_{\min}(H_{ba})$	Transverse minimum	Negative modes	KL
11	$2.513\,10^{-12}$	.0519187	+.00174653	0	.008426627
12	$4.369\,10^{-14}$	.0515476	−.00124838	1	.008136147
16	$3.982\,10^{-14}$	.0506760	−.00960756	1	.007557032
24	$3.617\,10^{-14}$	.0500081	−.01810993	1	.007228793
32	$3.951\,10^{-11}$	.0497429	−.02239912	1	.007134125
64	$3.200\,10^{-14}$	.0494177	−.02886195	1	.007055936
128	$9.837\,10^{-12}$	.0492813	−.03210040	1	.007039262
∞	$7.083\,10^{-12}$	.0491572	−.03533998	2	.007034194

All new points pass radial stationarity qualification. The integrated limit remains constrained to the collinear submodel; its positive risk is not a limit on the full two-factor model. The second negative mode at infinity was not pursued. Integer type separation is not a theorem about a unique branch, monotone transition, or universal critical K.

The direction $d_{\text{ang}}^{(12)} \in \mathbb{R}^8$ is the K12 endpoint's minimum angle eigenvector, mapped from a fixed Helmert basis, with the largest-magnitude component positive. Its rounded entries are $(-.5316182, -.3864170, -.2460486, -.0867412, .0867412, .2460486, .3864170, .5316182)$; the full-precision unit vector, orthogonal to $\mathbf{1}$, is used throughout. The angle eigengap is .0122054. This angle direction differs from the physical risk direction d_{risk} in I.8.5. Using the same $(d_{\text{ang}}^{(12)})^\top H_\phi d_{\text{ang}}^{(12)}$ in all four cells gives:

Parameters	F_{11}	F_{12}
θ_{11}	+.000542300179	−.000056581709
θ_{12}	+.000226560907	−.000394546595

Changing K first gives contributions $-.000598881888$ and $-.000337964886$; changing parameters first gives $-.000315739271$ and $-.000621107502$. Both total $-.000936846774$, with interaction $-.000022225614$. This is order-dependent four-cell algebra, not a unique causal decomposition. Cross cells retain the radial compensation because they are not stationary for their cross objective.

I.7. Direct noncollinear behaviour, numerical qualification, and limits

The two locked endpoints use (I.18) with the same $d_{\text{ang}}^{(12)}$. Primary amplitude is $|t| = .02; .05, .10$ are fixed sensitivities. Every negative and positive side is evaluated independently, with a shared zero baseline. Main-rule values are:

K	t	ΔF_K	ΔR	$2\Delta F_K/t^2$	$2\Delta R/t^2$
11	−.10	$+2.90960527\,10^{-6}$	$-3.62277988\,10^{-5}$	+.000581921053	−.00724555975
11	−.05	$+6.90272432\,10^{-7}$	$-9.10994254\,10^{-6}$	+.000552217946	−.00728795403
11	−.02	$+1.08777517\,10^{-7}$	$-1.45996822\,10^{-6}$	+.000543887585	−.00729984112
11	+.02	$+1.08777517\,10^{-7}$	$-1.45996822\,10^{-6}$	+.000543887585	−.00729984112
11	+.05	$+6.90272432\,10^{-7}$	$-9.10994254\,10^{-6}$	+.000552217946	−.00728795403
11	+.10	$+2.90960527\,10^{-6}$	$-3.62277988\,10^{-5}$	+.000581921053	−.00724555975
12	−.10	$-1.76055405\,10^{-6}$	$-3.91948957\,10^{-5}$	−.000352110811	−.00783897915
12	−.05	$-4.79905287\,10^{-7}$	$-9.85357108\,10^{-6}$	−.000383924230	−.00788285687
12	−.02	$-7.85692828\,10^{-8}$	$-1.57903202\,10^{-6}$	−.000392846414	−.00789516009
12	+.02	$-7.85692828\,10^{-8}$	$-1.57903202\,10^{-6}$	−.000392846414	−.00789516009
12	+.05	$-4.79905287\,10^{-7}$	$-9.85357108\,10^{-6}$	−.000383924230	−.00788285687
12	+.10	$-1.76055405\,10^{-6}$	$-3.91948957\,10^{-5}$	−.000352110811	−.00783897915

Both zero rows have differences zero by definition. At .02, $2\Delta R/t^2$ is $-.00729984112$ and $-.00789516009$. The finite-objective quadratic readouts differ from the saved curvatures by .293%/.431%; at .10 the deviations become 7.31%/10.76%.

Taylor agreement is not a qualification gate. The primary risk improvements are only about .017%/.019% of baseline KL: this is a mechanism check, not substantial recovery or a newly optimized two-factor solution.

Numerical protocols. PMFs and learned endpoints use GH40/60/80 and a GH72 tensor rule rotated by $\pi/7$: mass tolerance $2\,10^{-12}$, PMF difference $2\,10^{-7}$, risk difference $2\,10^{-6}$, and NLL/gradient tolerances $10^{-5}/10^{-4}$. Fixed-margin curvature controls compare normal-weighted Legendre384 on $[-10, 10]$ with Legendre512 on $[-11, 11]$, using log-u grids $[-24, 56]/.125$ and $[-28, 64]/.125$. Additional Hermite96 and Legendre256/$[-10, 10]$ checks both use $[-24, 48]/.25$. The matrix, spectrum, Ward and mass gates are the corresponding ones specified below. All fixed-anchor curvature qualifications passed.

Stationary solves use two fixed phases: Legendre128/$[-10, 10]$, log-u $[-24, 48]/.25$, at most 12 accepted steps/80 calls; then Legendre256/$[-10, 10]$, $[-28, 56]/.25$, at most 4/40. Gradient stops are $10^{-9}/10^{-10}$. Newton directions use $H_s = (H + H^\top)/2$ and shift $\max(0, 10^{-4} - \lambda_{\min}(H_s))I$, with infinity norm capped at .25. The initial multiplier is $\min(1, .99\alpha_{\text{box}})$, where α_{box} is the largest feasible multiplier, followed by at most 12 halvings. Armijo uses 10^{-4} and fixed loss slack $5\,10^{-13}$. Qualification uses the **unshifted** Hessian; symmetrization removes arithmetic skew without clipping eigenvalues.

Final stationary/continuation comparisons use Legendre256/$[-10, 10]$, log-u $[-28, 56]/.25$, against Legendre384/$[-11, 11]$, $[-32, 64]/.125$. Gates are 10^{-7} for loss, $2\,10^{-7}$ for gradients/matrices/Ward and raw coefficient-mass residuals, $2\,10^{-6}$ for spectra, and $2\,10^{-12}$ for normal mass. Own-objective signatures additionally require gradient infinity norm at most 10^{-8}, box margin at least 10^{-3}, positive radial minimum above 10^{-5}, and Cartesian Ward/rotation residuals at most $2\,10^{-7}$. Transverse signs use thresholds $\pm 10^{-5}$. All reported stationary, continuation and four-cell qualifications passed.

For direct paths the three independently specified rules are:

Rule	First Gaussian axis	Second axis	log-u interval / step
A	GH96	GH24	$[-28, 56]/.25$
B, main	GH144	GH32	$[-32, 64]/.125$
C, check	Normal-weighted Legendre384, $[-11, 11]$	GH40	$[-36, 72]/.125$

Paired ΔF requires B/C absolute difference $\le 10^{-10}$ and relative difference $\le 2\%$, and A/C $\le 10^{-9}$ and $\le 10\%$; corresponding risk limits are $10^{-11}/2\%$ and $10^{-10}/10\%$. Relative comparisons use the maximum of both magnitudes and 10^{-12}. Signs additionally require same B/C sign and minimum magnitude exceeding $\max(10^{-9}, 20|\Delta_B - \Delta_C|)$. Every rule checks norm error $\le 5\,10^{-14}$, marginal change $\le 2\,10^{-10}$, both effect-evenness residuals $\le 10^{-11}$, and PMF evenness $\le 2\,10^{-12}$.

Direct paths additionally require normal/model mass error $\le 2\,10^{-12}$, unchanged biases, and nonnegative PMFs. A/B-to-C absolute-loss, raw-KL and PMF limits are respectively $10^{-7}, 10^{-10}, 2\,10^{-10}$. Two-dimensional baselines are compared with the saved one-dimensional reference at loss/PMF tolerances $10^{-7}/2\,10^{-10}$. Direct-versus-subtracted interfaces have tolerance $5\,10^{-12}$. These prespecified screens are neither statistical confidence bounds nor rigorous integration bounds; the earlier 10^{-7} absolute-loss gate alone cannot resolve the primary effects.

All six baseline rules and twelve nonzero parameter points passed. Maximum B/C paired discrepancies were $5.620\,10^{-17}$ for ΔF and $1.551\,10^{-16}$ for ΔR; maximum marginal change was $3.886\,10^{-16}$. Independent execution checks reconstruct saved aggregations, solver decisions and qualification, with separately implemented selected-label integrals. For the direct paths, all fourteen C-rule PMFs were independently reconstructed; adaptive auxiliary integration checked labels 0,85,255 at positive $.02, .10$ for both anchors, with maximum label-difference discrepancy $2.776\,10^{-16}$. Negative-side integrals were not independently repeated, although every saved negative-side result entered aggregation and evenness checks. This is not full independent reoptimization or a rigorous Gaussian/auxiliary remainder certificate.

A retained training-record erratum concerns sixteen intermediate parameter fields aliased to GH-LAST. Contemporaneous path files and saved trajectories supplied the correct parameters; losses, gradients, qualification flags, updates, locked outputs and profile construction were unaffected. Original records and the failed audit attempt were preserved, with additive corrected metadata rather than rewritten science.

The four training streams remain the only independent learning replicates. Deterministic follow-ups do not create new

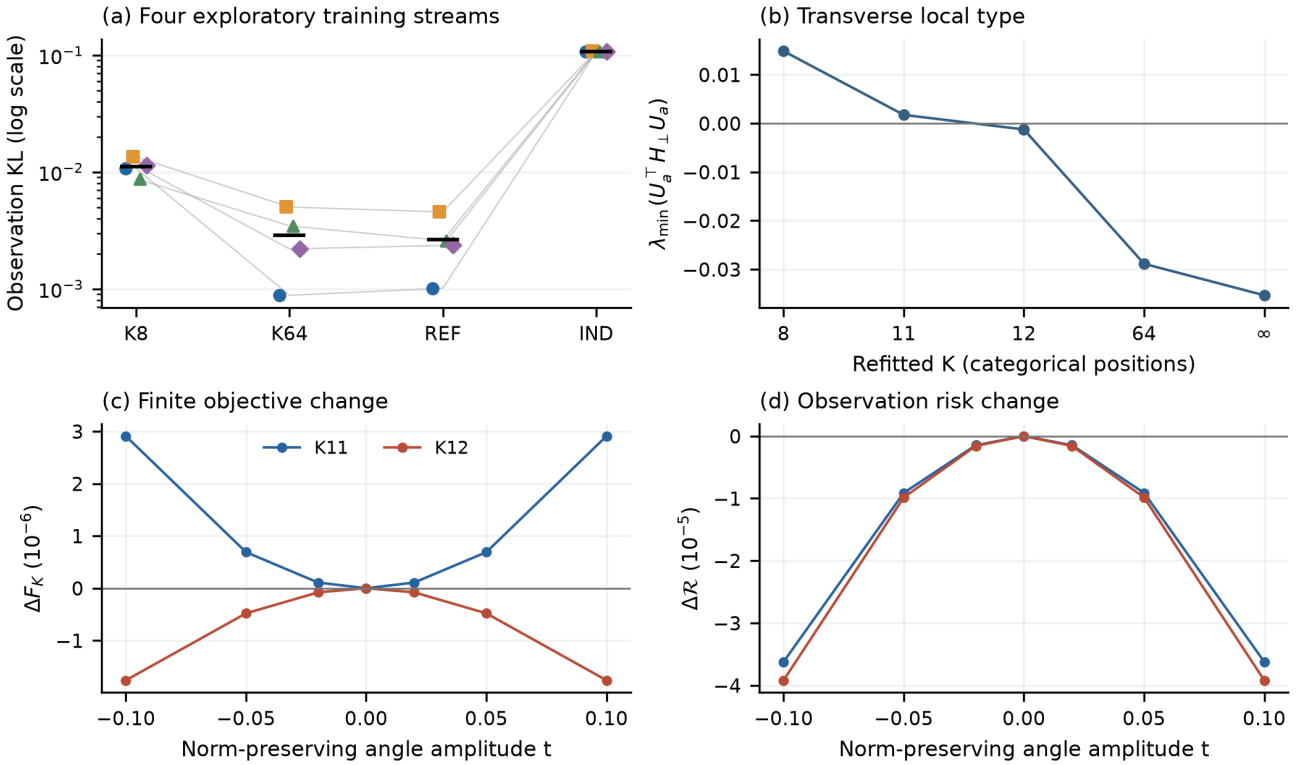


Figure S1. Learning and local geometry. (a) Observation KL for four independent streams on a logarithmic scale; black bars are means. (b) Minimum Cartesian transverse curvature, excluding common rotation, after refitting at each particle count; these are not angular eigenvalues or a continuous bifurcation. (c,d) K11/K12 objective and risk changes in nats per observation along $d_{\mathrm{ang}}^{(12)}$, with biases and row lengths fixed. These small numerical effects differ from the same-root K8 certificate (14)–(15).

statistical confidence families or inherit the older appendix coverage guarantees. The fixed finite-path panel closes this local mechanism test without additional two-dimensional optimization, second-mode searches or extra K/teacher/seed scans. Strict stationary existence remains unestablished for the four empirical fits and the K11/K12 continuation. The population K8 problem is certified below; global recovery, Adam attraction and independent-condition prediction remain unestablished.

I.8. Certified population local minimum and risk conflict

This section proves Theorem 3 for the exact population objective (I.1), at K8. The witness is a genuine stationary point of the expectation objective, located by inclusion rather than by treating the printed numerical endpoint as an exact root. The teacher, the centre and every norm in the argument refer to the same model.

Full quantitative statement of Theorem 3.

(i) Full-space local minimum and positive risk.

For the exact teacher in (13), let q_* be its observation law and $\mathcal{R}(\theta) = D(q_*\|Q_\theta)$. There exist a collinear interior parameter $\theta_* = (0, a_*, 0)$ with $a_{*,i} > 0$ and a fixed physical unit direction $d_{\mathrm{risk}} \in \mathbb{R}^{24}$ such that θ_* is a strict local minimum of $F_8^{q_*}$ in the full 24-dimensional parameter space modulo common latent rotations. For the exact dyadic center θ_0 specified in Appendix I.8,

$$\|\theta_* - \theta_0\|_2 < 1.805 \times 10^{-8},$$
$$\nabla F_8^{q_*}(\theta_*) = 0, \qquad D\mathcal{R}(\theta_*)[d_{\mathrm{risk}}] < -.0201723.$$

Every observation marginal is correct, $Q_{\theta_*}(Y_j = 1) = q_*(Y_j = 1) = 1/2$, while $0.0104335716 \le \mathcal{R}(\theta_*) \le 0.0104335744$. The fixed 10^{-7} step along d_{risk} raises F_8 and strictly lowers risk while preserving these observation

marginals; row lengths and Gaussian logit variances change.

(ii) Same-root angular conflict.

There is also a fixed $d = d^{(8)}_{\mathrm{ang}} \in \mathbf{1}^{\perp} \subset \mathbb{R}^8$, $\|d\|_2 = 1$, such that the path $\theta_{\mathrm{ang}}(t) = (0, W(t))$ satisfies

$$\begin{aligned} W_i(t) &= a_{*,i}\big(\cos(td_i), \sin(td_i)\big), \\ r(t) &= \mathcal{R}(\theta_{\mathrm{ang}}(t)), \\ r'(0) = 0, &\qquad r''(0) < -.00450377. \end{aligned}$$

There exists $\varepsilon > 0$ such that, for every $0 < |t| < \varepsilon$,

$$\begin{aligned} F_8^{q_*}(\theta_{\mathrm{ang}}(t)) &> F_8^{q_*}(\theta_*), \\ \mathcal{R}(\theta_{\mathrm{ang}}(t)) &< \mathcal{R}(\theta_*), \\ \operatorname{rank} W(t) &= 2. \end{aligned}$$

The path preserves all row lengths, logit marginals $N(0, a^2_{*,i})$ and fair observation marginals. Its unit normalization is in angle space, not in Cartesian path velocity. The preserved candidate logit variances differ from the teacher's $.8^2$.

I.8.1. Exact inputs, symmetry and the full parameter space

The teacher length and angles are exact rationals. The candidate centre is an exact dyadic interpretation of the paired symmetrization of the old population endpoint. Its positive first four row lengths are specified below, and the remaining four are their reversal. The centre is not reoptimized. Write the loading columns as u and v; vectors use physical Euclidean norms and matrices use operator norms.

$$\begin{aligned} W_{*,j} &= \tfrac{4}{5}(\cos\phi_{*,j}, \sin\phi_{*,j}), \quad b_* = 0, \quad \phi_* = (-\tfrac{7}{10}, -\tfrac{1}{2}, -\tfrac{3}{10}, -\tfrac{1}{10}, \tfrac{1}{10}, \tfrac{3}{10}, \tfrac{1}{2}, \tfrac{7}{10}), \\ \theta_0 &= (0, a_0, 0), \qquad a_{0,j} = a_{0,9-j}, \\ (a_{0,1}, \ldots, a_{0,4}) &= \left(\frac{4236614515730759}{9007199254740992}, \frac{1274451243427741}{2251799813685248}, \frac{181129089615167}{281474976710656}, \frac{775431105319425}{1125899906842624}\right). \end{aligned} \tag{I.21}$$

Let P reverse the eight observations. The teacher satisfies label-complement symmetry and reversal symmetry: the latter reverses the sign of its second latent coordinate. Simultaneously complementing labels and all latent particles proves whole-vector bias evenness, not separate evenness of each bias. Reflection of the second latent coordinate is valid for every target. Consequently, with four isometric coordinates,

$$\begin{aligned} q_*(\mathbf{1} - y) = q_*(y) = q_*(Py), &\qquad F_8^{q_*}(-b, W) = F_8^{q_*}(b, W), \\ F_8^{q}(b, u, -v) = F_8^{q}(b, u, v), &\qquad U_+ = \frac{1}{\sqrt{2}}(e_1 + e_8, e_2 + e_7, e_3 + e_6, e_4 + e_5), \\ a = U_+\xi, &\qquad f(\xi) = F_8^{q_*}(0, U_+\xi, 0), \qquad \nabla_b F_8^{q_*} = \nabla_v F_8^{q_*} = 0, \quad \nabla_a F_8^{q_*} = U_+ \nabla f. \end{aligned} \tag{I.22}$$

The last line holds on the paired zero-bias, collinear subspace. It gives equality of the four-coordinate and full 24-coordinate gradient norms. It does not reduce the local-type test to four directions: the test retains eight biases, four symmetric and four antisymmetric row-length directions, and seven nonrotation transverse directions. All Hessians below are differentiated in the unconstrained full model before evaluation at the paired centre.

For computation only, a label has four paired counts. Gaussian particle reflection identifies each count with its complement, leaving 41 lexicographically chosen representatives. Original teacher probabilities are summed into these classes. The teacher is not assumed exchangeable within individual pairs.

$$
\begin{aligned}
c_j = y_j + y_{9-j} \in \{0,1,2\}, \qquad A_c(z) &= \prod_{j=1}^{4} \sigma(a_{0,j}z)^{c_j}[1-\sigma(a_{0,j}z)]^{2-c_j}, \\
G_{\xi,c}(z) = \frac{z}{\sqrt{2}}\{c - 2\sigma(a_{0,1:4}z)\}, \qquad A_{2-c}(-z) &= A_c(z), \quad G_{\xi,2-c}(-z) = G_{\xi,c}(z), \\
\mu_c = -E\sum_{k=1}^{8} \pi_k G_{\xi,c}(Z_k) = \mu_{2-c}, \qquad \nabla f &= \sum_{r\in\mathcal{C}} w_r \mu_r, \\
\mathcal{C} = \{\min(c, 2-c) : c \in \{0,1,2\}^4\}, \quad |\mathcal{C}| = 41, \quad w_r &= \sum_{y:\min(c(y),2-c(y))=r} q_*(y).
\end{aligned}
\tag{I.23}
$$

The equality of class gradients uses the sign-reflected Gaussian bank, not an unchanged fixed bank. For Hessians the full label residual must be retained.

I.8.2. The Hessian density that is actually integrated

Set the likelihood scale to one and use the physical Laplace variable. With the full 16-coordinate score and conditional negative Hessian from (I.6), define the following inner moments; all expectations are one-dimensional standard Gaussian expectations at the collinear point. Here $B = B_y(z)$ is the single-particle conditional negative Hessian, whereas $\mathcal{B} = \mathcal{B}_y(v)$ is its t-weighted Gaussian moment; the label and auxiliary-coordinate dependence is suppressed in (I.24).

$$
\begin{aligned}
u = e^v, \quad t = uA_y(z)e^{-uA_y(z)}, \quad \Psi = Ee^{-uA_y(Z)}, \quad T_0 = Et, \\
\mathcal{R} = E[tG], \quad \mathcal{Q} = E[tGG^\top], \quad \mathcal{B} = E[tB], \quad \mathcal{Q}_p = E[trr^\top], \quad \mathcal{B}_p = E[t\,\mathrm{diag}\,\sigma'(a_0 Z)], \\
I_{ba,y}(v) = K\Psi^{K-1}\mathcal{B} - K(K-1)\Psi^{K-2}(T_0\mathcal{Q} - \mathcal{R}\mathcal{R}^\top), \\
I_{p,y}(v) = K\Psi^{K-1}\mathcal{B}_p - K(K-1)\Psi^{K-2}T_0\mathcal{Q}_p, \qquad K = 8.
\end{aligned}
\tag{I.24}
$$

Integrating these densities over the real auxiliary axis and averaging over labels gives the complete radial and transverse blocks. The first term is the conditional negative Hessian. The second term contains the same-particle contribution minus the distinct-particle score outer product. In the transverse block the latter vanishes because different particles have independent, centred second coordinates. The identities for the reciprocal and squared reciprocal in I.3 supply the displayed powers and factors; there is no additional scaling by K or u.

One must distinguish this same/distinct-particle density from direct parameter differentiation of the gradient density. For a real bank put the sum and likelihood weights below; the resulting covariance is positive semidefinite.

$$
\begin{aligned}
S = \sum_k A_k, \quad w_k = A_k/S, \quad s = e^v S, \\
\bar{B} = \sum_k w_k B_k, \quad \bar{G} = \sum_k w_k G_k, \quad \bar{Q} = \sum_k w_k G_k G_k^\top, \\
I_{ba}(v) = E[e^{-s}\{s\bar{B} - s^2(\bar{Q} - \bar{G}\bar{G}^\top)\}], \\
I_{\mathrm{dir}}(v) = E[e^{-s}\{s(\bar{B} - \bar{Q}) + s^2\bar{G}\bar{G}^\top\}], \\
I_{ba}(v) - I_{\mathrm{dir}}(v) = E[\partial_v(se^{-s})\bar{Q}], \\
\int_{\mathbb{R}} (I_{ba} - I_{\mathrm{dir}})\,dv = 0.
\end{aligned}
\tag{I.25}
$$

The boundary term is zero for each positive bank sum, with Gaussian score moments justifying exchange of expectation and integration. The densities have the same full real integral but different finite-grid values. The error envelopes below apply to the first density, which is the one used for the certificate.

For a complement-side label use its effective label complement, while the self-complement count is included once. This preserves the three diagonal blocks; the bias–radial mixed block changes sign and is zero only after using exact teacher symmetry. For each representative form weighted label moments and inner probability moments:

$$
\begin{aligned}
&W = \sum q_y, \quad M_i = \sum q_y \widetilde{y}_i, \quad S_{ij} = \sum q_y \widetilde{y}_i \widetilde{y}_j,\\
&T_l = E[tz^l], \quad P_{l,i} = E[tz^l p_i(z)], \quad PP_{l,ij} = E[tz^l p_i(z) p_j(z)], \quad p_i(z) = \sigma(a_{0,i} z),\\
&D_l = W \operatorname{diag}(P_l - \operatorname{diag} PP_l), \quad l = 0, 2,\\
&J_{bb} = W(T_0 PP_0 - P_0 P_0^\top),\\
&J_{aa} = (T_0 T_2 - T_1^2) S - T_0(M P_2^\top + P_2 M^\top) + T_1(M P_1^\top + P_1 M^\top) + W(T_0 PP_2 - P_1 P_1^\top),\\
&J_p = T_0(T_0 S - M P_0^\top - P_0 M^\top + W PP_0),\\
&I_{bb} = K\Psi^{K-1} D_0 - K(K-1)\Psi^{K-2} J_{bb}, \quad I_{aa} = K\Psi^{K-1} D_2 - K(K-1)\Psi^{K-2} J_{aa},\\
&I_p = K\Psi^{K-1} D_0 - K(K-1)\Psi^{K-2} J_p.
\end{aligned}
\tag{I.26}
$$

The label moments cancel algebraically in the first covariance difference, but remain in the other two. They are eight-coordinate vectors and matrices; no division by class mass is used. Four distinct probabilities reduce the inner scalar moment count to 36 including the Laplace transform, without reducing any Hessian dimension. This retains the information needed for broken-pair and antisymmetric row-length directions.

I.8.3. Interval quadrature, discrete tails and composition

The computation uses 192-bit midpoint-radius arithmetic (Johansson, 2017). Finite sums are enclosed by their exact dyadic endpoints. They are then enlarged by analytic infinite-grid and omitted-grid errors. Agreement between floating-point rules is not used as an error bound. For an analytic strip with vanishing contour edges and integrable boundary envelopes, Fourier contour shift followed by Poisson periodization gives (Trefethen & Weideman, 2014)

$$
\left\| \int_{\mathbb{R}} f(x)\, dx - h \sum_{k \in \mathbb{Z}} f(kh) \right\| \le \frac{M_+ + M_-}{e^{2\pi d/h} - 1}, \qquad M_\pm \ge \int_{\mathbb{R}} \|f(x \pm id)\|\, dx. \tag{I.27}
$$

The same argument holds for vector Euclidean and matrix operator norms. In this model all real-axis derivatives have Gaussian-polynomial or auxiliary rapid-decay envelopes, so the real functions are Schwartz; absolute convergence of the Fourier series also verifies the periodization identity. Analyticity alone would not suffice.

For the teacher the grid is the full-weight two-dimensional lattice on the stated rectangle. In its polystrip each logit avoids sigmoid poles, and the sum of absolute likelihoods over all labels factorizes. This bounds the complete probability vector without multiplying a per-label bound by 256.

Specifically, $\sum_y |A_y(x+it)| \le \prod_j \sec(B_j/2)$, and $|\varphi(x_i + it_i)| = e^{t_i^2/2}\varphi(x_i)$. Fourier shifts in the two coordinates give the product of their geometric alias sums; deleting the zero frequency gives the factor in the next formula.

$$
\begin{aligned}
&\theta_{\mathrm{T}} = (b_*, W_*), \quad h_t = \tfrac{1}{8}, \quad L_t = 10, \quad d_t = 1,\\
&q_y^f = h_t^2 \sum_{|k_1|,|k_2| \le 80} \varphi(h_t k_1)\varphi(h_t k_2) A_y(h_t k; \theta_{\mathrm{T}}),\\
&B_j = d_t(|W_{*,j1}| + |W_{*,j2}|) < \pi, \quad M_t = e^{d_t^2} \prod_j \sec(B_j/2), \quad \rho_t = e^{-2\pi d_t/h_t},\\
&\|q_* - q^f\|_1 \le \delta_t := M_t \frac{4\rho_t}{(1-\rho_t)^2} + 2(1 + h_t\varphi(0)) \frac{2\varphi(L_t)}{L_t} < 4.894\, 10^{-21}.
\end{aligned}
\tag{I.28}
$$

The second term is an omitted-lattice Gaussian mass bound: the infinite one-axis mass is at most one plus the central grid weight, and each omitted right tail is bounded by its adjacent continuous tail. All retained edges and corners have full weights. Finite-sum arithmetic radii are retained separately in the probability balls.

The candidate inner and outer grids, including full endpoint weights, are fixed as follows. Set the total loading sum, sector angle and secant constants below. Each sigmoid factor has argument between zero and its logit imaginary part, so the likelihood product has a continuous argument bounded by the sector angle. Hence its real part is positive in the inner strip, uniformly for every real auxiliary coordinate.

$$
\begin{aligned}
&h_z = \tfrac{1}{40},\ L_z = 11,\ d_z = \tfrac{1}{8}, \qquad h_v = \tfrac{1}{8},\ v_- = -32,\ v_+ = 64,\ d_v = \tfrac{1}{2},\ W_v = 96.125,\\
&C_a = \sum_i |a_{0,i}|, \quad \beta = C_a d_z < \pi/2, \quad c_i = \sec(|a_{0,i}| d_z/2),\ C_R^2 = \sum_i c_i^2,\ c_*^2 = \max_i c_i^2,\\
&\operatorname{Re} A_y(x+it) \ge |A_y(x+it)| \cos\beta > 0, \quad |e^{-uA_y}| \le 1, \quad |uA_y e^{-uA_y}| \le (e\cos\beta)^{-1},\\
&F_z = \frac{e^{d_z^2/2}}{e\cos\beta}, \quad J_z = 2 + d_z^2, \quad \rho_z = e^{-2\pi d_z/h_z}, \qquad \delta_X = \frac{2M_X\rho_z}{1-\rho_z} + T_X.
\end{aligned}
\tag{I.29}
$$

There are 881 inner nodes and 769 outer nodes. On the strip, residual norms and conditional-Hessian norms are bounded by the secants; Gaussian boundary integration gives the following envelopes. The middle column is a scalar, vector or operator boundary-integral bound as appropriate; the last column is an omitted-inner-grid bound.

The finite-sum implementation can enclose e^{-x} and xe^{-x} by $[0, e^{-128}]$ and $[0, 128e^{-128}]$ when the interval input proves $x = uA \ge 128$. These are conservative saturation intervals, not substituted zero values; their radii remain in the final finite-sum boxes.

Moment	M_X	T_X
Ψ	$e^{d_z^2/2}$	p_L
T_0	F_z	p_L/e
$\mathcal{R}$	$F_z\sqrt{C_R^2 J_z}$	$2\sqrt{2m}\varphi(L_z)/e$
$\mathcal{Q}$	$F_z C_R^2 J_z$	$m j_L/e$
$\mathcal{B}$	$F_z c_*^2 J_z/4$	$j_L/(4e)$
$\mathcal{Q}_p$	$F_z C_R^2$	$m p_L/e$
$\mathcal{B}_p$	$F_z c_*^2/4$	$p_L/(4e)$

The tail symbols are $p_L = 2\varphi(L_z)/L_z$ and $j_L = 2\varphi(L_z)(L_z + 2/L_z)$, with $m = 8$. They follow from monotone right-rectangle comparison beyond the retained endpoint, including the first and second Gaussian moments. Ordinary transposes, not conjugate transposes, occur in the analytic density; complex rank-one matrices still satisfy the operator bound by the squared vector norm.

Uniform inner errors propagate through the nested powers and matrix products. For this estimate only, let lower-case letters denote upper bounds on the true moment norms, and use the following product-error operator.

$$
\begin{aligned}
P &= 1 + \delta_\Psi, \quad \mathcal{E}_n(b, \delta b) = P^n \delta b + n P^{n-1} \delta_\Psi b,\\
C &= T_0\mathcal{Q} - \mathcal{R}\mathcal{R}^\top, \quad \|C\| \le tq + r^2, \quad \delta_C \le \delta_{T_0} q + (t + \delta_{T_0})\delta_\mathcal{Q} + 2r\delta_\mathcal{R} + \delta_\mathcal{R}^2,\\
C_p &= T_0\mathcal{Q}_p, \quad \|C_p\| \le tq_p, \quad \delta_{C_p} \le \delta_{T_0} q_p + (t + \delta_{T_0})\delta_{\mathcal{Q}_p},\\
E_{\mathrm{in},ba} &= W_v\{K\mathcal{E}_{K-1}(b, \delta_\mathcal{B}) + K(K-1)\mathcal{E}_{K-2}(tq + r^2, \delta_C)\},\\
E_{\mathrm{in},p} &= W_v\{K\mathcal{E}_{K-1}(b_p, \delta_{\mathcal{B}_p}) + K(K-1)\mathcal{E}_{K-2}(tq_p, \delta_{C_p})\}.
\end{aligned}
\tag{I.30}
$$

The factor $W_v = 769 h_v = 96.125$ is the total weight of the finite outer sum, including full endpoint weights. The terms $E_{\mathrm{in},ba}$ and $E_{\mathrm{in},p}$ compare exact and approximated inner moments within this finite sum; outer alias and omitted-grid tails are bounded separately.

The valid real-axis bounds are $t = 1/e$, $r = \sqrt{2m}/e$, $q = 2m/e$, $b = 1/(2e)$, $q_p = m/e$, and $b_p = 1/(4e)$. Telescoping powers and the matrix triangle inequality prove (I.30); no factor close to one is silently removed. The four-gradient computation uses the same argument for $D = E[tG_\xi]$, with $\|D\| \le E_G/e$ and the following vector envelope.

$$
\begin{aligned}
E_G &= \sqrt{m}\sqrt{2/\pi}, \quad M_D = F_z\sqrt{C_R^2(1 + d_z^2)}, \quad \delta_D = \frac{2M_D\rho_z}{1-\rho_z} + 2\sqrt{m}\varphi(L_z)/e,\\
E_{\mathrm{in},g} &= KW_v\{P^{K-1}\delta_D + (K-1)P^{K-2}\delta_\Psi E_G/e\}.
\end{aligned}
\tag{I.31}
$$

For the outer strip, real-bank likelihood weights remain real; only the auxiliary scalar becomes complex. Integrating the absolute bank densities after the change of variable to the bank-scaled auxiliary radius gives

$$\begin{aligned} M_{ba} &= (1+K)\{(4\cos d_v)^{-1} + m(\cos d_v)^{-2}\}, \\ M_p &= (4\cos d_v)^{-1} + m(\cos d_v)^{-2}, \quad M_g = KE_G/\cos d_v, \\ E_{\mathrm{alias},X} &= \frac{2M_X\rho_v}{1-\rho_v}, \qquad \rho_v = e^{-2\pi d_v/h_v}, \quad X \in \{ba, p, g\}. \end{aligned} \tag{I.32}$$

At the lower end, the densities and their derivatives decay exponentially. At the upper end split Gaussian banks at a latent threshold proportional to the auxiliary coordinate: on the bounded-latent part the likelihood sum gives double-exponential decay; on its complement Gaussian tails give rapid decay. Differentiating produces polynomials in the bank-scaled auxiliary radius times its damped exponential, so the same split applies at every fixed derivative order. This establishes the contour-edge and Schwartz conditions for the outer use of (I.27).

Omitted outer points require discrete-tail bounds. For the two bank kernels, the total variations below control a right-grid tail by its continuous integral plus a mesh-times-variation term. If the boundary is already past the peak, monotone right rectangles give the tighter continuous bound.

$$\begin{aligned} &f_1(v) = se^{-s}, \quad f_2(v) = s^2e^{-s}, \quad \int_{\mathbb{R}} |f_1'(v)|\,dv = 2/e, \quad \int_{\mathbb{R}} |f_2'(v)|\,dv = 8/e^2, \\ &l_1 = \frac{h_v e^{v_-}}{e^{h_v}-1}, \quad l_2 = \frac{h_v e^{2v_-}}{e^{2h_v}-1}, \quad E_{\mathrm{low},p} \le Kl_1/4 + mK^2 l_2, \quad E_{\mathrm{low},ba} \le (1+K)E_{\mathrm{low},p}, \\ &T = 4, \quad U = e^{v_+}, \quad \eta = 2^{-m}e^{-C_aT}, \quad U\eta > 2, \\ &p_T = 2\varphi(T)/T, \quad t_{2T} = 2\varphi(T)(T+1/T), \quad C_{\mathrm{bad}} = \tfrac{1}{4}(1+2h_v/e) + m(1+8h_v/e^2), \\ &G_p = \{\tfrac{1}{4} + m(1+U\eta)\}e^{-U\eta}, \quad E_{\mathrm{up},p} \le G_p + C_{\mathrm{bad}}p_T^K, \\ &E_{\mathrm{up},ba} \le (1+K)G_p + C_{\mathrm{bad}}(p_T^K + Kt_{2T}p_T^{K-1}). \end{aligned} \tag{I.33}$$

Here Z_k is the scalar first latent coordinate used by the collinear likelihood and its one-dimensional inner integration. Define $G = \{\min_{1\le k\le 8}|Z_k| \le T\}$ and $G^c = \bigcap_{k=1}^8 \{|Z_k| > T\}$. On G, the likelihood sum satisfies $S = \sum_k A_y(Z_k) \ge \eta$. On G^c, independence and double-sided Gaussian tails give $\Pr(G^c) \le p_T^K$ and $E[(1+M^2)\mathbf{1}_{G^c}] \le p_T^K + Kt_{2T}p_T^{K-1}$, where $M = \max_k |Z_k|$. Thus the mesh correction is retained on the bad set rather than applying a continuous Laplace tail to every omitted lattice point. For the gradient the corresponding bounds are

$$\begin{aligned} E_{\mathrm{low},g} &\le KE_G l_1, \\ E_{\mathrm{up},g} &\le \sqrt{m}\{\sqrt{K}e^{-U\eta} + (1+2h_v/e)2K\varphi(T)p_T^{K-1}\}. \end{aligned} \tag{I.34}$$

Sum the inner replacement, outer alias and the two discrete tails to obtain a uniform error for each label. The finite teacher mass is at most one plus its L1 error. The labelwise true gradient bound and conservative Hessian bounds below transfer teacher error only once per vector or matrix norm.

$$\begin{aligned} E_X^{\mathrm{unif}} &= E_{\mathrm{in},X} + E_{\mathrm{alias},X} + E_{\mathrm{low},X} + E_{\mathrm{up},X}, \\ \|g_{\xi,y}\| &\le \sqrt{m}E\max_k |Z_k| \le \sqrt{mK} = 8, \\ \|H_{ba,y}\| &\le (2m+\tfrac{1}{4})(1+K) = 146.25, \quad \|H_{p,y}\| \le m + \tfrac{1}{4} = 8.25, \\ E_g &\le (1+\delta_t)E_g^{\mathrm{unif}} + 8\delta_t, \quad E_{ba} \le (1+\delta_t)E_{ba}^{\mathrm{unif}} + 146.25\delta_t, \\ E_p &\le (1+\delta_t)E_p^{\mathrm{unif}} + 8.25\delta_t. \end{aligned} \tag{I.35}$$

For the Hessian, use each label's block-diagonal bias/radial principal blocks and its transverse block; folding preserves exactly these objects. The exact teacher mixed block is zero. Finite teacher probabilities need not satisfy a bitwise symmetry

test. The interval finite matrices retain their weight and arithmetic uncertainty; Frobenius upper bounds dominate operator norms when comparing them with saved dyadic matrices.

The final outward-rounded bounds are as follows. The saved matrices are exact dyadic interpretations of the previous symmetric binary64 matrices; their positivity is proved by interval LDL. A rational basis of the fixed transverse slice is tested against its Gram matrix, rather than assumed orthonormal.

The rational basis has columns $V_j = e_j - (a_{0,j}/a_{0,8})e_8$, $1 \le j \le 7$. The test is $V^\top (S_p - .014I)V \succ 0$, which certifies the physical curvature bound relative to $V^\top V$.

$$\begin{aligned} \|\nabla F_8^{q_*}(\theta_0)\|_2 &< 7.572\,10^{-10}, \\ \|H_{ba}(\theta_0) - S_{ba}\|_{\mathrm{op}} &< 1.138\,10^{-8} < .001, \\ \|H_p(\theta_0) - S_p\|_{\mathrm{op}} &< 2.531\,10^{-9} < .001, \\ S_{ba} \succ .050 I_{16}, &\qquad w^\top S_p w > .014\|w\|_2^2 \quad (0 \neq w \perp a_0). \end{aligned} \tag{I.36}$$

Contribution	Gradient L2	Radial Hessian operator	Transverse Hessian operator
Inner replacement	2.49106e−10	9.04057e−9	2.27090e−9
Outer alias	5.00387e−10	2.33628e−9	2.59587e−10
Outer lower tail	2.14647e−13	2.14005e−13	2.37783e−14
Total mathematical error, including teacher	<7.497076e−10	<1.137708e−8	<2.530514e−9
Finite-box norm or distance to saved matrix	<7.490176e−12	<1.456e−13	<2.663e−14

Table decimals summarize bounds; exact dyadic endpoints and rational comparisons determine (I.36). The omitted upper-tail contribution is included in the total. No 41-class, 256-label or matrix-entry multiplier is inserted into a bound already expressed in the corresponding vector or operator norm.

I.8.4. Root inclusion and 23-direction local minimality

A uniform full-parameter Hessian variation bound turns the centre certificate into a neighbourhood certificate. Let the bank envelope below include both latent coordinates. Conditional log-kernel derivatives of orders one, two and three are bounded by the stated powers. Differentiating log-sum-exp gives a mean third derivative, three covariance terms and a centred third moment; their bounds give the displayed constant. Gaussian moments justify differentiation under expectation.

$$\begin{gathered} q_Z = \sqrt{1 + \max_{k \le K} \|Z_k\|^2}, \\ \|D\ell_k\| \le \sqrt{m} q_Z, \quad \|D^2 \ell_k\| \le q_Z^2/4, \quad \|D^3 \ell_k\| \le q_Z^3/4, \\ L_F \le (\tfrac{1}{4} + \tfrac{3}{2}\sqrt{m} + 8m^{3/2})[1 + 2dK + dK(dK+2)]^{3/4} < 14069, \quad m = K = 8,\ d = 2, \\ \|H_F(\theta) - H_F(\theta')\|_{\mathrm{op}} \le 14069 \|\theta - \theta'\|_2. \end{gathered} \tag{I.37}$$

For example each covariance term is at most twice the product of the first- and second-derivative bounds, and the centred third moment is at most eight times the cube of the first bound. The bank's squared envelope is dominated by one plus a chi-square variable with sixteen degrees of freedom. The integer cap is checked by rational upper bounds, not fitted from numerical derivatives. With the preassigned physical radius,

$$\begin{aligned} r = 5\,10^{-7}, &\qquad m_r = m_{16} = .050 - .001 - 14069r = .0419655, \\ m_T = .014 - .001 - 14069r = .0059655, &\qquad \epsilon_g = 7.572\,10^{-10} < m_r r = 2.098275\,10^{-8}. \end{aligned} \tag{I.38}$$

The four-coordinate closed ball has a strongly convex restricted objective. On its sphere, integrating the Hessian along the centre-to-boundary line gives the outward inequality below. A closed-ball minimum cannot lie on the boundary, since an inward move would decrease the objective. It is therefore an interior restricted root, unique within this ball by strong convexity. Exact symmetry in (I.22) lifts it to a full stationary point. Strong monotonicity further gives its location.

$$\begin{aligned} \delta^\top \nabla f(\xi_0+\delta) &\geq -\epsilon_g r + m_r r^2 > 0 \quad (\|\delta\| = r), \\ \nabla F_8^{q_*}(\theta_*) &= 0, \qquad \theta_* = (0, a_*, 0), \qquad \|\theta_* - \theta_0\|_2 < \epsilon_g/m_r < 1.805\,10^{-8}. \end{aligned} \tag{I.39}$$

The fixed slice is $a_0^\top v = 0$. The derivative of this constraint along a common latent rotation is $a_0^\top u$, positive throughout the small neighbourhood. The implicit function theorem gives a unique small rotation taking any nearby point into the slice. At the true root, reflection makes the radial/transverse mixed blocks zero; (I.38) makes its 16 radial and seven slice-transverse directions positive. Taylor's theorem gives a strict slice minimum, and rotation invariance lifts it to a strict local minimum modulo common rotation. The 24-dimensional Hessian is singular along the common-rotation direction, as rotation invariance requires; the strict minimum is modulo that rotation, with the 16 bias/radial and seven transverse directions certified positive.

I.8.5. Risk direction, fixed small step and scope

The same centre and a fixed paired dyadic vector define the mathematical unit direction $d_{\mathrm{risk}} \in \mathbb{R}^{24}$. Candidate probabilities and their directional derivatives are enclosed by 177 one-dimensional full-weight nodes with spacing one eighth on the interval from minus eleven to eleven. Strip and Gaussian-tail errors are added; teacher probability errors transfer through the posterior ratios. Jensen's inequality supplies a uniform positive probability bound on the physical ball, and the posterior negative-Hessian identity bounds risk curvature.

The direction-defining vector h in (I.40) differs from the grid spacings h_t, h_z, h_v and the generic scalar mesh in (I.27). Its entries come from pair symmetrization of a numerical radial risk gradient; certification uses the fixed exact dyadics without assuming equality to the true gradient. Its first four components are the following exact dyadics; set $h_{9-j} = h_j$ and use the mathematical norm $\|h\|_2 = \sqrt{\sum_{j=1}^{8} h_j^2}$.

Component	Exact dyadic
h_1	$-8589911184852041/1152921504606846976$
h_2	$-2443582330405755/288230376151711744$
h_3	$-5282230754430109/576460752303423488$
h_4	$-2739850785036709/288230376151711744$

$$\begin{aligned} d_{\mathrm{risk}} &= (0, -h/\|h\|_2, 0), \\ B_{h,y} &= E[A_y(Z) Z \sum_i h_i \{y_i - \sigma(a_{0,i} Z)\}], \\ DR(\theta_0)[d_{\mathrm{risk}}] &= \sum_y q_*(y) \frac{B_{h,y}}{Q_y(\theta_0)\|h\|_2} \\ &\in [-.024564580310120128, -.024564580310120124], \\ Q_y(\theta) &\geq q_{\min} := 2^{-8} \exp\{-3r/2 - [r^2 + (\|a_0\|_2 + r)^2]/8\}, \\ \|H_R(\theta)\| &\leq M_R := 24/q_{\min} < 8784.371, \\ DR(\theta)[d_{\mathrm{risk}}] &< -\nu, \quad \nu = .0201723, \qquad \|\theta - \theta_0\|_2 \leq r. \end{aligned} \tag{I.40}$$

The numerator $B_{h,y}$ is the derivative of Q_y along $+(0, h, 0)$, so $DQ_y(\theta_0)[d_{\mathrm{risk}}] = -B_{h,y}/\|h\|_2$. The negative log probability in the risk contributes a second minus sign, giving the positive sign in (I.40).

For completeness, the candidate probability and numerator boundary norms in a width-one strip are at most the Gaussian shift factor times the product of loading secants, and this quantity times the secant-weighted absolute direction sum times

square root of two, respectively. Formula (I.27) bounds their lattice errors; omitted tails are at most twice the Gaussian endpoint density divided by the endpoint, and twice that density times the absolute direction sum. Positive probability lower bounds permit interval division. The teacher L1 allowance multiplies the largest absolute posterior ratio once.

To verify the curvature bound, a label's negative-log-probability Hessian is the posterior mean conditional negative Hessian minus the posterior score covariance. Both terms are positive semidefinite, so their difference has norm at most the larger individual norm. The likelihood is at most one; dividing prior score moments by the uniform probability lower bound gives the stated bound. The direction is not selected after evaluating the root.

The preassigned step remains inside the certified ball even when started at the unknown exact root. It preserves zero biases, hence each observed Bernoulli marginal is exactly fair by Gaussian central symmetry, while row logit variances change with the row lengths. It is not the fixed-row-norm angular path of I.7. Radial curvature and the uniform risk derivative give

$$\begin{aligned} t = 10^{-7}, \quad \|\theta_* - \theta_0\|_2 + t < r, \\ F_8^{q_*}(\theta_* + t d_{\mathrm{risk}}) - F_8^{q_*}(\theta_*) \geq \tfrac{1}{2} m_{16} t^2 = 2.098275\,10^{-16} > 0, \\ R(\theta_*) - R(\theta_* + t d_{\mathrm{risk}}) > \nu t = 2.01723\,10^{-9}, \qquad R(\theta_*) > 2.01723\,10^{-9}. \end{aligned} \tag{I.41}$$

This is an analytic consequence of the certified neighbourhood, not a subtraction of nearly equal float64 losses. It certifies a strict conflict at one population local minimum, not the size of a typical training improvement. The positivity of KL proves the final inequality.

Risk magnitude from stored probability intervals. The preceding small-step bound remains the core existence certificate. The teacher and candidate probability intervals also allow a deterministic KL enclosure at the centre and its transfer to the exact root. The observed score information is bounded by the complete-data information, so the following derivative bound uses only probability ratios.

$$\begin{aligned} \chi^2(q_* \| Q_0) = \sum_y \frac{(q_*(y) - Q_y(\theta_0))^2}{Q_y(\theta_0)}, \quad I_{\mathrm{obs}}(\theta_0) \preceq I_{\mathrm{complete}}(\theta_0) \preceq \tfrac{1}{4} I_{24}, \\ \|\nabla R(\theta_0)\|_2 \leq \tfrac{1}{2}\sqrt{\chi^2(q_* \| Q_0)}, \\ |R(\theta_*) - R(\theta_0)| \leq \tfrac{1}{2}\sqrt{\chi^2(q_* \| Q_0)}\,\eta_* + \tfrac{1}{2} M_R \eta_*^2, \quad \eta_* = \epsilon_g / m_r. \end{aligned} \tag{I.42}$$

Indeed the conditional complete-data score has cross-row information zero and within-row information at most one quarter times the expected outer product of the augmented Gaussian input, which is the identity. Conditional expectation gives the observed score, decreasing information by Jensen's inequality. Its model-probability-weighted mean is zero; Cauchy–Schwarz applied to the target-minus-model weights then yields (I.42). No target probability is renormalized, and no old floating-point gradient is substituted for the full gradient norm.

Endpoint-only interval composition gives a centre risk about 0.01043357299286302, chi-square divergence about 0.021791388777069481, a full gradient bound below 0.073809534, and a root-transfer allowance below 1.334e−9. Outward rational rounding yields the rigorous enclosure

$$R(\theta_*) \in [.0104335716,\ .0104335744]. \tag{I.43}$$

This additional bound is a deterministic composition of saved probability intervals and the certified root distance, with no new integral, fit or random sample. It explains why the conservative positive-risk lower bound in (I.41) should not be read as the actual risk magnitude.

The evidence consists of complete interval finite sums plus the analytic errors above. Independent alternative implementations recomputed eight selected teacher labels over their full grid and two of the 41 gradient/Hessian classes over their full grids; this is not independent recomputation of every class. All endpoint uncertainties enter the final certificates. The earlier empirical fits, weak-loading K2 precision failures, K11/K12 numerical continuation and mixed profile comparisons retain their original scope. The result does not certify a globally optimal solution, an optimizer's probability of entering the neighbourhood, a universal critical K, or numerical detectability of the tiny guaranteed finite step.

I.9. Qualitative stability under target perturbations

Corollary I.1 (qualitative teacher-law persistence). There is a neighbourhood of the exact teacher probability vector in the interior of the probability simplex such that every target in that neighbourhood has a nearby rank-one strict local minimum of its K8 objective modulo common rotation, with negative integrated-risk derivative along the same fixed direction $d_{\mathrm{risk}} \in \mathbb{R}^{24}$. The perturbed probabilities need not preserve complement or reversal symmetry. The conclusion also holds for the distributions induced by all sufficiently nearby teacher parameters in the original two-factor model. The neighbourhood is not quantified.

Proof. Use all 23 coordinates on the fixed slice $\mathcal{S} = \{a_0^\top v = 0\}$, not the four-coordinate symmetric branch. Let $E_{\mathcal{S}}$ have orthonormal columns spanning its tangent space and use affine coordinates for probabilities summing to one. The finite-particle objective is linear in the target and smooth in the parameters, with Gaussian polynomial domination permitting each derivative. At the certified root define

$$\begin{aligned} \mathcal{F}(s,q) &= E_{\mathcal{S}}^\top \nabla_\theta F_8^q(\theta_* + E_{\mathcal{S}} s), \\ \mathcal{F}(0,q_*) &= 0, \qquad D_s\mathcal{F}(0,q_*) = E_{\mathcal{S}}^\top H_F(\theta_*) E_{\mathcal{S}} \succ 0. \end{aligned} \tag{I.44}$$

The implicit function theorem gives a unique local smooth slice-critical branch $\theta(q)$. Take a sufficiently small slice neighbourhood centred at $\theta_* = (0, a_*, 0)$ that is invariant under $T(b,u,v) = (b,u,-v)$ and lies inside the theorem's uniqueness neighbourhood; shrink the target neighbourhood so that the branch remains there. For every target, latent reflection leaves the objective and the fixed slice invariant. The reflection of its slice-critical point is therefore another slice-critical point in the same neighbourhood. Uniqueness makes them equal. Continuity keeps $u(q)$ nonzero, so

$$\begin{aligned} F_8^q(T\theta) = F_8^q(\theta), \qquad T\mathcal{S} = \mathcal{S}, \qquad T\theta(q) = \theta(q), \\ v(q) = 0, \qquad \operatorname{rank} W(q) = 1. \end{aligned}$$

Slice criticality means that the ambient gradient is proportional to the slice normal. Rotation invariance makes that gradient orthogonal to the orbit tangent; the normal has nonzero product with this tangent because the original transverse condition persists. Thus the ambient gradient is zero, without imposing bias or paired-row constraints on the perturbed target. Positive slice curvature persists by continuity, giving strict local minimality modulo rotation.

$$\begin{aligned} \theta(q) &= \theta_* + E_{\mathcal{S}} s(q), \qquad \nabla F_8^q(\theta(q)) = 0, \\ R_q(\theta) &= D_{\mathrm{KL}}(q \| p_\theta), \qquad DR_q(\theta(q))[d_{\mathrm{risk}}] < 0. \end{aligned} \tag{I.45}$$

The final strict inequality follows by continuity of the directional risk derivative, since all probabilities remain positive near the interior teacher vector and the certified root. It also gives positive risk. For the correctly specified teacher subclass, each observed probability is continuous in the teacher parameters: its likelihood is continuous and bounded by one, so dominated convergence applies. The inverse image of the probability neighbourhood is thus a neighbourhood of the original teacher parameters, and the conclusion holds there. This does not assert that every nearby probability vector belongs to the original two-factor model.

This is a qualitative consequence of nondegeneracy, not a new teacher-grid experiment or a quantified applicability region. No additional K range is claimed. The nearby branch remains rank one, while its biases and paired row lengths need not retain their original symmetry. In particular, the fair observation marginals at the original root and along its certified risk step in I.8.5 are not asserted for perturbed teachers or their roots.

I.10. A fixed-marginal angle conflict at the same exact root

We strengthen the witness in I.8 using its same exact teacher q_*, centre $\theta_0 = (0, a_0, 0)$ and root $\theta_* = (0, a_*, 0)$. No root is refitted. The new path changes relative angles while keeping every row length fixed; it diagnoses the local minimum in the full model, rather than imposing a constraint on the original learning problem.

I.10.1. Direction and full path derivative

Define the integer vector and mathematical unit angle direction by

$$
\begin{aligned}
(u_1,u_2,u_3,u_4) &= (-8440215,-6758298,-4585241,-1673686),\\
u_{9-i} &= -u_i \quad (1\le i\le 4), \qquad S=\sum_i u_i^2 = 281474961915412,\\
d = d^{(8)}_{\mathrm{ang}} &= u/\sqrt{S}, \qquad \|d\|_2=1, \qquad \mathbf{1}^\top d = 0.
\end{aligned}
\tag{I.46}
$$

Saved numerical risk-angle information at the earlier P point motivated this direction. Its exact construction was fixed before the new integral evaluation and was not changed after evaluation; this is an exploratory existence witness, not a blindly selected direction. All quadratic coefficients use exact ratios such as $(u_i-u_j)^2/S$, without rounding $\sqrt{S}$. The angle normalization is distinct from unit Cartesian path speed. This direction is also distinct from both d_{risk} in I.8.5 and $d^{(12)}_{\mathrm{ang}}$ in I.6–I.7.

For a collinear loading a, let Z,X be independent standard Gaussians and set $\eta_i(t)=a_i\{Z\cos(td_i)+X\sin(td_i)\}$. Write $p_i(z)=\sigma(a_iz)$, $r_i=y_i-p_i(z)$, $D_i=p_i(1-p_i)$, $A_y(z)=\prod_i p_i^{y_i}(1-p_i)^{1-y_i}$ and $Q_y=EA_y(Z)$. For this paragraph define $U=\sum_i a_id_ir_i$, $V=\sum_i a_id_i^2r_i$, $G=\sum_i a_ir_i$ and $T=\sum_i a_i^2d_i^2D_i$. Differentiating the likelihood and integrating out X gives

$$
\begin{aligned}
W_i(t) &= a_i(\cos(td_i),\sin(td_i)), \qquad b_i(t)=0,\\
\dot\eta_i(0) &= a_id_iX, \qquad \ddot\eta_i(0) = -a_id_i^2Z,\\
Q_y'(0) &= 0, \qquad Q_y''(0) = EB_y(Z),\\
B_y(z) &= A_y(z)\{U^2-T-zV\}.
\end{aligned}
\tag{I.47}
$$

The first derivative vanishes by reflection in X. In particular, this is not a negative first-order risk direction. The acceleration term $-zV$ cannot be dropped: a Cartesian transverse Hessian alone differentiates a different path. Bounded real sigmoid derivatives and Gaussian polynomial envelopes justify these differentiations.

Set $P_y(z)=\sum_{i<j}a_ia_j(d_i-d_j)^2r_ir_j$. Expansion gives $P_y=GV-U^2$, $A_y'=A_yG$ and $V'=-T$. Consequently

$$
\begin{aligned}
B_y + A_yP_y &= (A_yV)' - zA_yV = \{\varphi A_yV\}'/\varphi,\\
Q_y''(0) &= -E[A_y(Z)P_y(Z)], \qquad N_y := E[A_y(Z)P_y(Z)],\\
\mathcal{H}_R(a;d) &:= \left.\frac{d^2R(\theta(t))}{dt^2}\right|_{t=0} = \sum_y q_*(y)\frac{N_y}{Q_y} = \sum_y q_*(y)E_{\pi_y}P_y,\\
\pi_y(dz) &= A_y(z)\varphi(z)\,dz/Q_y.
\end{aligned}
\tag{I.48}
$$

Here φA_yV vanishes at both real infinities because $A_y\le 1$ and V is bounded. Thus the first line is a Gaussian weighted total derivative, and the second line follows only after integration. The integrands B_y and $-A_yP_y$ are not asserted to be equal pointwise. Differentiating the negative log probability supplies the second minus sign; $Q_y'(0)=0$ removes its squared-first-derivative term. Constant d gives $P_y=0$, as required for common rotation.

I.10.2. Centre enclosure and an alternative path expression

At the exact centre in (I.21), $|a_{0,i}|<1$ and $\sum_{i<j}(d_i-d_j)^2=8$, so $|P_y|\le 8$. All 256 labels are evaluated individually. On the closed strip $|\operatorname{Im} z|\le 1$, set $c=\sec(1/2)$. The inequality $|1+se^{iv}|\ge(1+s)\cos(v/2)$ for $s\ge 0$, $|v|\le 1$, bounds each complex Bernoulli factor by c times its real-axis value. Hence $|A_y|\le c^8$, $|r_i|\le c$ and $|D_i|\le c^2/4$. The two numerator expressions have separate one-sided boundary-integral and omitted-lattice bounds:

$$
\begin{aligned}
h = 1/8, \quad N = 88, \quad L = Nh = 11, \quad c = \sec(1/2),\\
M_{\mathrm{pair}} = e^{1/2}c^8\, 8c^2, \qquad M_Q = e^{1/2}c^8,\\
M_{\mathrm{path}} = e^{1/2}c^8\{8c^2 + c^2/4 + c\sqrt{2}\},\\
T_{\mathrm{pair}} = 16\varphi(L)/L, \qquad T_Q = 2\varphi(L)/L,\\
T_{\mathrm{path}} = (8 + 1/4)2\varphi(L)/L + 2\varphi(L),\\
\epsilon_j = \frac{2M_j}{e^{2\pi/h} - 1} + T_j, \qquad j \in \{\mathrm{pair}, \mathrm{path}, Q\}.
\end{aligned}
\tag{I.49}
$$

For the pair expression the integrand is $\varphi A_y P_y$; for the alternative it is $-\varphi B_y$. On a strip boundary, $|U| \le \sqrt{8}c$, $|T| \le c^2/4$, $|V| \le c$ and $E|Z \pm i| \le \sqrt{2}$, which prove M_{path}. Contour translation gives Fourier decay $M_j e^{-|\xi|}$; summing the nonzero Poisson frequencies proves the alias term, as in (I.27). The Gaussian envelope makes the contour end boundaries vanish. On the real axis the two numerator factors are bounded by 8 and $8 + 1/4 + |z|$, respectively. The omitted full-weight nodes $|k| > N$ are bounded by adjacent continuous tails, using the monotonicity of $\varphi(x)$ and $x\varphi(x)$ for $x \ge L \ge 1$. The identities $\int_L^\infty x\varphi(x)dx = \varphi(L)$ and $\int_L^\infty \varphi(x)dx \le \varphi(L)/L$ give the stated tails.

Each 177-node sum includes both endpoints with full weight and uses 192-bit Arb arithmetic. Its outward arithmetic interval is retained separately from the analytic error. The pair error is approximately $1.44001098659947\,10^{-20}$ per label, not an error for the final curvature. The alternative implementation forms the complete path derivative in (I.47), retaining acceleration, with separate integer-label probability products and its own bounds in (I.49); it also recomputes Q_y.

Let $[F_y]$ be the pair finite-sum interval and $[Q_y]$ the original strictly positive true-probability enclosure used in I.8.5. Let $\widehat{q}_y$ be the exact midpoint of the two outward endpoints of the saved teacher finite-sum interval. No normalization is applied to these midpoints. The teacher certificate gives $\|q_* - \widehat{q}\|_1 < \epsilon_q := 4.894\,10^{-21}$. Interval composition is

$$
\begin{aligned}
[N_y] &= [F_y] + [-\epsilon_{\mathrm{pair}}^{\mathrm{up}}, \epsilon_{\mathrm{pair}}^{\mathrm{up}}],\\
[\mathcal{H}_0] &= \sum_y \widehat{q}_y [N_y]/[Q_y] + [-8\epsilon_q, 8\epsilon_q],\\
|N_y/Q_y| &= |E_{\pi_y} P_y| \le 8, \qquad 8\epsilon_q = 3.9152\,10^{-20}.
\end{aligned}
\tag{I.50}
$$

Here ϵ^{up} denotes a certified outward upper endpoint. The teacher allowance occurs once, through the true posterior ratio bound; the label-wise numerator error is propagated through division and weighting. The alternative centre enclosure substitutes its path sum and ϵ_{path}, but uses the same old $[Q_y]$. Its newly computed probabilities are a cross-check, not a different denominator for this composition.

After their respective analytic enlargements, all 256 pairs of numerator intervals overlap; all 256 recomputed probability intervals overlap their old counterparts. These checks compare enclosures of equal integrals, not identical finite sums, and do not replace the analytic error proof. The implementations share Arb, teacher inputs and the root certificate. A separate exact rational-arithmetic check recomposes saved endpoints only; it does not recompute transcendental functions or integrals. They are different-expression implementation checks, not independent end-to-end certifications with different arithmetic backends.

I.10.3. Scalar transfer to the certified root

We require a variation bound for the one scalar $\mathcal{H}_R(a; d)$ at fixed $q_*, b = 0, d$, not a bound on the full risk Hessian. Suppose $\max_i |a_i| \le A$ along the segment in question, with m rows and $\|d\|_2 = 1$. Integrating $(zA_y\varphi)'$ over the real line gives

$$
\begin{aligned}
E_{\pi_y} Z^2 &= 1 + E_{\pi_y}[Z\, \partial_z \log A_y], \qquad |\partial_z \log A_y| \le mA,\\
\mu_y^2 &\le E_{\pi_y} Z^2 \le 1 + mA\mu_y, \qquad \mu_y := E_{\pi_y}|Z|,\\
\mu_y &\le M_1 := \frac{mA + \sqrt{m^2A^2 + 4}}{2}.
\end{aligned}
\tag{I.51}
$$

The boundary vanishes by Gaussian decay, and Cauchy–Schwarz gives the second line. No numerical lower bound on Q_y

is needed for this posterior-moment estimate. Write $c_{ij} = (d_i - d_j)^2$, so $\sum_{i<j} c_{ij} \le m$ and $|P_y| \le mA^2$. For a unit loading direction h_0, direct differentiation gives

$$\begin{aligned} D_a P_y[h_0] &= \sum_{i<j} c_{ij}\big[(h_{0,i}a_j + a_i h_{0,j}) r_i r_j \\ &\qquad - z a_i a_j (D_i h_{0,i} r_j + r_i D_j h_{0,j})\big], \\ |D_a P_y[h_0]| &\le 2mA + (mA^2/2)|z|, \qquad |D_a \log A_y[h_0]| \le \sqrt{m}|z|, \\ D_a E_{\pi_y} P_y[h_0] &= E_{\pi_y} D_a P_y[h_0] + \mathrm{Cov}_{\pi_y}(P_y, D_a \log A_y[h_0]), \\ |\,\mathrm{Cov}_{\pi_y}(P_y, D_a \log A_y[h_0])| &\le 2m^{3/2} A^2 E_{\pi_y}|Z|. \end{aligned} \tag{I.52}$$

The covariance retains differentiation of the posterior denominator; its two terms are each bounded using $|P_y| \le mA^2$. Weighting by the fixed probability vector and integrating the derivative bound along the loading segment proves

$$\begin{aligned} |\mathcal{H}_R(a;d) - \mathcal{H}_R(a';d)| &\le L_{\mathrm{ang}} \|a - a'\|_2, \\ L_{\mathrm{ang}} &= 2mA + (mA^2/2 + 2m^{3/2}A^2) M_1, \\ L_{\mathrm{ang}}\big|_{m=8,A=1} &= 16 + (4 + 16\sqrt{8})(8 + \sqrt{68})/2 < 417. \end{aligned} \tag{I.53}$$

For an exact rational verification of the last cap, $\sqrt{8} < 2829/1000$ and $\sqrt{68} < 8247/1000$ give an upper bound $52024513/125000 < 417$. All coordinates of $a_0 \pm (1.805\,10^{-8})\mathbf{1}$ lie in $(0,1)$. By (I.39), the segment to the same true root therefore satisfies $A = 1$, with no bias or teacher variation. Thus

$$|\mathcal{H}_R(a_*;d) - \mathcal{H}_R(a_0;d)| < 417(1.805\,10^{-8}) = 7.52685\,10^{-6}. \tag{I.54}$$

This uses neither the risk-Hessian norm cap in (I.40) nor the finite-objective variation constant in (I.37) as a risk-curvature variation bound. Combining (I.50) and (I.54) yields the following conservative decimal enclosures, each containing both corresponding saved dyadic certificates:

Object	Lower endpoint	Upper endpoint
$\mathcal{H}_R(a_0;d)$	-0.004511298531248709	-0.004511298531248699
$\mathcal{H}_R(a_*;d)$	-0.004518825382	-0.004503771681

I.10.4. Fixed-marginal local conflict

Finally take $a = a_*$ in (I.47). The preceding enclosure and reflection give $R'(0) = 0$ and $R''(0) < -0.00450377$. Twice differentiability implies risk descent for every sufficiently small nonzero step on either side. Since d is nonconstant and all $a_{*,i} > 0$, some two-row minor is $a_{*,i}a_{*,j}\sin(t(d_j - d_i)) \ne 0$ for all sufficiently small nonzero t. Such rank-two points lie outside the common-rotation orbit of the rank-one root. Its full-space strict local minimality modulo rotation, established in I.8.4, then proves that for some $\varepsilon > 0$,

$$\begin{aligned} &R'(0) = 0, \qquad R''(0) < -0.00450377, \\ &0 < |t| < \varepsilon \quad\Longrightarrow\quad R(\theta(t)) < R(\theta_*), \\ &F_8^{q_*}(\theta(t)) > F_8^{q_*}(\theta_*), \quad \operatorname{rank} W(t) = 2. \end{aligned} \tag{I.55}$$

The entire path preserves every candidate logit law $N(0, a_{*,i}^2)$ and every observation marginal $\mathrm{Bernoulli}(1/2)$. The latter correctly match the teacher; the candidate logit variances need not match the teacher's $(4/5)^2$. Relative covariances change as $a_{*,i}a_{*,j}\cos(t(d_i - d_j))$, so this is a change of dependence, not a common rotation. It supplies the fixed-logit-marginal part of Theorem 3 at the original exact root, while preserving the distinct radial finite-step guarantee in (I.41).

No numerical ε, specified finite-step gain, full angle-spectrum certificate or K11/K12 transition is asserted. In particular the curvature bound is not a certified loss change at $t = .02$. I.9 remains a qualitative result for the radial direction; I.10

alone does not extend all its angle-path properties to perturbed teachers. I.11 gives that extension for a specified zero-bias family. Exact inputs, finite intervals and their evidence mapping are retained with the AngularWitness-v1 certificate; the proof here depends on the displayed scalar identity, integral envelopes and transfer argument rather than on execution logs.

I.11. Quantitative persistence in a changing teacher family

Corollary (quantitative local persistence in an explicit teacher family). Keep the teacher biases zero and all teacher row lengths $4/5$, and replace only its first angle by $-.7+\lambda$. Let q_λ be its observation law and $\mathcal{R}_\lambda(\theta) = D(q_\lambda \| Q_\theta)$. There is a smooth branch $\theta_\lambda = (0, a(\lambda), 0)$ on the entire interval $|\lambda| \le 10^{-3}$, connected to Theorem 3's root, such that every θ_λ is a strict local minimum of its own $F_8^{q_\lambda}$ in the full parameter space modulo common latent rotation. With the same fixed $d = d_{\mathrm{ang}}^{(8)}$ as in (14), set $\theta_\lambda(t) = (0, W_\lambda(t))$, $f_\lambda(t) = F_8^{q_\lambda}(\theta_\lambda(t))$ and $r_\lambda(t) = \mathcal{R}_\lambda(\theta_\lambda(t))$. Then

$$\begin{aligned} W_{\lambda,i}(t) &= a_i(\lambda)(\cos(td_i), \sin(td_i)), \\ f'_\lambda(0) &= r'_\lambda(0) = 0, \\ f''_\lambda(0) &> .0011, \qquad r''_\lambda(0) \le -.004351070278 < 0. \end{aligned}$$

For each λ, some $\varepsilon_\lambda > 0$ makes every $0 < |t| < \varepsilon_\lambda$ raise $F_8^{q_\lambda}$, lower its own true risk $\mathcal{R}_\lambda$ and give $\operatorname{rank} W_\lambda(t) = 2$. The path preserves the candidate logit marginals $N(0, a_i(\lambda)^2)$ and the correct fair observation marginals.

I.11.1. Teacher expansion with a joint remainder

Keep the teacher of (13), except for its first angle, and write $\rho = 4/5$, $\phi = (-.7, -.5, -.3, -.1, .1, .3, .5, .7)$. For $z \in \mathbb{R}^2$, let $A_y^*(z;\lambda)$ denote its Bernoulli product. Differentiating only the first factor gives

$$\begin{aligned} t_i^* &= \rho\{\cos(\phi_i + \lambda \mathbf{1}_{i=1})z_1 + \sin(\phi_i + \lambda \mathbf{1}_{i=1})z_2\}, \\ \partial_\lambda A_y^* &= A_y^*(y_1 - \sigma(t_1^*))\partial_\lambda t_1^*, \qquad q_\lambda(y) = E A_y^*(Z;\lambda), \\ q_\lambda &= q_0 + \lambda q_0' + e_\lambda, \qquad \sum_y e_{\lambda,y} = 0, \\ \|q_\lambda''\|_1 &\le \frac{\rho^2}{3\sqrt{3}} + \frac{\rho}{2}\sqrt{2/\pi} < .442323, \qquad \|e_\lambda\|_1 \le .2211615\lambda^2. \end{aligned} \tag{I.56}$$

The second-derivative bound holds for all real λ: sum the two first-bit derivatives before the other seven factors, use $\sup|\sigma''| = 1/(6\sqrt{3})$, $\sup \sigma' = 1/4$, $E(\partial_\lambda t_1^*)^2 = \rho^2$ and $E|\partial_\lambda^2 t_1^*| = \rho\sqrt{2/\pi}$. Exact probability mass gives the zero sum. Gaussian reflection preserves complement symmetry and fair marginals, without preserving reversal symmetry.

All 256 components of q_0, q_0' are enclosed by the two-dimensional full-weight trapezoidal rule with spacing $\delta = 1/8$, cutoff $L = 10$ and 192-bit arithmetic. For complex coordinate shifts of modulus at most $\eta = 1$, put $\beta_i = \rho\eta(|\cos\phi_i| + |\sin\phi_i|)/2$. The Bernoulli modulus sum is at most $\sec\beta_i$; the derivative factor has bound $\rho\sec^2\beta_1/2$ times a linear Gaussian envelope. With $s = e^{-2\pi\eta/\delta}$ and $\kappa = |\sin\phi_1| + |\cos\phi_1|$, explicit boundary and omitted-node bounds give

$$\begin{aligned} M_q &= e^{\eta^2}\prod_i \sec\beta_i, \\ M_{q'} &= e^{\eta^2}\frac{\rho}{2}\sec^2\beta_1\left(\prod_{i>1}\sec\beta_i\right)\kappa(\sqrt{2/\pi} + \eta), \\ S_0 &= 1 + \delta\varphi(0), \quad S_1 = \sqrt{2/\pi} + 4\delta\varphi(1), \\ T_0 &= 2\varphi(L)/L, \quad T_1 = 2\varphi(L), \\ T_q &= 2S_0T_0, \qquad T_{q'} = (\rho/2)\kappa(T_1S_0 + T_0S_1), \\ \epsilon_j &\le M_j\frac{4s}{(1-s)^2} + T_j, \qquad j \in \{q, q'\}, \\ \|q_0 - q_0^f\|_1 &< 4.894\times10^{-21}, \qquad \|q_0' - (q_0')^f\|_1 < 5.927\times10^{-21}. \end{aligned} \tag{I.57}$$

The alias coefficient sums both single-axis and double-axis frequencies. Discrete tails use monotonic Gaussian tails outside the cutoff and the total variation of $|z|\varphi(z)$ for its whole-axis moment. These are joint label-L^1 bounds, paid once. Arithmetic intervals remain separate; the resulting component enclosures are neither averaged nor normalized.

I.11.2. Eight free candidate row lengths

Reversal pairing cannot reduce the new candidate. The eight row-length parameters a_i can vary separately. We apply the Laplace representations in I.3 and I.8.3 with all eight row lengths free. At $u = e^v$, let $p_i = \sigma(a_i z)$, $\widetilde{p}_i = p_i - 1/2$, $\widetilde{y}_i = y_i - 1/2$, $t = uA_y e^{-uA_y}$ and $\Psi = Ee^{-uA_y}$. The complete centered moments are

$$\begin{aligned} &T_l = E[tz^l], \quad \widetilde{P}_l = E[tz^l\widetilde{p}], \quad \widetilde{PP}_l = E[tz^l\widetilde{p}\widetilde{p}^\top], \\ &R_1 = T_1\widetilde{y} - \widetilde{P}_1, \\ &\Gamma_l = T_l\widetilde{y}\widetilde{y}^\top - \widetilde{y}\widetilde{P}_l^\top - \widetilde{P}_l\widetilde{y}^\top + \widetilde{PP}_l, \\ &D_l = \operatorname{diag}(T_l/4 - \operatorname{diag}\widetilde{PP}_l), \qquad c_K = K\Psi^{K-1}, \quad d_K = K(K-1)\Psi^{K-2}. \end{aligned} \tag{I.58}$$

Use $l = 0, 1, 2$ for $T, \widetilde{P}$ and $l = 0, 2$ for $\widetilde{PP}$, retaining every matrix entry by symmetry. Conditional gradient and Hessian blocks are the integrals over all real v of

$$\begin{aligned} g_y(v) &= -c_K R_1, \\ H_{bb,y}(v) &= c_K D_0 - d_K(T_0\widetilde{PP}_0 - \widetilde{P}_0\widetilde{P}_0^\top), \\ H_{aa,y}(v) &= c_K D_2 - d_K(T_0\Gamma_2 - R_1R_1^\top), \\ H_{\perp,y}(v) &= c_K D_0 - d_K T_0\Gamma_0, \qquad K = 8. \end{aligned} \tag{I.59}$$

Centering is an exact algebraic change of moments. The transverse block lacks a mean-score square because the second Gaussian coordinate has zero mean. Reflection $A_{1-y}(z) = A_y(-z)$ makes these four quantities complement-invariant, allowing 128 representatives weighted by $q_y + q_{1-y}$. No 41-class paired reduction is used. The I.8 contour and discrete-tail bounds apply with $\sum_i \bar{a}_i$ and $\sum_i \sec^2(\bar{a}_i d_z/2)$ over all eight individual upper endpoints. Their uniform gradient and operator errors are denoted $\epsilon_g, \epsilon_H, \epsilon_P$; interval dependence and arithmetic radii remain in the finite matrices.

I.11.3. One root box for the entire interval

Let a_0 be the exact center (I.21), $h = .001$, $X = a_0 + \prod_i[-r_i, r_i]$, $r = (.00075, .00015, \ldots, .00015)$ and $t_h = .2211615h^2$. Write g_y^f, H_y^f for the finite conditional enclosures. Preserve the signed teacher derivative by first forming

$$\begin{aligned} G_1^f(X) &= \sum_y [q'_{0,y}] g_y^f(X), \\ H_{j,\mathrm{fin}}(X) &= \sum_y [q_{0,y}] H_{j,y}^f(X) + [-h, h] \sum_y [q'_{0,y}] H_{j,y}^f(X), \\ E_H &= \epsilon_H(1 + hQ_1) + 146.25t_h, \qquad E_P = \epsilon_P(1 + hQ_1) + 8.25t_h, \quad Q_1 \geq \|q'_0\|_1. \end{aligned} \tag{I.60}$$

Here $j = bb, aa, \perp$; E_H applies to both principal blocks of the full bias/radial Hessian. Teacher component intervals already contain (I.57). The global conditional caps are those of I.8: 8, 146.25 and 8.25. Multiplying each linearized weight by an unrelated interval for λ before summation would discard useful sign cancellation.

Use the certificate's fixed rational symmetric preconditioner A, whose positive definiteness is checked by strict interval LDL, and $T_\lambda(a) = a - AG(a, \lambda)$. For row A_i, let b_i be the midpoint of the joint range of $(Ag_y^f(X))_i$. Zero remainder mass permits subtraction of this same constant from every label. Thus

$$\begin{aligned} \widehat{C}_i &= \min\left\{8\|A_i\|_2, \max_y \sup|(Ag_y^f(X))_i - b_i| + \|A_i\|_2\epsilon_g\right\}, \\ Y_i &= 7.572 \times 10^{-10}\|A_i\|_2 + h\sup|(AG_1^f(X))_i| + h\|A_i\|_2\epsilon_g Q_1 + t_h\widehat{C}_i. \end{aligned} \tag{I.61}$$

The old center residual is retained. Although G_1^f covers X, Y_i is needed only for the displacement at a_0. For the derivative of T_λ, an operator-error allowance gives

$$\begin{aligned} Z_i &= \sum_j \sup|(I - AH_{aa,\mathrm{fin}}(X))_{ij}|r_j + \|A_i\|_2 E_H \|r\|_2, \\ \max_i Z_i/r_i &< .264019, \qquad \max_i (Y_i + Z_i)/r_i < .753428, \\ \min_i (r_i - Y_i - Z_i) &> 8.37210 \times 10^{-5}. \end{aligned} \tag{I.62}$$

Banach's theorem supplies a unique radial root in X for every λ, uniformly in the radius-weighted maximum norm. Invertibility of A equates fixed points with roots. Uniform contraction gives continuity. The old true root is inside X by (I.39), so uniqueness identifies $a(0) = a_*$. Smoothness follows from the implicit function theorem and the uniformly invertible radial Hessian below; local branches agree by uniqueness.

I.11.4. Full stability from original-matrix uncertainty

For each original symmetric 8×8 finite interval matrix with exact endpoints l_{ij}, u_{ij}, form the midpoint before projection:

$$\begin{aligned} (M_0)_{ij} &= (l_{ij} + u_{ij})/2, \qquad B_{ij} = (u_{ij} - l_{ij})/2, \\ \|H_{j,\mathrm{fin}} - M_0\|_{\mathrm{op}} &\le \|B\|_F \le \bar\rho_F, \qquad \sum_{i,j} B_{ij}^2 \le \bar\rho_F^2. \end{aligned} \tag{I.63}$$

The final squared comparison is checked with exact rational arithmetic, including all 64 entries. Let U_0 be the last seven columns of the Householder matrix for $u_0 = a_0/\|a_0\|$, $v_0 = u_0 + e_1$. Its exact orthogonality and $U_0^\top a_0 = 0$ give unit operator norm. Strict interval LDL proves, with each block's own $M_0, \bar\rho_F$,

$$\begin{aligned} M_{0,bb} - (.1 + E_H + \bar\rho_{F,bb}) I_8 &\succ 0, \\ M_{0,aa} - (.02 + E_H + \bar\rho_{F,aa}) I_8 &\succ 0, \\ U_0^\top M_{0,\perp} U_0 - (.005 + E_P + \bar\rho_{F,\perp}) I_7 &\succ 0. \end{aligned} \tag{I.64}$$

Bounding original-matrix uncertainty avoids the extra dependency loss from projecting an interval matrix first. The thresholds remain unchanged. Complement and latent-coordinate reflections, as in I.2, make the other gradient components and mixed blocks zero. At the root, rotation invariance supplies the Ward identity:

$$\begin{aligned} \nabla F_8^{q_\lambda}(0, a(\lambda), 0) = 0, \qquad H_\perp a(\lambda) = 0, \qquad a_0^\top a(\lambda) > 0, \\ H_{bb} \succ .1 I_8, \qquad H_{aa} \succ .02 I_8, \qquad U_0^\top H_\perp U_0 \succ .005 I_7. \end{aligned} \tag{I.65}$$

The fixed slice is transverse to the rotation direction throughout X; its positivity therefore gives full-space strict local minimality modulo rotation. The number .005 is a bound on $a_0^\perp$, and is not asserted unchanged on the moving $a(\lambda)^\perp$.

I.11.5. Each teacher's own angular risk

Use the fixed angular direction $d = d_{\mathrm{ang}}^{(8)}$ defined in (I.46), with the full path from I.10, including its acceleration. For every $a \in X$, independently enclose all 256 quantities

$$\begin{aligned} Q_y(a) &= E A_y(Z; a) > 0, \\ N_y(a) &= E[A_y(Z; a) P_y(Z; a, d)], \qquad \rho_y(a) = N_y(a)/Q_y(a), \\ \sum_y A_y(z; a) P_y(z; a, d) &= 0, \qquad \sum_y N_y(a) = 0. \end{aligned} \tag{I.66}$$

The node identity follows by summing independent Bernoulli residual cross-products. The 177-node rule and separate probability/numerator bounds in (I.49) apply directly to every member of X because $0 < a_i < 1$. This encloses candidate

movement without a 417-times-displacement transfer. Positive denominators, analytic errors and arithmetic intervals are all included. Set B_* to the midpoint of the joint $\rho_y(X)$ range. Then

$$\begin{aligned} R_0(a) &= \sum_y q_{0,y}\rho_y(a) = \sum_y (q_{0,y} - Q_y(a))\rho_y(a), \\ [r''_\lambda(0)] &\subset [R_0(X)] + [-h,h]\sum_y [q'_{0,y}]\rho_y(X) \\ &\quad + [-t_h,t_h]\max_y \sup|\rho_y(X) - B_*| \\ &\subset [-.004671548834, -.004351070278]. \end{aligned} \tag{I.67}$$

Intersecting the two valid enclosures of R_0 tightens its bound; their overlap alone would not certify arbitrary inputs. Both centerings use exact zero-mass identities, without changing target probabilities. Reflection gives $r'_\lambda(0) = 0$. For finite-objective curvature put $v = \mathrm{diag}(a)d$, $c = (a_0^\top v)/(a_0^\top a)$ and $w = v - ca$. From (I.65),

$$\begin{aligned} w \in a_0^\perp, \qquad & f''_\lambda(0) = v^\top H_\perp v = w^\top H_\perp w \\ & > .005\|w\|^2 \geq .005 a_{\min}^2 \|d - c\mathbf{1}\|^2 \geq .005 a_{\min}^2 > .0011, \\ a_{\min} = \min_i(a_{0,i} - r_i), \qquad & \|d - c\mathbf{1}\|^2 = 1 + 8c^2. \end{aligned} \tag{I.68}$$

Stationarity gives $f'_\lambda(0) = 0$. Twice differentiability now proves the teacher-family corollary and its consequences for each teacher's own sufficiently small two-sided steps. Positive lengths and nonconstant d give rank two, while fixed lengths preserve candidate Gaussian logits and fair observation marginals. No common numerical step radius or family-wide KL interval is inferred.

I.11.6. Nontriviality and certification boundary

Taylor's inequality and the enclosed $\|q'_0\|_1$ show a genuine law change:

$$\begin{aligned} .159347929859 \leq \frac{\|q_\lambda - q_0\|_1}{|\lambda|} \leq .159790252860 \quad (0 < |\lambda| \leq .001), \\ \|q_{\pm.001} - q_0\|_1 \in [.000159347929, .000159790253]. \end{aligned} \tag{I.69}$$

The seven first-row spin-pair derivatives remain strictly positive after the allowance $.442323h$; the joint law of the other seven observations is unchanged. Hence the formerly equal $(1,2)$ and $(8,7)$ moments separate for every nonzero λ, breaking reversal symmetry. To enclose the implicit derivative $a' = -H_{aa}^{-1}G_\lambda$, choose g_C coordinatewise as the midpoint of the joint conditional-gradient range, and set $C_g = \min\{8, \max_y \sup\|g_y^f(X) - g_C\|_2 + \epsilon_g\}$. Since $q'_\lambda - q'_0$ has zero mass and L^1 norm at most $.442323h$, the error in approximating G_λ by G_1^f is at most $\epsilon_g Q_1 + C_g\, .442323h$. Enlarge every entry of the finite H_{aa} enclosure by E_H and perform interval linear solution. This gives

$$\begin{aligned} a'_1(\lambda) - a'_8(\lambda) &\in [.429945195966, .565673464386], \\ \frac{a_1(\lambda) - a_8(\lambda)}{\lambda} &\in [.429945195966, .565673464386] \quad (\lambda \neq 0), \\ |a_1(\pm.001) - a_8(\pm.001)| &\in [.000429945195, .000565673465]. \end{aligned} \tag{I.70}$$

Here $a_1(0) = a_8(0)$ comes from the old root's exact symmetry. The branch therefore leaves the paired-length subspace. The three prescribed box halfwidths retain the following outcomes; "unresolved' ' means insufficient enclosure, not a counterexample.

Halfwidth h	Original root inclusion	Original $bb/aa/\perp$ LDL	Own-risk sign	After midpoint spectral bound
.01	Unresolved	Pass / unresolved / unresolved	Negative	Unresolved
.001	Pass	Pass / pass / unresolved	Negative	Full certificate
.0001	Pass	Pass / pass / pass	Negative	Full certificate

The midpoint argument was a subsequent deterministic refinement of saved matrices; it preserves the original failures and all thresholds. Seven floating-point development points reaching $\pm.1$ do not certify that interval. The endpoint law change in (I.69) locates the proved result at a small quantitative local scale. Exact intervals and their expression checks reside in R03TeacherFamily-v1; no development trajectory is needed for this proof.